Does the Expectations Hypothesis Fail Economically? Evidence from US Repo Rates*

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Abstract

This paper asks whether the expectations hypothesis (EH) fails in an economically meaningful way at the short end of the US repo curve once no parametric model of expectations is imposed, that is, whether a forecast earns the bid–ask spread it pays. Ten maturity pairs are forecast out of sample from 1994 to 2026 with 23 linear and nonlinear models, from the vector autoregression (VAR) of Della Corte, Sarno, and Thornton (2008; DST) to neural networks, under inference that controls for the search across them. After the search, only overnight versus one week rejects the EH. The rejection lives in same-day quotes: without them, the out-of-sample R^2 of DST's VAR falls from 20% to 2%. Perfect sign foresight is worth at most 2.6 to 5.6 basis points a year, and after spreads no forecast traded on its sign earns a significant gain.

Keywords: Expectations hypothesis, Term structure, Term premium, Repo rates, Forecast evaluation, Economic value, Machine learning
JEL: E43, G12, C22, C53

1. Introduction

The expectations hypothesis (EH) holds that a long-term rate equals the average expected short-term rate over its life, up to a constant premium. In the repo market, lending at the one-week rate and rolling overnight loans for a week should then earn the same expected return. Della Corte et al. (2008) (DST) reject this restriction for every pair of US repo maturities from overnight to three months, where Longstaff (2000) could not, yet in sample, trading on the rejections is economically unimportant. Their tests, however, impose a linear vector autoregression (VAR) for expectations, so they test the EH jointly with it (Froot, 1989; Guidolin and Thornton, 2018); DST read the gap between their statistical and economic results as doubt about this model and ask whether "tiny data imperfections" drive the

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rejection. A misspecified model of expectations could create spurious rejections or hide economically important ones.

This paper asks whether the EH fails in an economically meaningful way at the short end once no parametric model of expectations is imposed, where meaningful means that a forecast earns more than the bid–ask spread it pays. The EH does not fail: a forecast traded on its sign is worth at most a few basis points a year per unit of notional, and the statistical rejection that remains is confined to one pair and lives in same-day quotes. The answer matters to anyone who pays the spread to switch daily between term and overnight repo. The paper tests the EH as a restriction on the excess return of rolling short repos over holding a long one, out of sample on daily dealer quotes, with forecasts from DST's VAR to flexible machine-learning models and with inference that accounts for the search across them; the economic answer rests on a bound that holds for any such forecast.

Three findings follow. First, linear and nonlinear forecasts find the same rejection: of the ten pairs, only overnight versus one week rejects the EH out of sample after the search across 23 forecasts and the adjustment for testing ten pairs (Holm-adjusted $p = 0.016$). Second, the rejection is a same-day phenomenon: without the quotes of the forecast date, the out-of-sample R^2 of DST's VAR falls from 20.5% to 1.8%, a drop that holds for every model and stale-quote rule. Third, perfect foresight of the sign bounds the gross value of any forecast in DST's strategy at 2.55 to 5.55 basis points a year per unit of notional (the benchmark of Sarno et al., 2016); it breaks even at spreads of 3 to 9 basis points, against quoted spreads of 4 to 6, and after spreads no forecast traded on its sign earns a significant net gain over 1994–2026. Treasury studies found that predictability need not create economic value (Thornton and Valente, 2012; Sarno et al., 2016), survey-based tests reject the EH without assuming rational expectations (Crump et al., 2024), and the repo premium is positive once the repo spread is counted (Kopchak, 2013); this paper shows, out of sample and after the search, that at the very short end only one of DST's ten rejections survives, carried by the quotes of the day.

The rest of the paper is organised as follows. Section 2 describes the repo quotes, the bid–ask spreads, and the construction of the excess returns. Section 3 states the testable restriction and the out-of-sample design: the information sets, the 25 linear and nonlinear forecasts, the inference that accounts for the search across them, and the trading strategy with its perfect-foresight bound. Sections 4 to 6 follow the answer: where the statistical rejection survives, why it is tied to same-day quotes, and why it is worth little after the spreads a participant pays. Section 7 collects the robustness checks of each finding (Table 21 maps each finding to its checks), Section 8 the size and power of the tests, and Section 9 concludes. Appendix Appendix A gives the details needed for replication.

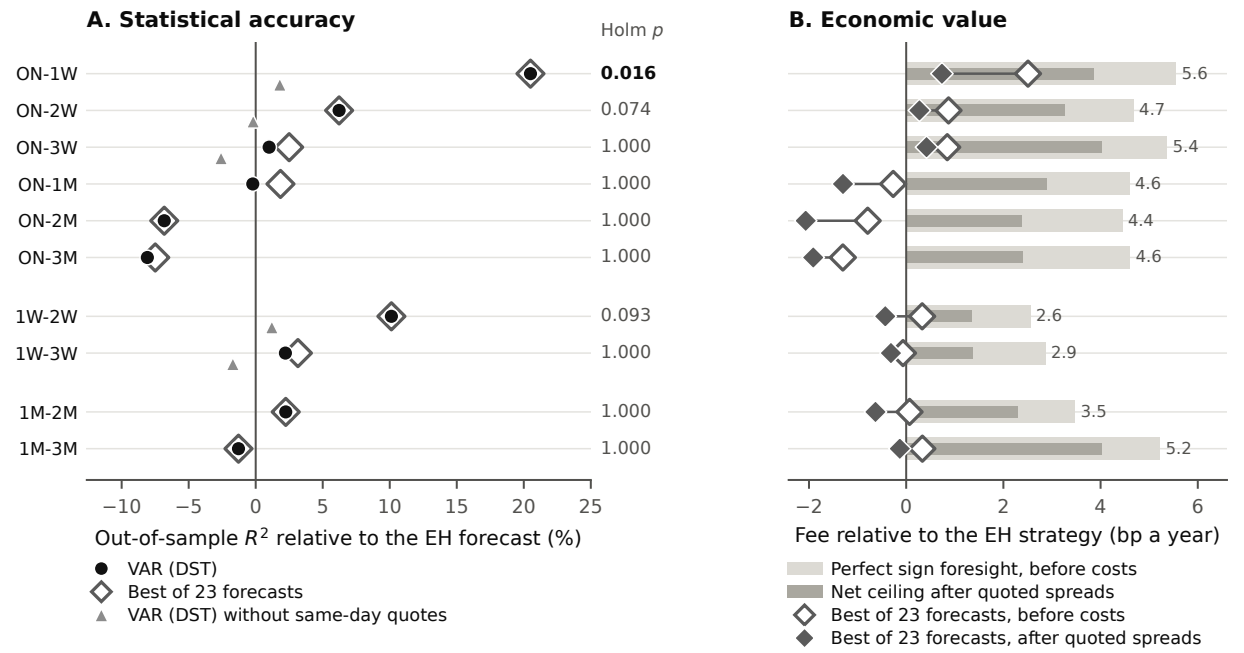


Figure 1: **Statistical Accuracy and Economic Value by Maturity Pair.** Panel A: out-of-sample R^2 (%) against the EH forecast of DST's VAR (circles), the most accurate of the 23 forecasts (diamonds), and the VAR without same-day quotes (triangles); Holm-adjusted SPA p in bold below 5%. Panel B: fee of the sign strategy (bp a year), best of the 23 forecasts before costs (open) and after quoted spreads (filled); light bars: perfect sign foresight; dark bars: net ceiling. Numbers: Table 4; best forecasts: Tables 7 and 16.

2. Data

The repo series combines two Bloomberg pricing sources day by day, follows the policy rate over 1991–2026, and its gaps are documented rather than filled. All series are daily on US business days; rates are simple ACT/360 rates in percent per annum, the quoting convention of the repo market.

2.1. Repo quotes and pricing sources

The study uses Bloomberg closing quotes of US general-collateral (GC) Treasury repo rates for seven maturities: overnight (ON, ticker USRG1T), one, two, and three weeks (1W, 2W, 3W: USRG1Z, USRGCG2Z, USRGCG3Z), and one, two, and three months (1M, 2M, 3M: USRGCGA, USRGCGB, USRGCGC), the maturities of Della Corte et al. (2008) (DST) and Longstaff (2000). Bloomberg does not document the time at which the closing quotes are taken; they are end-of-day dealer indications and can differ from the morning tri-party rate (on 17 September 2019 the overnight close was 3.93% against a TGCR of 5.25%), whereas overnight repo trades mostly before 9 a.m. (Afonso et al., 2021, p. 16). Bloomberg does not document the settlement convention of the quotes either; the construction assumes that each repo starts on the date of its quote. The quotes were retrieved on 18 October 2026.

Bloomberg reports each ticker from two contributor pricing sources, ICUS and PREB. ICUS starts on 21 May 1991 for all maturities; PREB starts on 23 February 2004 for ON and 1W and on 6 June 2013 for the other maturities. ICUS term quotes are unchanged from one day to the next on 35–93% of days, depending on the year, and the ICUS overnight quote on 69–89% of days from 2022; PREB is unchanged on 3–21% of days for the term maturities in most years, but on up to 66% in 2020–2024 and on 46–92% of days for ON and 1W in 2004–2007. Neither source is uniformly better, so the series are combined day by day.

A quote of a source is stale if it lies in a run of more than five identical consecutive values, except for the first value of the run, which is usually a genuine quote. On each day and for each maturity, the combined rate is the PREB quote if it exists and is not stale, otherwise the ICUS quote if it exists and is not stale, and otherwise missing. The quotes are not averaged and not adjusted for the small level difference between the sources (on all days with both quotes, stale ones included, ICUS is on average 2 to 2.5 basis points higher for ON and 1W and about 1 basis point lower for 2W to 3M). On days on which both sources have fresh quotes, they differ by 4.0 to 5.4 basis points in absolute value on average (standard deviation 8 to 11), with mean differences (ICUS minus PREB) of -2.2 to 0.6 basis points by maturity: the source matters for single days, not for the level. The rule was chosen by comparing four constructions with the transaction-based tri-party general collateral rate (TGCR) of the Federal Reserve Bank of New York, available from April 2018: the daily changes of the combined overnight rate correlate 0.79 with those of TGCR under the PREB-first rule, 0.68 for the average of the two sources, 0.56 when the more recently updated source is chosen, and 0.38 for ICUS alone. Averaging mixes stale ICUS quotes into the series and creates jumps when one source drops out: on days on which the source switches, the daily change of the one-week rate is about twice as large as on other days.

A term quote that lies more than 1 percentage point outside the range of the two neighbouring maturities on the same date (for 3M, the two shorter maturities; neighbours carried forward for at most five days) is treated as an error, and the adjacent quotes of the same episode are masked as long as they lie more than 0.5 percentage points outside that range. The filter masks four episodes: the 3M quote from 13 April to 7 May 1993, which drifts between 0.98% and 6.03% while the other maturities are at about 3%; the 1M quote of 18 January 2001 (4.345% against about 5.8% for its neighbours); the 1W quote of 18 and 19 September 2008; and the 3M quote of 30 September 2008. The overnight quote is exempt from this filter, because its spikes on settlement dates and at period ends are genuine and mirror the federal funds rate.

A first delivery repeated the one-month series under the overnight ticker; the construction now stops if two maturities coincide. Two periods are weak. In 2005–2007 only ICUS exists

for 2W to 3M, and about half of its days are stale and therefore missing. From 2025 the PREB overnight quote is unchanged more often, so 35 overnight business days are missing in 2026, and the ON-3M and ON-2M excess returns, which need every overnight quote of their life, end with origins in February and March 2026. No other source is used to fill these gaps.

2.2. Construction of the analysis sample

Table 1 follows the quotes of each maturity through the construction. US federal holidays and Good Friday are removed from the calendar for simplicity, including the 19 Fridays before a Saturday holiday, although repo settles on some of them (the New York Fed published rates on 18 June 2021, 31 December 2021, and 10 November 2023) (a few one-off market closures, such as 30 October 2012, are kept). Isolated reverting spikes of more than 2 percentage points are removed, among them two overnight quotes (2.60% on 9 November 1999 and 1.95% on 31 December 2007, a year-end print reported by both sources), which revert on the next business day like the other isolated spikes, and in runs of identical values longer than 20 business days of the combined series all but the first value are masked (after the source rule, this masks no quote). A predictor uses the last quote of its maturity within five business days, as an investor would at t; excess returns use quoted days only. The combined series covers all seven maturities from 21 May 1991 to 16 October 2026; between 4,430 and 4,743 quotes per maturity fall in the period from July 2007, after the weak years 2005–2007. Figure 2 plots the series, and Table 2 describes the rates and the excess returns.

Stale runs and spikes are identified ex post, from the whole run or the quote after the spike, as in the construction of any cleaned data set; a forecaster at t could not always have made this classification, which decides which quotes are missing and, on some days, which pricing source is used. Section 7.4 repeats the key results with a real-time classification.

The mean excess return on the out-of-sample dates is negative for every pair and significant for all but ON-1W: holding the long repo earned more than rolling (Table 2, Panel C). It is small, 0.2 to 5.0 basis points a year, against an average absolute deviation of the excess return from the EH forecast of 3.3 to 10.0 basis points a year (Table 16).

2.3. Excess returns

The seven maturities form the ten pairs of DST in which the long maturity is an integer multiple of the short one (Figure 3): ON-1W, ON-2W, ON-3W, ON-1M, ON-2M, ON-3M, 1W-2W, 1W-3W, 1M-2M, and 1M-3M. Let $i_t^{(n)}$ be the rate at date t on a repo with a maturity of n business days, and let the short repo have maturity m, with $n = km$ for an integer k. Roll $j = 0, \ldots, k-1$ starts on the business day $t_j = t + jm$ and lasts d_j calendar days, and the long repo lasts $D_t = \sum_j d_j$ calendar days. Holding the long repo earns r_t^f,

Table 1: **Construction of the Repo Rate Series: From Raw Quotes to the Analysis Sample**

	ON	1W	2W	3W	1M	2M	3M
Step 1: Quotes of each Bloomberg pricing source, as delivered							
PREB quotes	5,499	5,348	3,479	3,478	3,478	3,484	3,484
first PREB quote	Feb 2004	Feb 2004	Jun 2013	Jun 2013	Jun 2013	Jun 2013	Jun 2013
stale (run of more than five identical values)	655	975	175	114	71	12	5
ICUS quotes (from May 1991)	8,874	8,839	8,829	8,827	8,838	8,814	8,859
stale (run of more than five identical values)	893	2,388	2,694	2,908	2,904	3,138	3,156
Step 2: Combined series, PREB first, then ICUS, no averaging							
Quoted business days	8,989	8,345	8,108	8,138	8,229	8,230	8,341
from PREB	4,844	4,373	3,304	3,364	3,407	3,472	3,479
from ICUS	4,145	3,966	4,803	4,766	4,816	4,755	4,859
other	0	6	1	8	6	3	3
Step 3: Cleaning in the analysis							
Quotes on holidays removed	243	206	188	196	209	209	226
Isolated reverting spikes (>2 pp) removed	2	0	0	0	0	0	0
Term quotes outside the curve removed	0	2	0	0	1	0	20
Runs of more than 20 identical values masked	0	0	0	0	0	0	0
Quotes used (21 May 1991 to 16 October 2026)	8,744	8,137	7,920	7,942	8,019	8,021	8,095
of which from 1 July 2007	4,743	4,472	4,430	4,508	4,531	4,542	4,542
Variant: stale quotes classified in real time (Section 7.4)							
Quotes used	8,800	8,525	8,379	8,384	8,426	8,393	8,418

Counts of daily closing quotes by maturity (Bloomberg pull of 18 October 2026). Stale: all values of a run except the first. Spikes: blocks of at most five quotes more than 2 percentage points from agreeing neighbours. Outside the curve: term quotes more than 1 percentage point outside the range of the neighbouring maturities, with the adjacent quotes of the episode (Section 2.1). Real-time variant: masked only from the sixth identical value of a run on.

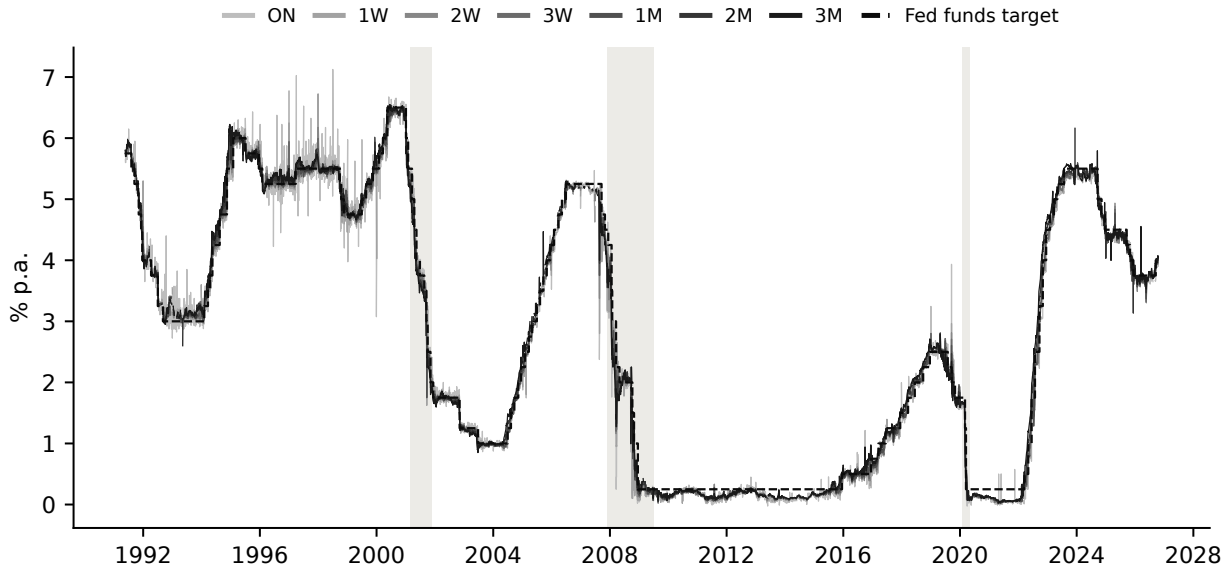


Figure 2: **Repo Rates and the Federal Funds Target.** Percent per annum, 21 May 1991 to 16 October 2026; grey lines from light (ON) to dark (3M); dashed: federal funds target (upper bound of the range from December 2008). Gaps: missing or stale quotes. Shaded: NBER recessions.

known at t; rolling the short repo k times earns r_t, which depends on future short rates. The

Table 2: **Descriptive Statistics**

	ON	1W	2W	3W	1M	2M	3M
Panel A: Repo rates (% p.a.)							
Mean	2.71	2.65	2.65	2.65	2.68	2.72	2.75
Standard deviation	2.16	2.15	2.15	2.16	2.16	2.17	2.16
Minimum	−0.03	−0.01	0.02	0.02	0.02	0.02	0.03
Maximum	7.12	6.72	6.53	6.51	6.50	6.53	6.55
First-order autocorrelation	0.998	0.999	1.000	1.000	1.000	1.000	1.000
Std. dev. of daily changes (bp)	14.1	8.2	6.0	5.7	5.4	5.1	4.8
Quotes without change (%)	13.1	19.3	19.9	19.8	19.2	18.4	17.3
Business days with a quote (%)	99.0	92.1	89.6	89.9	90.8	90.8	91.6
Observations	8,744	8,137	7,920	7,942	8,019	8,021	8,095

Panel B: Excess returns of rolling over holding (bp a year)							
Pair	n	Mean	Std. dev.	Minimum	Maximum	$\rho(1)$	Obs.
ON-1W	5	−0.33	11.9	−221.7	133.4	0.711	7,956
ON-2W	10	−1.14	11.7	−153.2	116.1	0.858	7,626
ON-3W	15	−1.50	11.9	−155.8	106.2	0.875	7,543
ON-1M	21	−2.16	12.8	−171.1	99.6	0.903	7,495
ON-2M	42	−3.55	14.6	−181.8	80.9	0.939	7,134
ON-3M	63	−5.07	17.8	−191.8	78.3	0.961	6,892
1W-2W	10	−0.80	7.2	−113.6	89.5	0.651	7,394
1W-3W	15	−1.26	8.4	−100.9	81.4	0.720	7,128
1M-2M	42	−1.31	8.2	−83.8	69.8	0.842	7,331
1M-3M	63	−2.77	12.7	−95.5	50.0	0.928	6,979

Panel C: Mean excess return on the out-of-sample dates, bp a year (HAC t-statistic)				
Pair	All	ZLB	Away from ZLB	From Apr 2018
ON-1W	−0.24 (−0.8)	−0.43 (−1.2)	−0.17 (−0.4)	−2.75 (−6.5)
ON-2W	−1.10 (−2.1)	−1.00 (−2.4)	−1.15 (−1.7)	−3.35 (−6.5)
ON-3W	−1.47 (−2.4)	−1.09 (−2.5)	−1.63 (−2.0)	−3.76 (−6.1)
ON-1M	−2.08 (−2.8)	−1.34 (−2.8)	−2.38 (−2.4)	−4.21 (−4.6)
ON-2M	−3.42 (−3.3)	−1.61 (−3.3)	−4.17 (−3.0)	−6.41 (−3.7)
ON-3M	−5.01 (−3.3)	−1.82 (−3.0)	−6.34 (−3.1)	−7.68 (−2.8)
1W-2W	−0.84 (−3.8)	−0.46 (−3.3)	−0.98 (−3.4)	−0.67 (−3.1)
1W-3W	−1.20 (−3.6)	−0.71 (−3.8)	−1.39 (−3.2)	−1.18 (−3.1)
1M-2M	−1.45 (−3.4)	−0.44 (−2.4)	−1.82 (−3.3)	−1.53 (−2.0)
1M-3M	−2.97 (−3.1)	−0.55 (−1.2)	−3.85 (−3.2)	−2.75 (−1.8)

US business days, 21 May 1991 to 16 October 2026. Panel B: excess return of Equation (1) on all dates with a quoted target; n: horizon in business days; $\rho(1)$: first-order autocorrelation. Panel C: Newey–West t in parentheses; ZLB: federal funds target range 0–0.25%.

annualised excess return of rolling over holding is

$$xr_t^{(n,m)} = \left(r_t - r_t^f\right)\frac{360}{D_t}, \qquad r_t^f = i_t^{(n)}\frac{D_t}{360}, \qquad r_t = \prod_{j=0}^{k-1}\left(1 + i_{t_j}^{(m)}\frac{d_j}{360}\right) - 1. \tag{1}$$

One month is 21 business days, so the horizon n is 1, 5, 10, 15, 21, 42, and 63 business days for ON to 3M; the long repo is assumed to run n business days, which differs from its calendar term by 0.3 days on average for 1W and 1.7 days for 3M. Maturities and roll dates are counted in positions of the business-day calendar, and the rolls accrue actual calendar days. Nine weekdays that are neither holidays nor Good Friday are missing from the delivered quotes (14 November 1991; 24 and 27 to 29 April, 6 July, and 28 July 1992;

12 and 13 September 2001); the gaps of 1991–92 fall in the first estimation window only. Excess returns are reported in basis points a year. The excess return is known only at $t+n$, so consecutive excess returns overlap by $n-1$ days.

The study works at the daily frequency of DST; sampling once a month would leave about 380 observations per pair and would measure mainly the month-end behaviour of the overnight rate. The target is the excess return, not the level of a rate: a random walk beats the ten-year mean of the level with an out-of-sample R^2 of about 99%, which measures persistence, not a departure from the EH.

2.4. Other data

The three- and six-month Treasury bill rates (FRED series DTB3 and DTB6, secondary market) and the one- and two-year constant-maturity yields (DGS1, DGS2) from the Federal Reserve's H.15 release measure market expectations of the policy-rate path. Bill rates are quoted on a discount basis d^b and are converted to money-market (ACT/360) yields,

$$b = \frac{d^b}{1 - d^b\, T_b/360}, \qquad T_b \in \{91, 182\} \text{ days}, \tag{2}$$

the basis of repo rates; the one- and two-year yields are used on their published bond-equivalent basis. H.15 posts the values of day $t-1$ on day t, so they enter with a lag of one business day.

Bloomberg reports bid and ask quotes for each ticker from the same two pricing sources. On each date and maturity the spread is the bid minus the ask (repo is quoted with the bid above the ask) of the source that the combined series uses on that date, so stale quotes do not enter. Crossed quotes are treated as missing, and a missing spread carries the last quoted spread of the maturity forward. From 17 September 2019 the PREB overnight quote carries a fixed band of 100 basis points, which is not a tradable spread; on every date with a PREB overnight spread of 50 basis points or more (the band and three days in September 2008) the ICUS overnight spread of that date is used. The spread of a pair on date t is the average of the spreads of its two legs (Section 3.6); over the out-of-sample period it averages 4.4 to 5.9 basis points per pair, and 2 to 4 since 2016. The spreads are quoted in whole basis points, and since 2015 as fixed widths per maturity: 2 basis points for 1W and 2W, 3 for 3W to 2M, and 4 for 3M on 98% to 100% of days. They are quoted, not transacted, spreads.

The transaction-based overnight rates SOFR, BGCR, and TGCR of the Federal Reserve Bank of New York (from April 2018) validate the overnight quote (Section 5.2). The scheduled FOMC meetings from 1991 to 2027 (Federal Reserve Board calendars), the effective federal funds rate (FRED series DFF), and the NBER recession months (USREC) enter no

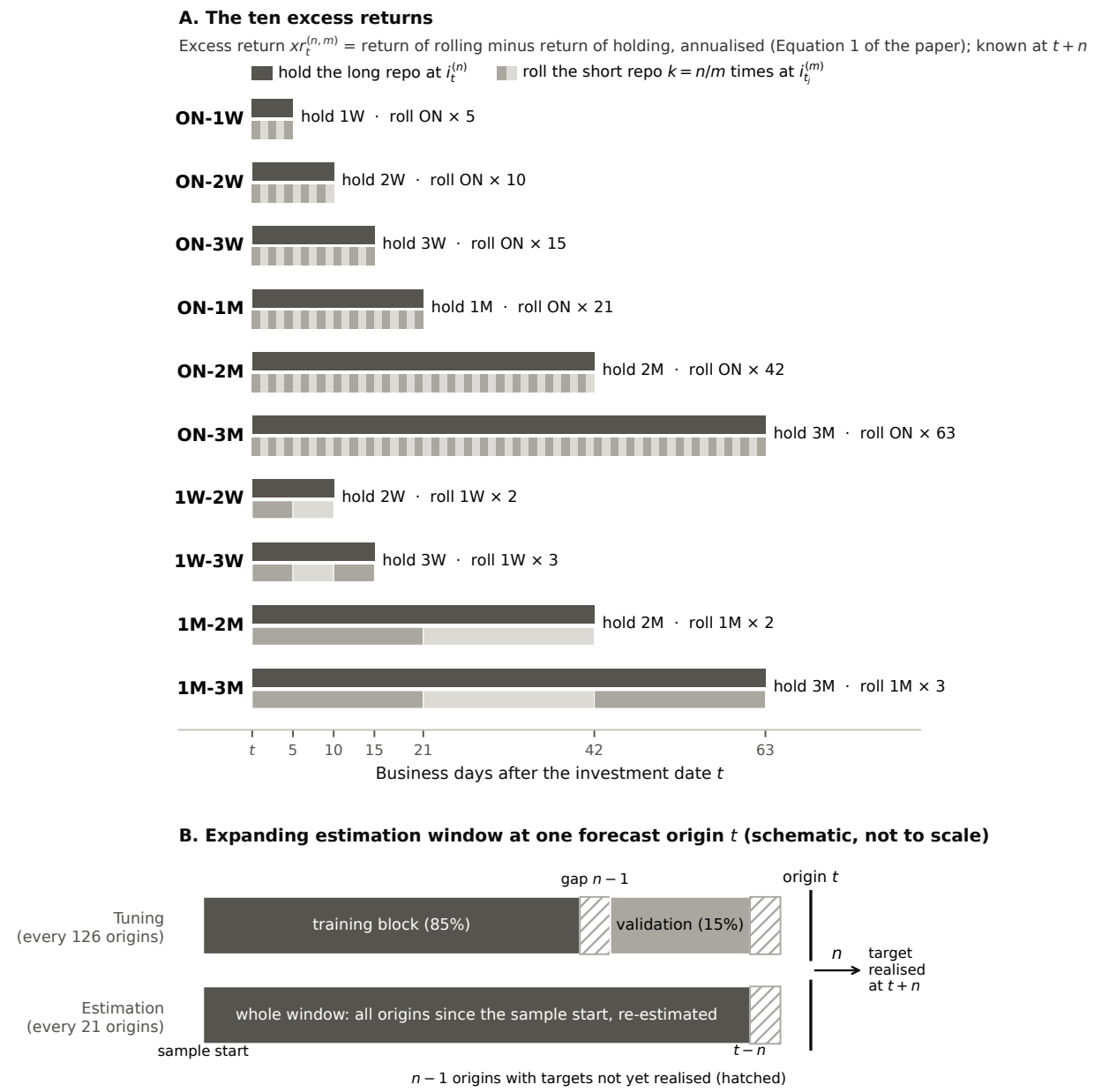


Figure 3: **Construction of the Excess Returns and the Real-Time Estimation Window.** Panel A: holding the long repo for n business days (dark bar) or rolling the short repo $k = n/m$ times (segments). Panel B: expanding window at origin t: estimation on origins up to $t - n$, tuning every 126 origins on the last 15% of the window, separated by $n - 1$ origins; re-estimation every 21 origins.

reported forecast; the NBER months enter only the conditional evaluation. The 126 monthly series of FRED-MD (McCracken and Ng, 2016) enter a robustness set only (Appendix Appendix A.3).

3. Forecasting Design and Inference

The test of the EH is forecast-based, out of sample, and adjusted for the search across forecasts, and it imposes no parametric model of expectations; this section defines the restriction, the protocol, the forecasts, the tests, and the trading strategy. Unless an exhibit states otherwise, exhibits use DST's information set, the full out-of-sample period, expanding windows, and the evaluation benchmark of Equation (12).

3.1. The testable restriction

The framework follows Bianchi et al. (2021b), adapted to simple ACT/360 repo rates and to the rolling strategy of Equation (1), which is linear in the rates up to the compounding of the rolls (about 1 basis point a year for ON-1M at a rate of 5%, up to about 5 for ON-3M at the highest rates in the sample). Define the term premium π_t as the long rate minus the calendar-weighted expected short rates, $\pi_t = i_t^{(n)} - \sum_j (d_j/D_t) E_t[i_{t_j}^{(m)}]$, with the expected compounding term included, and the expectation error η_{t+n} as the weighted realised minus expected short rates. Then, exactly,

$$xr_t^{(n,m)} = -\pi_t + \eta_{t+n}, \qquad E_t[\eta_{t+n}] = 0, \qquad \text{so} \qquad E_t\big[xr_t^{(n,m)}\big] = -\pi(\mathcal{S}_t) \equiv g(\mathcal{S}_t), \tag{3}$$

where $\mathcal{S}_t$ is the state at t and g an unknown function, as in Equation (2) of Bianchi et al. (2021b). Up to the compounding term, π_t is the premium $c(n,m)$ in Equation (1) of DST, with calendar weights d_j/D_t in place of $1/k$. The EH with a constant premium restricts g to a constant, and by the law of iterated expectations $E[xr_t^{(n,m)} \mid z_t] = -\pi^{(n,m)}$ for any predictors z_t dated t or earlier.

Under the restriction the best forecast of the excess return from any information at t is a constant, which the mean of past excess returns estimates in real time. A forecast that beats this mean out of sample, on origins it was not estimated on, therefore shows that the conditional mean varies with information dated t or earlier, whatever its functional form. Unlike the in-sample Wald test of DST's VAR restrictions (Section 4.4), the test does not rely on the VAR being the correct model of expectations, so it avoids the joint test of the EH and the expectations model (Guidolin and Thornton, 2018). The converse does not hold: a forecast that fails to beat the mean does not establish the EH, because estimation noise can mask a small departure (Section 8.2).

With essentially affine prices of risk (Duffee, 2002), g is affine in the state, and with expectations linear in lagged rates this is DST's case. At the short end, three features can make g nonlinear: the zero lower bound, policy steps on scheduled meeting dates, and calendar dates such as the settlement days of the reserve maintenance period (Hamilton, 1996)

and month, quarter, and year ends. The flexible forecasts are the out-of-sample counterpart of the in-sample nonlinearity tests of Section 4.4.

The predictors and the excess return are built from the same Bloomberg quotes, the channel of measurement error that Cochrane and Piazzesi (2005) address with lagged forward rates. Let the observed quote be $\hat{i}_s^{(\ell)} = i_s^{(\ell)} + \nu_s^{(\ell)}$, $\ell \in \{m, n\}$, with noise $\nu_s^{(\ell)}$ of mean zero that is independent of the true rates and over time. To first order, the measured excess return is

$$\widehat{xr}_t \approx xr_t^{(n,m)} - \nu_t^{(n)} + \frac{d_0}{D_t}\,\nu_t^{(m)} + \sum_{j=1}^{k-1} \frac{d_j}{D_t}\,\nu_{t_j}^{(m)}. \tag{4}$$

The noise of today's long quote enters one-for-one with the opposite sign, and that of today's short quote with the weight d_0/D_t of the first roll. Because the true rates are persistent, a jump of today's quote relative to its lags is partly noise, so a high same-day long quote predicts a low measured excess return even under the EH. Predictors dated $t-1$ to $t-5$ contain none of this noise and cannot predict under the EH. The R^2_{oos} without the quotes of day t therefore measures approximately the predictability that does not come from noise, and the decline the noise part, if the genuine and the noise-driven parts of the forecast are not negatively correlated. The closing overnight quote's gap to the TGCR has a first-order autocorrelation of 0.37 (Section 5.2), so part of the noise reaches the lagged predictors and the lagged R^2_{oos} slightly overstates the predictability that does not come from noise.

A rejection is a statement about measured excess returns. It becomes tradable predictability only if the information is available before a position is taken and positions can be traded at the quoted rates net of the bid–ask spread, which Sections 5 and 6 examine.

3.2. Protocol and information at the origin

Let $Y_t = (i_t^{(m)}, i_t^{(n)})'$ stack the two rates of a pair and $y_t = xr_t^{(n,m)}$ the target. Forecasts are made in real time from an expanding window. At origin t a model is estimated on

$$\mathcal{E}_t = \big\{(X_s, y_s) :\ s \le t - n\big\}, \qquad X_t = \big(Y_t', \ldots, Y_{t-4}'\big)', \qquad X_t^{lag} = \big(Y_{t-1}', \ldots, Y_{t-5}'\big)', \tag{5}$$

so that every target used is realised by t (DST's VAR is estimated on the rates through t); X_t is DST's information set with same-day quotes and X_t^{lag} the same set lagged by one business day. A model is first estimated once 500 targets are realised, so the forecasts start in April 1994 to May 1995 depending on the pair. Models are re-estimated every 21 forecast origins, and hyperparameters are updated every 126 origins on a validation block of the last 15% of the window, chronologically after the training block and separated from it by $n-1$ origins, so that no training target overlaps a validation target. Within each window the predictors and the target are standardised with the mean and standard deviation of that window only,

standardised predictors are clipped at ± 5, and forecasts are transformed back to the units of the excess return. All forecasts of a pair and information set are evaluated on the origins common to all of them.

Only targets and forecast errors realised by t enter estimation, tuning, combination weights, and positions; repo predictors are closing quotes of day t and earlier, Treasury-curve predictors values of day $t-1$. Variables not available in real time, such as the NBER recession dates, enter only the evaluation. Table 3 lists the protocol, the date from which each input is available at the origin, and the variables used only for evaluation or robustness.

3.3. Information sets and specification choices

DST's information set is X_t of Equation (5), the ten inputs of the bivariate VAR(5) of DST, who choose five lags for the day-of-the-week pattern of overnight rates (Longstaff, 2000). The rates enter in levels: the comparison then isolates the functional form, the term spread and deviations from the usual level are linear combinations of the inputs, and repo rates stay in a bounded range. This set is the baseline for the test of the EH and the comparison with DST.

Under the EH the long repo rate is the expected average of future short rates, so information about the expected path of the policy rate bears directly on the restriction (cf. Altavilla et al., 2014, who measure it with federal funds futures). The second information set adds four such measures from the short Treasury curve, a deep market separate from the repo quotes:

$$X_t^{exp} = \left(X_t',\ b_{t-1}^{3M} - i_t^{(n)},\ b_{t-1}^{6M} - b_{t-1}^{3M},\ \gamma_{t-1}^{1Y} - b_{t-1}^{3M},\ \gamma_{t-1}^{2Y} - \gamma_{t-1}^{1Y} \right)', \tag{6}$$

with b the bill yields of Equation (2) and γ^{1Y}, γ^{2Y} the one- and two-year yields, all of day $t-1$.

Four further sets are estimated with all models except the networks. Calendar information (day-of-week dummies, first and last trading days of month, quarter, and year, and whether the investment window crosses a month, quarter, or year end) captures balance-sheet and settlement effects at period ends. A pair-level analogue of DST's state vector, which uses the overnight rate and its spreads to all six longer maturities (Della Corte et al., 2008, App. A.3), is built from the pair's short rate and spread and their daily changes; it is an invertible rotation of the ten lagged levels that leaves least squares unchanged and matters for models that are not invariant to rotations. The stationary form, the term spread and the daily change of the short rate at lags 0 to 4, lets trees and nearest neighbours, which predict only within the range of the levels in their window, use the same information. DST's set plus the 126 FRED-MD predictors asks whether macro-financial information, which predicts

Table 3: **Out-of-Sample Forecasting Protocol and Information Available at the Origin**

Element	Specification (every forecast of the paper and of this supplement unless stated)
Panel A: Forecasting protocol	
Target	Annualised excess return $xr_t^{(n,m)}$ of rolling the short repo over holding the long repo, Equation (1), for the ten pairs of DST
Forecast origin	Every US business day t on which all predictors of the information set are available
Horizon	n business days, the maturity of the long repo: 5 (1W), 10 (2W), 15 (3W), 21 (1M), 42 (2M), and 63 (3M); the target of origin t is realised at $t+n$
Estimation window	Expanding: all pairs (X_s, y_s) of predictors and target with $s \leq t-n$, so that every target used is realised by t
Minimum training observations	500 realised targets; first forecasts between April 1994 (ON-1W) and May 1995, depending on the pair; the latest estimate is kept on the rare dates with fewer usable observations
Re-estimation frequency	Every 21 forecast origins (about one month); the latest estimate is applied in between
Hyperparameter update	Every 126 forecast origins (about six months), on the validation block; grids in Table A.33
Validation scheme	Last 15% of the estimation window, chronologically after the training block and separated from it by $n-1$ origins, so that no training target overlaps a validation target; objective: mean squared error; the networks also stop early on this block
Standardisation	Predictors and target standardised with the mean and standard deviation of the current estimation window only; standardised predictors clipped at ± 5; forecasts transformed back to the units of the excess return
Missing predictors	A predictor uses the last quote of its maturity within five business days; origins with a missing predictor are not forecast and windows use only complete observations; no other imputation. Stale runs and isolated spikes are classified ex post (Section 2.2)
Common evaluation sample	All forecasts of a pair and information set are evaluated on the origins at which every forecast and the realised target exist, from the first forecast to the last origin with a realised target (February 2026 for ON-3M to October 2026 for ON-1W); supporting subsample: origins from 1 January 2016
Panel B: Information available at the origin t (predictors and estimation)	
Repo quotes (Bloomberg, PREB/ICUS)	Closing quotes of the pair's two maturities on days t to $t-4$ (DST's set), available at the close of day t (Bloomberg does not document the closing time). Same-day component: the quotes of day t, which also fix the first roll of the target and the return of holding the long repo. Lagged specification: days $t-1$ to $t-5$ (Section 5.1)
Treasury curve (H.15: DTB3, DTB6, DGS1, DGS2)	Values of day $t-1$, published on day t; bills converted to a money-market yield; market-expectations set only; no same-day component
Realised targets	Only $y_s = xr_s^{(n,m)}$ with $s+n \leq t$ (known on day $s+n$), in estimation, tuning, early stopping, the real-time prevailing mean, and the EWMA
Forecast errors in combination weights	Only errors of origins $s \leq t-n$ (DMSPE weights; equal weights during the first 250 out-of-sample days); position sizes use volatilities of realised excess returns only
Calendar variables	Day of week, period ends, and turns in the window; known in advance; calendar specification only (Table 29, Panel B)
FRED-MD series	Macroeconomic series of reference month M from the business day after the end of $M+3$; financial series (rates, exchange rates, S&P 500, VIX) from the business day after the end of M; final vintage 2026-09; robustness specification only (Section 7.5)
Panel C: Variables used only for evaluation or robustness, never as predictors	
NBER recessions, zero-lower-bound periods	Dated ex post; conditional evaluation only (Table 14)
Bid–ask spreads	The quoted bid–ask spread of the pair on each trade date (average of its two maturities, from the Bloomberg bid and ask quotes of the pricing source used that day); cost of the trading strategies only
TGCR, BGCR, SOFR	Transaction-based overnight rates of the New York Fed from April 2018, published the next business day; validation of the overnight leg only (Section 5.2)
Real-time stale variant	Stale runs classified from past quotes only; robustness check of the ex-post classification (Table 25)

Origin t: the business day on which the excess return realised at $t+n$ is forecast. Lagged specification: no quote of day t; with market expectations every predictor is then lagged by one business day.

one-year holding-period returns on bonds (Ludvigson and Ng, 2009; Bianchi et al., 2021b), predicts repo premia. DST's own sample, the bias correction of the VAR, and a rolling window on DST's sample are varied in Section 4.4; window lengths, validation shares, and re-estimation frequencies on the full sample are not.

The information sets and the window were fixed in a design study on forecast origins whose targets were realised by the end of 2015: the market-expectations set scored best of seven candidate additions to DST's set (the calendar and state-vector sets also passed; FOMC, zero-lower-bound, and liquidity sets did not), and the expanding window was chosen by the same score. The SPA and Holm p-values do not adjust for these choices, and because the specification was finalised after full-sample results had been examined, the origins from 2016 are supporting subsample evidence, not an independent holdout. The network specification (activation, bias terms, architecture, optimiser, learning rate, dropout, and input scaling) was chosen on 18 October 2026 in five screening rounds on static splits whose test blocks, 2017/18–2026 and 2007–2013, overlap the evaluation period; the SPA tests do not adjust for this choice. Re-run on the current data one factor at a time, the screening shows that the choice does not drive the results (Table 28, Section 7.5).

3.4. Forecasts

The benchmark model is a bivariate VAR(5) in the levels of the short and the long rate,

$$Y_t = c + \sum_{l=1}^{5} A_l Y_{t-l} + u_t, \tag{7}$$

estimated by least squares with a constant, which is equivalent to DST's demeaning within the window. DST correct the coefficients for small-sample bias by a bootstrap (their App. A.3). Here they are corrected analytically with the first-order bias of Pope (1990) in companion form; following Kilian (1998), no correction is applied if the least-squares estimate itself has a root of modulus one or more, and if the corrected estimate has such a root, the correction is scaled down in steps of 0.01 until the largest root is below one, with the constant reset so that the unconditional mean is unchanged. In a Monte Carlo check with a persistence of 0.95 and 200 observations, least squares estimates 0.927 and the corrected estimator 0.948. The corrected VAR is iterated forward to the expected path of the short rate (DST, App. A.1), and

$$\widehat{\mu}_t^{VAR} = \left[\prod_{j=0}^{k-1} \left(1 + \widehat{E}_t\big[i_{t_j}^{(m)} \big] \frac{d_j}{360} \right) - 1 - i_t^{(n)} \frac{D_t}{360} \right] \frac{360}{D_t}, \tag{8}$$

with the known rate $i_t^{(m)}$ for the first roll, the actual calendar days of the rolls, and the expected rates in place of the rates (the convexity term is negligible). The direct least-squares regression of the excess return on the ten rates, the reduced form of the VAR when expectations are linear, is a separate model (OLS, direct).

The learners are six regularised linear models (ridge, lasso, elastic net, principal-component regression, partial least squares, linear support vector regression) and seven nonlinear models

(SVR with a radial basis function kernel, nearest neighbours, a regression tree, a random forest of 300 trees (Breiman, 2001), gradient boosting (Friedman, 2001), bagging of 100 trees, and ridge on 100 to 5,000 random Fourier features (Kelly et al., 2024)), with hyperparameters selected on the validation block (grids in Table A.33).

A network with H hidden layers maps the standardised predictors $\tilde{x}_t$ of the information set (Equation (5)) to the standardised forecast

$$\widehat{\mu}_t^{NN} = w'_{H+1} h_{H,t}, \qquad h_{l,t} = \tanh\left(W_l\, h_{l-1,t}\right), \quad l = 1, \ldots, H, \qquad h_{0,t} = \tilde{x}_t, \tag{9}$$

without bias terms, so that a network adds no constant to the EH forecast and uses only the variation of the predictors; the forecast is transformed back to the units of the excess return. Without bias terms, and with predictors and target standardised and clipped symmetrically, a network is an odd function of the departure of the predictors from their window means: it forecasts symmetric departures with equal and opposite signs. It can represent saturation, but not symmetric, threshold, or convex nonlinearities, which the kernel, nearest-neighbour, tree-based, and random-feature forecasts can represent; an odd function captures about 7% of the variance of the nonlinear signal of Section 8.2, so the networks cannot recover it. Five architectures are used, with hidden layers of (16), (32), (16-8), (32-16), and (32-16-8) units, and their equal-weighted ensemble (NN avg.). They are compact because the daily targets overlap by $n-1$ days and the first windows hold 500 realised targets; the architectures follow Gu et al. (2020) and lie within the range of the return-prediction literature, including Bianchi et al. (2021b). The hyperbolic tangent bounds the response to inputs outside the range of the estimation window, unlike the piecewise-constant trees. The weights are initialised with the normal variant of the scheme of Glorot and Bengio (2010) and estimated by minimising the mean squared error plus an L2 penalty, with dropout (Srivastava et al., 2014) after each hidden layer, by the Nadam optimiser (Dozat, 2016; Kingma and Ba, 2015) at a learning rate of 0.001, in batches of 64 or 256 for at most 500 epochs. Each network is fitted on the first 85% of its estimation window; training stops when the error on the remaining 15%, chronologically after the training block, has not improved for 20 epochs, and the best weights are kept. Each forecast averages ten independently initialised networks trained jointly with a common stopping epoch, a seed ensemble in the spirit of Bianchi et al. (2021b), who average the ten best of 100 networks, and each re-estimation starts from the previous weights. The penalty, the dropout rate, and the batch size are tuned on the validation block every 126 origins with a single network trained from scratch (grids in Table A.33).

The combinations are the equal-weighted combination of all models (Rapach et al., 2010), as Guidolin and Timmermann (2009) combine forecasts of short rates; the combination

shrunk half-way toward the EH forecast, $0.5\,\widehat{\mu}_t^{comb} + 0.5\,\widehat{\mu}_t^{EH}$, with the weight fixed without tuning, a direct form of the shrinkage toward the historical average that Rapach et al. (2010) document for the mean combination; and a real-time combination of the models and the EH forecast with discounted mean squared forecast error (DMSPE) weights (Rapach et al., 2010, SSRN version, p. 7),

$$\widehat{\mu}_t^{DMSPE} = \sum_j \omega_{j,t}\widehat{\mu}_{j,t}, \qquad \omega_{j,t} = \frac{\left(\sum_{s\le t-n} e_{j,s}^2\right)^{-1}}{\sum_l \left(\sum_{s\le t-n} e_{l,s}^2\right)^{-1}}, \tag{10}$$

where $e_{j,s}$ is the error of forecast j at origin s, the discount factor is one, and the weights are equal until 250 errors are realised. The real-time EH forecast and the exponentially weighted mean of past excess returns (EWMA), a recency-weighted benchmark motivated by Nagel (2025), are

$$\widehat{\mu}_t^{EH} = \frac{1}{N_t}\sum_{s\le t-n} y_s, \qquad \widehat{\mu}_t^{EWMA} = \frac{\sum_{s\le t-n}\lambda^{t-n-s} y_s}{\sum_{s\le t-n}\lambda^{t-n-s}}, \qquad \lambda = 2^{-1/126}, \tag{11}$$

with N_t the number of realised targets, so that the EWMA has a half-life of 126 business days. In the backtests the real-time EH forecast is updated at each re-estimation, every 21 origins.

Twenty-five forecasts are reported. Twenty-three form the forecast-search family over which the SPA tests search: DST's VAR, OLS (direct), the 13 learners, the five networks and their ensemble, the equal-weighted combination, and the EWMA. The shrunk and the DMSPE combinations include the EH forecast and nest the benchmark, so they are reported but not searched. The family is the same for every pair and information set; with market expectations, DST's VAR on DST's set is among the 23. All forecasts of a pair receive the same information, so they differ in functional form, not in information.

3.5. Inference and search control

The inference follows the research question in three steps. Whether the EH is rejected once the search across forecasts is accounted for is decided by the SPA test within each pair and the Holm adjustment of the ten pairs' SPA p-values; the Clark–West test of each forecast is a supporting diagnostic. Whether the rejection is tied to same-day quotes is decided by comparing each pre-specified model with and without the quotes of the forecast date on the same dates (Section 5.1). Whether a forecast earns more than the EH strategy after costs is decided by the SPA test on the net returns of the 23 forecasts, with Newey–West t-statistics and upper bounds; the perfect-foresight bound requires no test (Section 3.6). Diebold–

Mariano and encompassing tests against the VAR, and the conditional tests of Giacomini and White (2006), describe the forecasts but do not enter the answer.

The loss is the squared error $L_{j,t} = (y_t - \widehat{\mu}_{j,t})^2$ of the annualised excess return. Accuracy relative to the EH is the out-of-sample R^2 of Campbell and Thompson (2008), as in Bianchi et al. (2021b), against the mean of all earlier targets from the start of the sample,

$$R^2_{oos} = 1 - \frac{\sum_t (y_t - \widehat{\mu}_{j,t})^2}{\sum_t (y_t - \bar{y}_{t-1})^2}, \qquad \bar{y}_{t-1} = \frac{1}{t-1} \sum_{s<t} y_s. \tag{12}$$

With overlapping targets the evaluation benchmark $\bar{y}_{t-1}$ contains slightly more information than the real-time $\widehat{\mu}^{EH}_t$ of Equation (11), which makes it harder to beat and the R^2_{oos} conservative (for DST's VAR by 0.1 to 0.5 percentage points for the pairs with horizons up to three weeks and by 0.7 to 1.9 for the pairs with one- to three-month long legs; against the real-time EH forecast ON-1M and 1M-3M turn positive, 0.5% and 0.2%). With the real-time EH forecast as benchmark the SPA test gives the same answer: after the Holm adjustment ON-1W remains ($p = 0.017$), and ON-2W (0.074) and 1W-2W (0.099) stay borderline. In accuracy statements, "the EH forecast" denotes this evaluation benchmark. The Clark–West and SPA tests use the same benchmark $\bar{y}_{t-1}$; wherever a forecast is traded or combined, the EH forecast is $\widehat{\mu}^{EH}_t$.

The test of Clark and West (2007) asks whether forecast j is more accurate than the EH forecast, adjusting for the noise of estimating a model that nests it:

$$f_{j,t} = \big(y_t - \bar{y}_{t-1}\big)^2 - \Big[\big(y_t - \widehat{\mu}_{j,t}\big)^2 - \big(\bar{y}_{t-1} - \widehat{\mu}_{j,t}\big)^2\Big], \qquad CW_j = \frac{\sqrt{T}\,\bar{f}_j}{\widehat{\omega}_{f,j}}, \tag{13}$$

one-sided, with $\widehat{\omega}_{f,j}$ the Bartlett long-run standard deviation of Newey and West (1987) with the automatic bandwidth of Andrews (1991) and at least n lags. The test of Diebold and Mariano (1995) uses the same statistic on the loss differential against DST's VAR, $\varrho^{VAR}_{j,t} = L_{VAR,t} - L_{j,t}$, two-sided, and the encompassing test of Harvey et al. (1998) asks whether a learner adds information to the VAR.

The test for superior predictive ability (SPA) of Hansen (2005) asks whether the best of the 23 forecasts beats the EH forecast once the search is accounted for. With $\varrho_{j,t} = L_{EH,t} - L_{j,t}$ and $L_{EH,t} = (y_t - \bar{y}_{t-1})^2$,

$$T^{SPA} = \max\Big(\max_j \frac{\sqrt{T}\,\bar{\varrho}_j}{\widehat{\omega}_j},\, 0\Big), \tag{14}$$

where $\widehat{\omega}_j$ is the bootstrap standard deviation of $\sqrt{T}\,\bar{\varrho}_j$, and its p-value is the share of

stationary-bootstrap statistics (Politis and Romano, 1994) at least as large, with each $\bar{\varrho}_j$ recentred at $\bar{\varrho}_j \mathbf{1}\{\sqrt{T}\bar{\varrho}_j/\widehat{\omega}_j \leq -\sqrt{2\log\log T}\}$ and a mean block length of $\max(20, 2n)$. Because the forecasts nest the EH forecast, the test is conservative, by analogy with the undersized MSPE tests for nested models of Clark and West (2007).

The headline inference applies the step-down procedure of Holm (1979) to the ten SPA p-values, ordered $p_{(1)} \leq \cdots \leq p_{(K)}$ with $K = 10$; the false-discovery rate of Benjamini and Hochberg (1995) is reported alongside:

$$p_{(i)}^{Holm} = \max_{l \leq i} \min\left\{1, (K - l + 1)\, p_{(l)}\right\}, \qquad p_{(i)}^{BH} = \min_{l \geq i} \min\left\{1, \frac{K}{l}\, p_{(l)}\right\}. \tag{15}$$

The Clark–West p-values are adjusted in the same way within a pair (23 forecasts) and across pairs (230). The model confidence set of Hansen et al. (2011) retains the forecasts that cannot be separated from the most accurate one at the 10% level.

SPA, Holm, and Benjamini–Hochberg p-values of the forecasts use 10,000 bootstrap draws (50,000 for ON-1W with long blocks and for the band rule over pairs and widths), as do those of the band rule; the other bootstrap results, among them the SPA tests of net fees under the sign rule and the bootstrap intervals of R^2_{oos}, use 1,000.

Under the EH, the Clark–West test rejects at the 5% level in 4% to 6.5% of simulated samples for the overnight pairs up to two months and for 1W-2W, and in 9% to 13% for the pairs with the longest horizons relative to their short leg (ON-3M, 1W-3W, 1M-2M, 1M-3M; Section 8.1); its rejections for these four pairs overstate the evidence. The size and power of the SPA test and its Holm adjustment are not simulated.

The conditional test of Giacomini and White (2006) asks whether the VAR's gain over the EH forecast is zero on average and conditional on states fixed in advance (windows that cross a month end, the zero lower bound defined here by a short rate below 0.25% rather than by the target range of Table 2, NBER recessions, and 63-day volatility of the short rate above its median up to t), following Engsted et al. (2013) and Bianchi et al. (2021b, Sec. 4.1), as Sarno et al. (2016, Sec. 5.4) condition economic value on states of macroeconomic uncertainty; the recession state is dated ex post, so that test is descriptive. The test of Pesaran and Timmermann (1992) asks whether the VAR and the network ensemble predict the sign of the departure from the EH, on non-overlapping origins. For economic value, an SPA test asks whether any of the 23 forecasts earns a higher mean net return than the EH strategy (Section 3.6).

3.6. Trading strategy and economic value

The economic evaluation translates forecasts into positions, following Section 5 of DST. The participant chooses between holding the long repo, riskless over its life, and rolling the

short repo k times; the position $w_{j,t}$ is the weight on rolling relative to holding. Being long or short the roll needs access to both sides of the market, such as a dealer lending at term and borrowing overnight or the reverse; dealers intermediate between repo segments (Anbil et al., 2021). The EH strategy trades on $\widehat{\mu}_t^{EH}$.

Let $x_t = xr_t^{(n,m)} D_t/360$ be the excess return over the life of the long repo and $\widehat{\sigma}_t$ the standard deviation of the x_s realised by t. The sign strategy uses DST's target-volatility weights,

$$w_{j,t} = \frac{\sigma_t^*}{\widehat{\sigma}_t} \operatorname{sign}\big(\widehat{\mu}_{j,t}\big), \tag{16}$$

with σ_t^* the average of $\widehat{\sigma}_s$ up to t, so that the median position is about one unit of notional; with one risky asset, DST's mean–variance problem (their Eq. 14) reduces to this rule. Two alternatives are reported: DST's annual targets of 1% and 2%, which, applied to x_t, imply positions of 8 to 62 times the notional on average (DST infer that their own positions stay between -1 and 2, p. 171); and mean–variance weights that scale with the forecast (Bianchi et al., 2021b, Section 5.1; bounds in Internet Appendix D.2),

$$w_{j,t}^{MV} = \frac{\widehat{\mu}_{j,t} D_t/360}{\delta\, s_{j,t}^2} \in [-1, 2], \tag{17}$$

where $s_{j,t}^2$ is an exponentially weighted average (smoothing parameter 0.05) of the squared errors of forecast j realised by t, for the excess return over the life of the long repo, as in Thornton and Valente (2012).

A tranche of $1/n$ of the notional is opened every business day and held for n days, and the bid–ask spread is charged over the life of the tranche on the change of position against the tranche it replaces. The spread of the trade date is

$$\tau_t = \tfrac{1}{2}\big(\tau_t^{(m)} + \tau_t^{(n)}\big), \qquad \tau_t^{(\ell)} = \hat{i}_t^{\,bid,(\ell)} - \hat{i}_t^{\,ask,(\ell)}, \tag{18}$$

the average of the two legs' spreads of the source used on day t (Section 2.4); the literature reference is a constant $\tau_t = 4$ basis points, DST's value. The gross return of the tranche opened at t is

$$R_{j,t} = 1 + r_t^f + w_{j,t}\, x_t - \tau_t \left| w_{j,t} - w_{j,t-n} \right| \frac{D_t}{360}. \tag{19}$$

The participant has the quadratic utility of West et al. (1993) with relative risk aversion $\delta = 5$. The certainty-equivalent (performance) fee $\mathcal{F}_j$ of Fleming et al. (2001) equates the

average utility of the forecast-based strategy, net of $\mathcal{F}_j$, with that of the EH strategy:

$$\frac{1}{T}\sum_{t=1}^{T}\Big[(R_{j,t}-\mathcal{F}_j)-\frac{\delta}{2(1+\delta)}(R_{j,t}-\mathcal{F}_j)^2\Big]=\frac{1}{T}\sum_{t=1}^{T}\Big[R_{EH,t}-\frac{\delta}{2(1+\delta)}R_{EH,t}^2\Big]. \tag{20}$$

All measures are annualised with $360/\bar{D}$, where $\bar{D}$ is the mean of D_t over the origins, and reported in basis points a year per unit of notional. The break-even spread τ_j^* (as in Thornton and Valente, 2012, Section 3.3) is the constant spread at which the fee vanishes, $\mathcal{F}_j(\tau_j^*)=0$ (zero if the gross fee is not positive). A participant who only lends, switching between lending short and lending long, holds $w_{j,t}^{lend}=(\sigma_t^*/\widehat{\sigma}_t)(1+\operatorname{sign}\widehat{\mu}_{j,t})/2\in\{0,\sigma_t^*/\widehat{\sigma}_t\}$, half the position change of the sign strategy, and earns half its gross fee, up to the negligible quadratic-utility adjustment.

Perfect sign foresight takes the same absolute position as every forecast but always the right sign,

$$w_t^{PF}=\frac{\sigma_t^*}{\widehat{\sigma}_t}\operatorname{sign}(x_t),\qquad\text{so}\qquad w_{j,t}x_t\le w_t^{PF}x_t=\big|w_t^{PF}\big|\,|x_t|\quad\text{for every } j. \tag{21}$$

Before costs, its gain over the EH strategy is $2|w_t^{PF}x_t|$ on the origins on which the EH strategy is on the wrong side and zero elsewhere, so the gross fee of any forecast in the sign strategy is at most

$$B=\frac{360}{\bar{D}}\,\frac{1}{T}\sum_{t=1}^{T}2\,\big|w_t^{PF}x_t\big|\,\mathbf{1}\big\{\operatorname{sign}(\widehat{\mu}_t^{EH})\neq\operatorname{sign}(x_t)\big\}\;\approx\;E|xr|-|E[xr]|, \tag{22}$$

up to the quadratic-utility adjustment, which is negligible here, and the approximation holds with unit positions when the EH strategy takes the sign of the mean. The bound, stated for a long/short sign strategy by Sarno et al. (2016, fn. 12), holds whatever the information or functional form of the forecast, and clipping positions to $[-1,2]$ or $[0,1]$ after the sign is set leaves it in force. It applies to gross fees and to the sign strategy, not to mean–variance weights, and the strategies trade at the same quotes that enter the forecasts, so untradeable noise in the quotes would inflate both the fees and the bound.

The band rule follows a forecast only when it deviates from the EH forecast by more than a multiple κ of the day's spread,

$$\widehat{\mu}_{j,t}^{band}=\begin{cases}\widehat{\mu}_{j,t} & \text{if } \big|\widehat{\mu}_{j,t}-\widehat{\mu}_t^{EH}\big|>\kappa\,\tau_t,\\ \widehat{\mu}_t^{EH} & \text{otherwise,}\end{cases}\qquad\kappa=1, \tag{23}$$

and is traded with Equation (16); $\kappa \in \{0.5, 1.5, 2\}$ are sensitivity checks. The threshold is set ex post; Table 18 adjusts for the search over $\kappa \in \{1, 1.5, 2\}$ jointly with the ten pairs.

After costs, perfect sign foresight is not a bound, because it changes position whenever the sign changes. The net ceiling is the exact maximum of the mean net return of a strategy that knows the excess return and pays the spread on each change of position. With $a_t = \sigma_t^*/\widehat{\sigma}_t$, it solves

$$\max_{s_t \in \{-1,1\}} \sum_t \Big[s_t a_t x_t - \tau_t \big| s_t a_t - s_{t-n} a_{t-n} \big| \frac{D_t}{360} \Big], \tag{24}$$

by backward recursion over each chain of tranches $t, t+n, t+2n, \ldots$, which do not interact; a variant also allows $s_t = 0$. The resulting positions enter the fee of Equation (20) like those of a forecast.

4. Statistical Rejection at Short Horizons

Search-adjusted evidence against the EH lies at horizons of one to three weeks: within pairs the SPA test rejects for ON-1W, ON-2W, and 1W-2W, and with market expectations also for 1W-3W; across the ten pairs only ON-1W survives the Holm adjustment, and ON-2W is borderline.

4.1. Evidence across pairs

Table 4 summarises the evidence for the ten pairs: the accuracy of DST's VAR, of the network ensemble, and of the most accurate forecast, the SPA and Holm-adjusted p-values, the VAR's accuracy without same-day quotes, and the economic value of the sign strategy against its perfect-foresight bound. Figure 1 plots the same numbers.

Table 5 collects, for each pair, the in-sample test, the out-of-sample accuracy of representative forecasts on both information sets, the split into same-day and lagged information, and the supporting subsample from 2016; Table 6 lists every inference check for the three pairs that survive the search. DST's VAR beats the EH forecast for six of the ten pairs (eight against the real-time EH forecast; Section 3.5), with an R^2_{oos} of 20.5% for ON-1W, 10.1% for 1W-2W, and 6.2% for ON-2W, and 1.0% to 2.2% for ON-3W, 1W-3W, and 1M-2M. Within pairs, the SPA test rejects for ON-1W ($p = 0.002$), ON-2W ($p = 0.008$), and 1W-2W ($p = 0.012$) and for no other pair ($p \geq 0.36$). After the Holm adjustment across the ten pairs, ON-1W remains ($p = 0.016$; 0.03 to 0.05 with mean blocks of 50 to 200 days), ON-2W is borderline (0.07 to 0.09; 0.04 to 0.05 with the false-discovery rate), and 1W-2W does not survive (0.09 to 0.29). For ON-1W the evidence does not depend on overlapping returns, on the asymptotic distribution of the tests, or on the set of forecasts; for ON-2W and 1W-2W it is weaker on non-overlapping dates (Section 7.2).

Table 4: **Statistical Rejection and Economic Value by Maturity Pair**

	Statistical					Costs and best forecast		Perfect sign foresight		
	VAR (DST) R^2_{oos} (%)	NN ensemble (%)	SPA p, 23 forecasts	Holm p, ten pairs	VAR without same-day quotes (%)	Quoted spread (bp)	Net fee (bp/yr)	Gross fee (bp/yr)	Break-even spread (bp)	Net ceiling (bp/yr)
ON-1W	20.5***	18.6***	0.002	0.016	1.8***	4.4	0.74	5.55	6.87	3.85
ON-2W	6.2***	5.3***	0.008	0.074	−0.2	4.9	0.27	4.68	7.25	3.27
ON-3W	1.0**	2.2***	0.772	1.000	−2.6	5.2	0.42	5.36	8.85	4.02
ON-1M	−0.2*	1.1**	0.910	1.000	–	5.2	−1.30	4.59	7.12	2.90
ON-2M	−6.8	−9.8	1.000	1.000	–	5.1	−2.07	4.45	5.74	2.38
ON-3M	−8.1	−12.6	1.000	1.000	–	5.5	−1.91	4.61	6.00	2.40
1W-2W	10.1***	7.4***	0.012	0.093	1.2**	4.8	−0.43	2.55	3.04	1.34
1W-3W	2.2***	2.8***	0.362	1.000	−1.7	5.1	−0.31	2.86	3.51	1.37
1M-2M	2.2***	−6.9	0.628	1.000	–	5.7	−0.63	3.48	4.06	2.29
1M-3M	−1.3*	−7.2	1.000	1.000	–	5.9	−0.13	5.22	6.69	4.02

Pairs: short-long maturity (ON: overnight, W: weeks, M: months); out-of-sample origins from April 1994–May 1995 to February 2026–October 2026. R^2_{oos} against the EH forecast; stars: Clark–West test at the 10, 5, and 1% level (it adjusts for estimation noise, so it can reject with a negative R^2_{oos}), which overstates the evidence for ON-3M, 1W-3W, 1M-2M, and 1M-3M (Section 8.1). NN ensemble: average of five networks. SPA p: no forecast of the 23 beats the EH forecast; Holm p: adjusted across the ten pairs. VAR without same-day quotes: today's quotes replaced by their projection from $t-1$ (–: not computed beyond three weeks). Fees: bp a year per unit of notional, sign strategy against the EH strategy; net fee: best of the 23 after quoted spreads. Perfect sign foresight, break-even spread, and net ceiling (the most a strategy that knows the excess return nets after quoted spreads): Section 3.6.

4.2. Accuracy of every forecast

Table 7 reports all 25 forecasts on DST's information set (Panel A) and with market expectations (Panel B). On DST's set the VAR is the most accurate forecast or close to it for every pair on which any forecast predicts. For ON-2M and ON-3M it is less accurate than the EH forecast (−6.8% and −8.1%), and for ON-1M and 1M-3M about as accurate (−0.2% and −1.3%). The term spread at t alone attains 18.7% for ON-1W, and the direct regression on the ten rates is slightly less accurate than the iterated VAR for every pair (19.9% for ON-1W).

Several regularised linear models come close to the VAR on the pairs with one- and two-week long legs, never above it. With the rates in levels, the tree-based models and the nearest-neighbour forecasts are less accurate than the EH forecast for every pair, because they split on the level of the rates and predict a constant outside the range of levels in their window; in stationary form the trees predict (Section 7.5). The equal-weighted combination lies below the VAR on the six pairs on which the VAR predicts, except 1W-3W; shrunk half-way toward the EH forecast it is positive for ON-1W (9.7%), ON-2W (1.3%), 1W-2W (5.7%), and 1W-3W (2.7%). The EWMA attains at most 2.5% (ON-2W), and the zero forecast, DST's own benchmark after demeaning, is less accurate than the EH forecast for every pair except ON-1W.

On DST's information set the five networks forecast almost alike, and their ensemble comes close to the VAR on the three pairs where the VAR predicts most, but not above it: 18.6% for ON-1W (VAR 20.5%), 5.3% for ON-2W (6.2%), and 7.4% for 1W-2W (10.1%);

Table 5: **Evidence Overview Corresponding to the Main Findings**

	Short leg ON						Short leg 1W		Short leg 1M	
	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: In-sample Wald test of DST's VAR restrictions										
Asymptotic p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
p-value simulated under the EH	0.055	0.065	0.085	0.080	< 0.005	0.010	0.225	0.255	< 0.005	< 0.005
Rejection rate of a true EH at 5% (%)	52.0	43.0	41.0	38.5	24.5	14.0	79.0	66.0	16.0	15.0
Panel B: R^2_{oos} (%), DST's information set										
Term spread only	18.7***	6.4***	2.7***	2.7***	−2.5	−4.4	7.1***	1.7**	−2.3	−1.1
VAR (DST)	20.5***	6.2***	1.0**	−0.2*	−6.8	−8.1	10.1***	2.2***	2.2***	−1.3*
Ridge	18.9***	1.3***	0.3**	−1.3*	−10.5	−9.4	6.8***	0.0***	−4.3	−7.8
Random forest	−13.9	−22.2	−24.5	−30.4	−35.5	−38.9	−10.2*	−13.7*	−9.4	−20.5
Neural-network ensemble	18.6***	5.3***	2.2***	1.1**	−9.8	−12.6	7.4***	2.8***	−6.9	−7.2
Best of 20 learners	19.9***	5.8***	2.5***	1.8**	−8.1	−7.5	9.5***	3.1***	−0.5	−5.5
Combination, 50% to the EH	9.7***	1.3**	−0.6	−1.2	−5.9	−6.6	5.7***	2.7***	−0.9	−2.8
SPA p-value: no forecast beats the EH	0.002	0.008	0.772	0.910	1.000	1.000	0.012	0.362	0.628	1.000
Panel C: R^2_{oos} (%), DST's set and market expectations										
OLS	23.1***	10.8***	5.3***	7.6***	−8.6	−11.4	17.2***	11.5***	−0.1***	−10.3**
Ridge	22.0***	7.7***	2.9***	3.2***	−6.6	−6.5	15.3***	9.3***	−1.7***	−3.7**
Neural-network ensemble	18.0***	8.7***	5.4***	9.6***	−5.7*	−10.6	16.9***	14.4***	1.2***	−15.3*
Best of 20 learners	23.1***	10.8***	7.4***	11.3***	−1.2**	−2.7	17.2***	14.7***	1.2***	−3.7**
Combination, 50% to the EH	11.1***	5.4***	4.0***	4.0***	−3.0	−3.3	10.0***	7.9***	3.3***	0.1*
SPA p-value: no forecast beats the EH	0.005	0.016	0.100	0.118	1.000	1.000	0.006	0.008	0.679	1.000
Panel D: What the predictability consists of, R^2_{oos} (%)										
OLS direct, with same-day quotes	19.9	3.3	−2.0	−1.8	−9.6	−10.4	7.6	0.3	−3.6	−7.5
OLS direct, without same-day quotes	2.0***	−2.6	−4.3	−3.4	−8.8	−9.7	0.0**	−2.4	−7.8	−7.4
Slowly moving premium (EWMA)	0.7***	2.5**	1.3*	−0.6*	−10.6	−14.1	2.1**	1.5*	−5.5	−6.5*
SPA p-value: no forecast beats the EWMA	0.002	0.306	0.666	0.575	0.544	0.333	0.011	0.594	0.085	0.432
Panel E: Supporting subsample: forecast origins from 1 January 2016, R^2_{oos} (%)										
VAR (DST)	38.8***	22.1***	13.6***	5.7***	0.0	−2.5	14.8***	8.4***	7.9**	−1.5**
OLS direct without same-day quotes	9.6***	6.6***	1.9*	1.1*	−0.1	−1.2	6.6**	2.1	−11.9	−7.3**
Combination, 50% to the EH, DST's set	22.7***	11.2***	6.8***	4.1*	0.3	−1.2	9.6**	6.0**	−0.8	0.7*
Combination, 50% to the EH, with expectations	15.6***	10.3***	8.6***	10.2***	6.9***	5.5**	13.2***	11.4***	7.2***	5.7**
Out-of-sample observations, DST's set	7,258	6,878	6,806	6,776	6,399	6,154	6,686	6,444	6,622	6,282

Panel A: Wald test of Equation (25); simulated: null model of Equation (30). Stars: Clark–West test at 10, 5, and 1%. Best-of rows ignore the selection. Learners: OLS direct, the 13 learners, the five networks, and their ensemble. Panel C: with market expectations. Panel D: OLS direct with and without same-day quotes, on common dates.

for 1W-3W, ON-1M, and ON-3W the ensemble attains 2.8%, 1.1%, and 2.2%, slightly and not significantly above the VAR (2.2%, −0.2%, and 1.0%), and for the pairs with two- and three-month long legs it is less accurate than the EH forecast (−6.9% to −12.6%). Its Clark–West test rejects for six pairs. No network is significantly more accurate than the VAR; for 1W-2W, 1M-2M, and 1M-3M the VAR is significantly more accurate than the ensemble (Table 26). Flexibility thus adds no detectable accuracy to the two rates of the pair; for the networks, which are odd functions of their inputs (Section 3.4), this holds for departures of that form only.

With market expectations, accuracy rises for linear and nonlinear forecasts alike: the best forecast attains 23.1% for ON-1W, 10.8% for ON-2W, 17.2% for 1W-2W, and 14.7% for 1W-3W. Here the networks gain most: the ensemble attains 18.0%, 8.7%, 16.9%, and 14.4% for these pairs and 9.6% for ON-1M, above least squares on the extended set for 1W-3W (11.5%) and ON-1M (7.6%) and below it for ON-1W (23.1%) and ON-2W (10.8%), and for

Table 6: **Inference Checks for the Three Pairs That Survive the Search**

Check (null: no gain over the EH)	Statistic	Multiplicity	ON-1W	ON-2W	1W-2W
VAR beats the EH forecast	Clark–West, HAC	none (pre-specified)	< 0.001	< 0.001	< 0.001
Best of 23 forecasts beats the EH	SPA, bootstrap	search within the pair	0.002	0.008	0.012
adjusted across ten pairs	SPA, Holm	ten pairs (FWER)	0.016	0.074	0.093
adjusted across ten pairs	SPA, Benjamini–Hochberg	ten pairs (FDR)	0.016	0.039	0.039
EH forecast in the 90% MCS, 24 forecasts	MCS p of the EH	built in	0.031	0.631	0.130
EH forecast in the 90% MCS, five forecasts	MCS p of the EH	built in (power check)	0.001	0.062	0.080
VAR beats the EH, non-overlapping origins	largest Clark–West p over n offsets	worst of n offsets	0.002	0.104	0.259
VAR R^2_{oos} under the simulated EH	simulated p (share $\geq$ data)	none	< 0.005	< 0.005	< 0.005
95% interval of the VAR's R^2_{oos} (%)	stationary bootstrap	none	[13.9, 27.2]	[3.1, 10.7]	[4.4, 17.0]
OLS direct without same-day quotes beats the EH	Clark–West, HAC	none	< 0.001	0.242	0.011
95% interval of its R^2_{oos} (%)	stationary bootstrap	none	[-0.9, 5.2]	[-7.3, 2.9]	[-4.1, 4.7]
Net fee above the EH strategy, sign rule	SPA, blocks of 126 / 250 days	23 forecasts	0.173 / 0.193	0.506 / 0.451	1.000 / 1.000
Net fee above the EH strategy, band rule	SPA, blocks $\max(20, 2n)$ / 250 days	23 forecasts	0.003 / 0.007	0.473 / 0.597	1.000 / 1.000
adjusted across ten pairs	SPA, Holm	ten pairs (FWER)	0.026 / 0.072	1.000 / 1.000	1.000 / 1.000
Out-of-sample origins			7,258	6,878	6,686
Origins per non-overlapping offset			1,451	687	668
Simulated samples under the EH			200	200	200

p-values unless stated; null in each row: the forecast, or the best of the 23 forecasts, does not beat the EH forecast or strategy. MCS: below 0.10 means outside the 90% set. Non-overlapping: every n-th origin, for each of the n offsets. Simulated EH: Equation (30). Net fees after quoted spreads; band rule: Equation (23).

1W-2W it is significantly more accurate than DST's VAR on DST's set (Diebold–Mariano $p = 0.03$), as is least squares on the extended set. The gain comes from the information; flexibility helps only once the information is there, and not consistently.

The gains accrue in a few episodes (Figure 4). For the short pairs the cumulative gain of the VAR over the EH forecast rises in a few steps, the largest in 2007–08; for ON-2M, ON-3M, and 1M-3M the VAR loses in the early 2000s and around 2008 and does not recover. Section 7.3 shows that the accuracy nevertheless comes mostly from calm periods.

4.3. Search-adjusted tests

Table 8 reports the tests across the 23 forecasts for both information sets (Panel A) and the sign and state tests of the VAR (Panel B). On DST's set the Clark–West test rejects at the 5% level for 18, 15, and 18 of the 23 forecasts of ON-1W, ON-2W, and 1W-2W with a positive R^2_{oos}, and after a Holm adjustment within the pair for 18, 13, and 16; for 1W-3W it rejects for 12 such forecasts and 11 after the adjustment, and for ON-3W and ON-1M for 11

Table 7: **Out-of-Sample Predictability of Departures from the EH**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: DST's information set										
VAR (DST)	20.5***	6.2***	1.0**	−0.2*	−6.8	−8.1	10.1***	2.2***	2.2***	−1.3*
OLS (direct)	19.9***	3.3***	−2.2*	−1.6**	−9.5	−10.4	7.4***	0.1***	−3.2**	−7.4
Ridge	18.9***	1.3***	0.3**	−1.3*	−10.5	−9.4	6.8***	0.0***	−4.3	−7.8
Lasso	18.8***	1.6***	0.0**	−1.0	−9.0	−8.1	7.6***	0.8***	−5.8	−9.0
Elastic net	18.8***	1.4***	0.1**	−0.9	−8.8	−8.2	7.8***	0.7***	−5.3	−8.8
PCR	16.3***	−0.5**	−0.4**	−2.2	−10.7	−9.4	6.3***	−0.7**	−7.5	−8.5
PLS	18.6***	0.9***	−0.9*	−2.5	−11.3	−10.0	5.3***	−1.0**	−5.7	−8.1
Linear SVR	18.5***	1.4***	0.0**	−2.3	−8.1	−7.5	9.5***	0.8***	−0.5	−7.6
RBF SVR	10.7***	−0.1***	−4.0	−6.0	−13.1	−21.6	5.5***	−0.7**	−5.7	−14.2
KNN	−1.6***	−21.8	−23.0	−21.6	−45.7	−41.9	−5.1**	−12.0*	−27.1	−55.3
Tree	−31.2	−50.8	−36.9	−47.0	−65.3	−46.9	−16.5**	−24.8*	−7.0	−15.1
Random forest	−13.9	−22.2	−24.5	−30.4	−35.5	−38.9	−10.2*	−13.7*	−9.4	−20.5
Boosting	−55.4	−30.4	−44.8	−40.0	−26.8	−40.1	−47.5*	−52.3*	−5.9	−23.8
Bagging	−27.7***	−38.8	−49.2	−55.4	−73.2	−79.6	−40.4**	−60.1***	−62.9	−86.8
Random features	6.6***	−18.6	−20.1	−22.8	−23.5	−49.9	2.4***	−2.9	−6.0	−17.0
NN (16)	18.5***	5.2***	1.5***	1.0**	−8.2	−12.1	4.6***	1.6***	−6.7	−11.0
NN (32)	18.8***	4.5***	1.5***	0.2**	−10.9	−15.6	7.0***	1.2***	−6.7	−11.0
NN (16-8)	18.1***	5.8***	2.4***	1.1**	−10.6	−13.3	7.9***	2.3***	−8.2	−6.6
NN (32-16)	17.8***	4.6***	2.5***	−0.1***	−9.9	−13.9	7.5***	3.1***	−7.5	−7.6
NN (32-16-8)	18.5***	5.2***	1.5***	1.8**	−11.5	−13.1	7.7***	2.5***	−7.5	−5.5
NN ensemble	18.6***	5.3***	2.2***	1.1**	−9.8	−12.6	7.4***	2.8***	−6.9	−7.2
Combination	14.8***	0.4**	−2.8	−3.9	−12.4	−13.2	7.9***	2.8***	−2.6	−7.3
Combination, 50% to EH	9.7***	1.3**	−0.6	−1.2	−5.9	−6.6	5.7***	2.7***	−0.9	−2.8
Combination, DMSPE	15.8***	1.0**	−2.0	−2.9	−11.0	−11.9	8.2***	3.1***	−2.2	−5.9
EWMA	0.7***	2.5**	1.3*	−0.6*	−10.6	−14.1	2.1**	1.5*	−5.5	−6.5*
Pure EH (zero)	0.0	−0.9	−1.5	−2.6	−5.7	−7.7	−1.3	−2.0	−2.7	−5.2
Spread only	18.7***	6.4***	2.7***	2.7***	−2.5	−4.4	7.1***	1.7**	−2.3	−1.1
Obs.	7,258	6,878	6,806	6,776	6,399	6,154	6,686	6,444	6,622	6,282
Panel B: DST's set and four market expectations										
OLS	23.1***	10.8***	5.3***	7.6***	−8.6	−11.4	17.2***	11.5***	−0.1***	−10.3**
Ridge	22.0***	7.7***	2.9***	3.2***	−6.6	−6.5	15.3***	9.3***	−1.7***	−3.7**
Lasso	21.2***	6.1***	1.9***	3.1***	−2.6	−2.7	15.6***	8.2***	−2.8***	−8.1
Elastic net	20.9***	6.1***	2.1***	1.4***	−3.5	−3.1	15.4***	8.6***	−2.2***	−7.6
PCR	4.4**	7.1***	3.9***	1.7**	−4.8	−8.0	7.9***	5.3***	−4.7*	−5.6**
PLS	10.2***	5.0***	4.2**	−3.1***	−6.7	−8.5	8.9***	7.9***	−0.5***	−7.2**
Linear SVR	21.0***	6.5***	4.2***	7.0***	−1.2**	−5.9	15.1***	7.7***	1.2***	−6.3**
RBF SVR	9.5***	0.8**	−4.1*	−4.7*	−37.8	−8.2*	−5.7	−2.7**	−3.0***	−5.7***
KNN	−0.3***	−6.8**	−10.5	−11.9	−24.1	−17.0	−2.9***	−6.5	−23.8	−41.3
Tree	−3.4***	−41.2	−30.9**	−11.4**	−48.5	−32.2	−37.7***	−18.2**	−15.5	−61.7
Random forest	−5.8***	−12.6	−9.2**	−24.1	−29.0	−16.9	−9.8***	−5.7**	−14.2	−22.3
Boosting	−14.4***	−17.1**	−24.5	−30.6	−51.1	−37.4	−17.7***	−7.1**	−11.3**	−14.5
Bagging	−27.8***	−68.8*	−37.0	−55.5	−55.7	−83.3	−57.5***	−45.3**	−22.7**	−55.3
Random features	−0.8***	−10.0	−23.6	−23.0	−36.8	−34.4	−7.2	−19.9	−15.5	−17.0*
NN (16)	17.3***	10.7***	7.4***	11.3***	−3.5***	−14.6	16.3***	14.7***	−0.1***	−11.6**
NN (32)	16.7***	5.2***	3.3***	4.4***	−14.2	−20.9	16.5***	13.4***	−0.9***	−26.7
NN (16-8)	16.4***	6.9***	2.3***	10.5***	−7.1	−9.2	14.6***	12.7***	0.8***	−11.2**
NN (32-16)	16.5***	8.7***	4.0***	7.6***	−6.2**	−13.5	16.3***	13.9***	−2.6***	−19.7
NN (32-16-8)	18.8***	8.2***	4.8***	5.9***	−8.9	−3.2*	16.5***	12.3***	−2.1***	−21.8
NN ensemble	18.0***	8.7***	5.4***	9.6***	−5.7*	−10.6	16.9***	14.4***	1.2***	−15.3*
Combination	17.7***	7.4***	5.2***	5.1***	−7.8	−7.4	14.4***	11.2***	2.6***	−3.9**
Combination, 50% to EH	11.1***	5.4***	4.0***	4.0***	−3.0	−3.3	10.0***	7.9***	3.3***	0.1*
Combination, DMSPE	17.8***	7.7***	5.3***	5.4***	−6.9	−6.5	15.0***	11.4***	3.0***	−2.9**
EWMA	0.7***	2.5**	1.3*	−0.6*	−10.6	−14.1	2.1**	1.5*	−5.5	−6.5*
Pure EH (zero)	0.0	−0.9	−1.5	−2.6	−5.7	−7.7	−1.3	−2.0	−2.7	−5.2
Obs.	7,258	6,878	6,806	6,776	6,399	6,154	6,686	6,444	6,622	6,282

Stars: Clark–West test at 10, 5, and 1% (possible with a negative R^2_{oos}). Rows: the 23 forecasts of the search family, two combinations with the EH forecast, and the zero forecast; NN: hidden layers; spread only: regression on the long minus the short rate at t. Panel B: OLS is a learner. Common dates within a panel and pair.

and 5 but none after it. Because the test adjusts for the estimation noise of the larger model, it also rejects for some forecasts with a negative R^2_{oos}, which these counts leave out. The model confidence set is less decisive: it excludes the EH forecast only for ON-1W ($p = 0.031$)

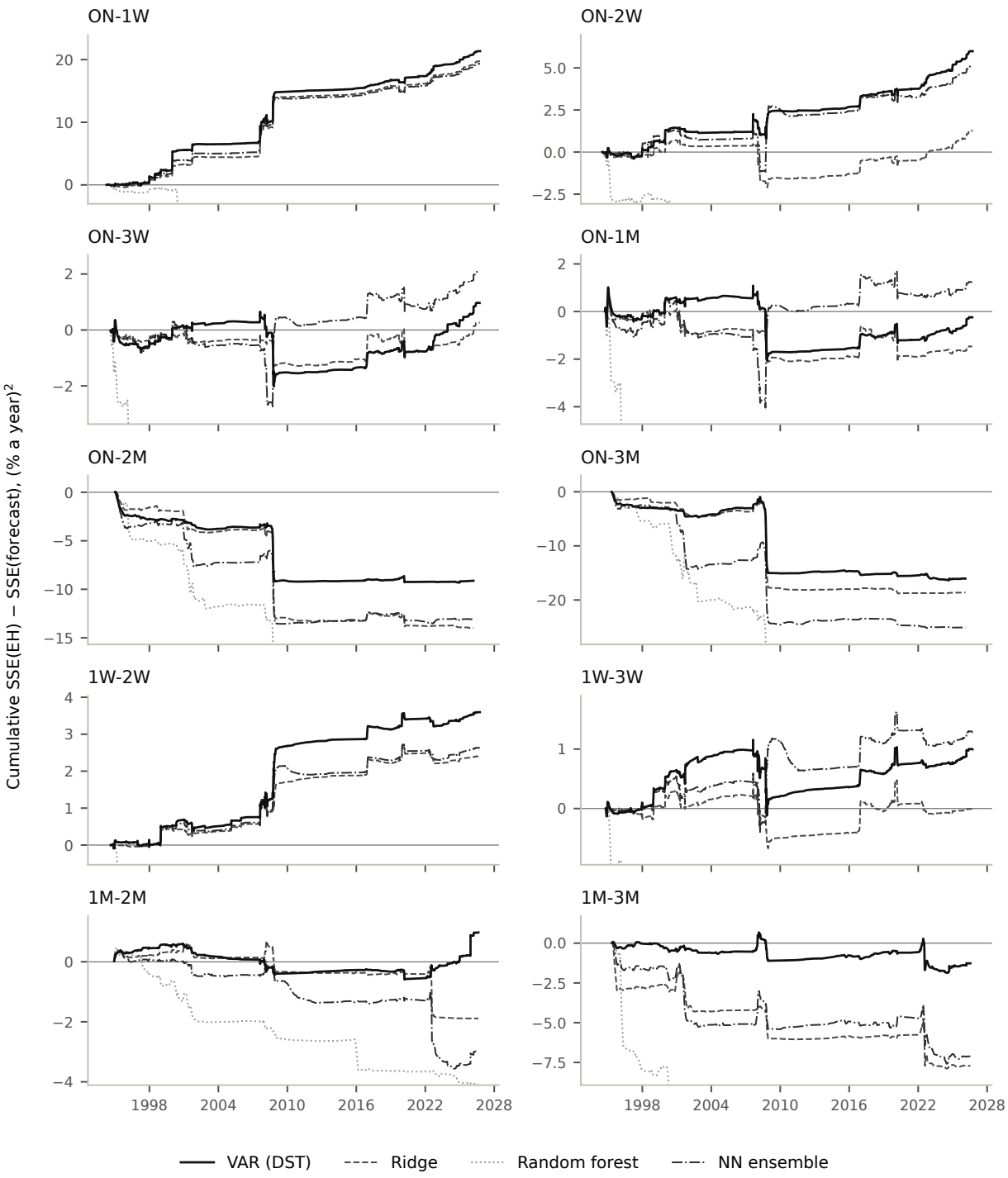


Figure 4: **Cumulative Forecast Gains Relative to the EH.** Cumulative squared errors of the EH forecast minus those of each forecast, squared percent a year; rising: more accurate than the EH forecast. The random forest is cut off where it leaves the axis.

and retains the VAR for every pair. It eliminates forecasts one at a time among all 24, the EH forecast included, and retains forecasts with very negative R^2_{oos}, which lowers its power

(Section 7.2).

With market expectations the SPA test rejects for ON-1W ($p = 0.005$), ON-2W ($p = 0.016$), 1W-2W ($p = 0.006$), and 1W-3W ($p = 0.008$), and at about the 10% level for ON-3W and ON-1M. Across the ten pairs none survives the Holm adjustment at 5%: ON-1W reaches 0.054 at the paper's block length (0.08 to 0.10 with longer blocks), 1W-2W 0.055, and 1W-3W 0.06 to 0.11 (Table 22). The Clark–West test rejects after the Holm adjustment within the pair for 10 to 23 forecasts on the pairs up to one month and for 15 on 1M-2M. Expectations thus add 1W-3W to the pairs that survive the search within the pair, but not across pairs; the choice of this set in the design study is not adjusted for (Section 3.3).

The VAR also predicts the direction of the departure from the EH forecast and gains in specific states (Panel B). On non-overlapping dates it predicts the sign in 54.0% to 67.7% of cases for seven pairs (all but ON-2M, ON-3M, and 1M-3M), significantly in the test of Pesaran and Timmermann (1992), and the network ensemble in 56.2% to 65.2% of cases on seven pairs. The conditional test with a month-turn dummy rejects a zero expected gain of the VAR for ON-1W, ON-2W, and 1W-2W, and the month-turn coefficient is significant for 1W-2W ($p = 0.034$) and at the 10% level for ON-2W; for the pairs with long legs of two and three months every window crosses a month end, so this test is not defined. With dummies for the zero lower bound, NBER recessions, and high volatility of the short rate, the test rejects for most pairs.

4.4. Relation to DST's in-sample rejection

Let z_t stack the short rate, the long rate, and their first four lags, so that $z_{t+1} = Gz_t + \varepsilon_{t+1}$ is the VAR(5) of Equation (7) in companion form, and let a_1 and a_2 select the short and the long rate from z_t. The EH with a constant premium and expectations formed by the VAR imply $i_t^{(n)} = k^{-1} \sum_{j=0}^{k-1} E_t[i_{t+jm}^{(m)}]$ up to a constant, and therefore the ten nonlinear cross-equation restrictions

$$a_2' = a_1' \frac{1}{k} \sum_{j=0}^{k-1} G^{mj}. \tag{25}$$

The long rate they imply, $i_t^{(n)*} = a_1' k^{-1} \sum_j G^{mj} z_t$, is the theoretical long rate.

DST test Equation (25) on daily repo rates from May 1991 to December 2005 with Lagrange-multiplier and distance-metric statistics and bootstrap p-values, and reject the EH for every pair (their Table 5); they avoid the Wald statistic for its poor small-sample properties (their footnote 7). This paper reports the Wald test on the sample to October 2026 and measures its size by simulation (Table 9, Panel A). With heteroskedasticity-robust covariances every asymptotic p-value is below 0.001, yet the theoretical long rate is close to the actual one: their correlation is 0.999 to 1.000, and the ratio of their standard deviations

Table 8: **Joint Tests across Forecasts**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: Forecasts against the EH forecast										
DST's information set										
SPA p-value (best forecast vs EH)	0.002	0.008	0.772	0.910	1.000	1.000	0.012	0.362	0.628	1.000
MCS p-value of the EH forecast	0.031	0.631	0.879	0.959	1.000	1.000	0.130	0.823	0.819	1.000
MCS p-value of the VAR	1.000	1.000	0.980	0.959	0.885	0.749	1.000	0.991	1.000	0.884
Forecasts in the 90% MCS	17/24	21/24	21/24	22/24	22/24	22/24	19/24	19/24	22/24	22/24
Clark–West rejections, unadjusted	20/23	17/23	12/23	7/23	0/23	0/23	21/23	17/23	2/23	0/23
Clark–West rejections, Holm within pair	20/23	14/23	0/23	0/23	0/23	0/23	16/23	12/23	0/23	0/23
Clark–West rejections, Holm across pairs	17/23	2/23	0/23	0/23	0/23	0/23	16/23	1/23	0/23	0/23
DST's set and market expectations										
SPA p-value (best forecast vs EH)	0.005	0.016	0.100	0.118	1.000	1.000	0.006	0.008	0.679	1.000
SPA p-value (best forecast vs DST's VAR)	0.354	0.473	0.223	0.142	0.109	0.167	0.036	0.066	0.824	0.758
MCS p-value of the EH forecast	0.023	0.262	0.812	0.750	1.000	1.000	0.042	0.153	0.990	1.000
MCS p-value of the VAR	0.857	0.913	0.926	0.750	0.846	0.815	0.216	0.344	0.990	0.987
Forecasts in the 90% MCS	14/24	22/24	24/24	23/24	24/24	24/24	16/24	22/24	24/24	24/24
Clark–West rejections, unadjusted	23/23	19/23	17/23	15/23	3/23	0/23	21/23	20/23	17/23	9/23
Clark–West rejections, Holm within pair	23/23	15/23	10/23	10/23	0/23	0/23	20/23	15/23	15/23	0/23
Clark–West rejections, Holm across pairs	16/23	11/23	9/23	5/23	0/23	0/23	15/23	15/23	9/23	0/23
Panel B: Direction and state dependence of the VAR's gain, DST's information set										
Hit rate of the VAR (%)	65.8***	62.8***	54.0**	61.3***	47.1	59.2	67.0***	67.7***	67.1***	56.0
Hit rate of the NN ensemble (%)	65.2***	62.4***	56.2***	59.4***	60.1**	59.2	62.2***	60.7***	51.3	54.0
GW p-value: month turn, joint	0.001	0.003	0.157	0.451	–	–	0.002	0.282	–	–
GW p-value: month-turn coefficient	0.218	0.061	0.114	0.265	–	–	0.034	0.263	–	–
GW p-value: zero lower bound	0.001	0.002	0.895	0.969	0.064	0.240	0.003	0.000	0.001	0.001
GW p-value: NBER recession	0.000	0.001	0.014	0.086	0.056	0.361	0.002	0.002	0.009	0.826
GW p-value: high volatility	0.000	0.005	0.543	0.641	0.038	0.342	0.000	0.141	0.296	0.788
Obs., DST's information set	7,258	6,878	6,806	6,776	6,399	6,154	6,686	6,444	6,622	6,282

Panel A: MCS: in the 90% set if at least 0.10; Clark–West rejections at 5%, unadjusted and Holm; with market expectations: SPA against DST's VAR over the 24 other forecasts. Panel B: hit rate on non-overlapping origins, stars: Pesaran–Timmermann test; GW: Giacomini–White tests; –: the state does not vary.

0.989 to 1.002. Against its distribution under a simulated EH, the statistic remains significant at the 5% level for four pairs only, ON-2M, 1M-2M, and 1M-3M ($p < 0.005$) and ON-3M ($p = 0.010$), and at the 10% level for ON-1W, ON-2W, ON-1M, and ON-3W ($p = 0.055$, 0.065, 0.080, and 0.085), because the asymptotic test rejects a true EH in 14% to 79% of the samples (Section 8.1). In sample, the evidence against the EH therefore depends on the null model, in line with Bekaert and Hodrick (2001).

The in-sample projection of the excess return on the ten lagged rates has an R^2 of 0.6% (ON-3M) to 22.8% (ON-1W), with jointly significant slopes for every pair (Panel B). The RESET test of Ramsey (1969) rejects its linearity at the 5% level for six pairs, the neural-network test of Lee et al. (1993), repeated for ten draws of its random weights, for none after a Bonferroni adjustment (Panel C).

Out of sample, DST's own data confirm their rejection pointwise but not after the search (Panel D and Table 10). On their sample, 21 May 1991 to 9 December 2005, the VAR's in-sample R^2 is at most 19.2% and the Wald test rejects for every pair; with the protocol of this paper the VAR beats the EH forecast for six pairs with an expanding window (ON-1W 15.7% on DST's sample; 15.5% on the origins of the main run with targets realised by 2005,

Table 10) and for six with a rolling window of 1,250 days. On origins whose targets are realised by the end of 2005, the SPA test over the 23 forecasts rejects within the pair only for 1W-2W ($p = 0.016$) and for ON-1W and 1W-3W at the 10% level ($p = 0.077$ and 0.087), no pair survives the Holm adjustment across pairs, and without same-day quotes the VAR's R^2_{oos} is negative for all five pairs with horizons up to three weeks (ON-1W −0.5%; Table 10). Of DST's ten in-sample rejections, none survives out of sample on their own period, and one, ON-1W, on the full sample. The iterated VAR with the small-sample correction is more accurate than the iterated VAR without it and than the direct regression for every pair (Panel E), consistent with the small-sample bias of persistent predictors (Stambaugh, 1999).

Table 9: **In-Sample Tests and DST's VAR**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: Wald test of DST's VAR restrictions, in sample										
Wald statistic	383.7	164.4	105.9	88.4	120.4	35.8	546.7	203.7	241.1	117.5
Asymptotic p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
$\rho(i^{(n)}, i^{(n)*})$	0.9995	0.9997	0.9998	0.9999	1.0000	1.0000	0.9999	1.0000	1.0000	1.0000
$\sigma(i^{(n)*})/\sigma(i^{(n)})$	1.0016	1.0003	0.9991	0.9968	0.9934	0.9885	0.9991	0.9982	0.9967	0.9922
Obs.	7,491	7,200	7,279	7,360	7,407	7,541	7,062	7,018	7,214	7,214
Panel B: In-sample projection on DST's information set										
R^2 (%)	22.8	10.0	7.3	7.5	2.5	0.6	14.6	8.4	4.9	4.2
Wald test that all slopes are zero, p-value	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Panel C: Tests for neglected nonlinearity, p-values										
Neural-network test, median	0.924	0.922	0.882	0.476	0.413	0.255	0.962	0.936	0.251	0.235
Neural-network test, Bonferroni	0.926	0.771	0.131	0.605	1.000	1.000	1.000	1.000	1.000	1.000
RESET test	0.009	0.006	0.003	0.010	0.136	0.169	0.417	0.086	0.001	0.000
Obs. (Panels B and C)	7,762	7,387	7,320	7,296	6,940	6,716	7,195	6,958	7,163	6,844
Panel D: DST's VAR on DST's sample, 21 May 1991 to 9 December 2005										
In-sample R^2 of the VAR forecast (%)	19.2	7.0	4.6	5.5	2.2	2.8	14.2	9.8	−1.1	3.8
In-sample R^2 of the direct regression (%)	17.5	7.6	5.9	8.0	3.1	3.8	14.9	10.2	7.3	8.4
Wald test of the EH restrictions, asymptotic p-value	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
R^2_{oos} (%), expanding window	15.7***	3.5***	0.9**	1.7**	−6.0	−8.4	6.4***	7.4***	−0.4**	−1.3
R^2_{oos} (%), rolling window of 1,250 days	14.5***	1.2**	1.4***	2.6**	−15.1	−24.0	5.2***	7.8***	−13.8	−10.6
Obs.	3,278	3,166	3,069	3,057	2,870	2,764	3,162	3,054	3,033	2,922
Panel E: Bias correction and the route through rate forecasts, R^2_{oos} (%)										
Direct regression on the ten rates	19.2	3.0	−1.5	−1.0	−8.2	−9.1	6.6	0.5	−3.1	−7.5
Iterated VAR, least squares	19.6	7.1	2.5	1.4	−7.9	−7.6	10.1	4.0	1.4	−3.3
Iterated VAR, bias-corrected (specification of the paper)	19.9	7.5	3.1	1.7	−6.7	−6.3	10.3	4.1	2.4	−1.7
Obs.	6,996	6,615	6,575	6,541	6,215	6,028	6,446	6,261	6,416	6,103

Daily data, 21 May 1991 to 16 October 2026; in sample unless stated. Stars: Clark–West test at 10, 5, and 1%. Panel A: heteroskedasticity-robust Wald test of Equation (25), $\chi^2(10)$; $i^{(n)*}$: long rate implied by the EH and the VAR. Panel C: Lee et al. (1993), Bonferroni over ten draws, and RESET (Ramsey, 1969). Panel D: out of sample: expanding or rolling (1,250 days) window. Panel E: separate recursion on common dates.

Table 10: **Out-of-Sample Evidence on DST's Period**

	Short leg ON						Short leg 1W		Short leg 1M	
	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
VAR (DST), R^2_{oos} (%)	15.5***	3.7***	1.0**	1.9**	−10.5	−6.8	6.5***	7.2***	0.7***	−1.7
SPA p, 23 forecasts	0.077	0.249	0.610	0.810	1.000	1.000	0.016	0.087	0.966	1.000
Holm across the ten pairs	0.693	1.000	1.000	1.000	1.000	1.000	0.161	0.696	1.000	1.000
VAR (DST) without same-day quotes (%)	−0.5*	−1.9	−3.4	–	–	–	−0.7	−0.6*	–	–
Direct regression (%)	12.6***	3.0***	−0.2**	–	–	–	5.2***	6.9***	–	–
without same-day quotes	0.0*	−1.2*	−3.0	–	–	–	0.5**	−1.9	–	–
Out-of-sample origins	2,788	2,667	2,559	2,544	2,326	2,195	2,662	2,537	2,473	2,316

Origins whose targets are realised by 31 December 2005, the end of DST's sample. Stars: Clark–West test at 10, 5, and 1%. SPA: 10,000 draws, mean block $\max(20, 2n)$. Without same-day quotes: the five pairs with horizons up to three weeks (–: not computed).

5. A Short-Lived Signal in Same-Day Quotes

A one-day lag removes most of the predictability: for the overnight pairs the signal is today's overnight rate against today's term quote, which prices the overnight rate but not the persistence of its deviation, and for 1W-2W a month-end and quarter-end premium in two closing term quotes.

5.1. Lagging the predictors by one day

Table 11 summarises the comparison for the five pairs with horizons of one to three weeks; Table 12 reports it for every model and variant.

Table 11: **The Same-Day Signal and Its Robustness**

	ON-1W	ON-2W	ON-3W	1W-2W	1W-3W
Panel A: Which same-day quote carries the signal (DST's VAR, 1994–2026)					
All quotes, lags t to $t-4$	20.5	6.3	1.1	10.4	2.3
Without today's short-leg quote	3.0	0.3	−1.7	2.4	−0.1
Without today's long-leg quote	10.6	4.0	−1.5	1.3	−2.0
Without same-day quotes	1.8	−0.2	−2.6	1.2	−1.7
Panel B: With / without same-day quotes					
Direct regression	19.9 / 2.0	3.3 / −2.6	−2.0 / −4.3	7.6 / 0.0	0.3 / −2.4
Neural-network ensemble	18.6 / 0.9	5.4 / 0.7	2.2 / −0.3	7.7 / 0.6	3.0 / 0.3
Ridge regression	19.0 / 0.3	1.3 / −1.2	0.3 / −1.4	7.0 / −0.4	0.1 / −0.7
With market expectations	23.1 / 2.9	11.0 / 3.6	–	17.6 / 6.2	11.5 / 5.0
From 2016	34.2 / 9.6	18.8 / 6.6	11.3 / 1.9	13.7 / 6.6	6.7 / 2.1
Stale quotes: runs of more than 3	18.6 / 1.5	2.7 / −2.0	–	8.4 / −1.6	–
Stale quotes: runs of more than 10	20.5 / 2.9	5.8 / 0.1	–	16.0 / 3.7	–
Overnight rolls at the closing quote, from 2020	39.9 / 1.6	12.3 / −6.8	0.5 / −16.8	–	–
Overnight rolls at the TGCR, from 2020	36.0 / 8.1	21.6 / 8.1	15.3 / 3.4	–	–

Out-of-sample R^2 (%) against the EH forecast, for the five pairs with horizons of one to three weeks, on the dates common to the variants compared (up to 0.4 points from Table 4, which uses all dates of each forecast). Panel A: DST's VAR without today's short-leg, long-leg, or both quotes, each replaced by its VAR projection. Panel B: first value with all quotes, second without the quotes of the forecast date (lags $t-1$ to $t-5$); direct regression unless stated; stale thresholds rebuild the series; closing quote and TGCR: overnight rolls at the closing quote or at the tri-party general collateral rate, evaluated from 2020. –: not computed.

Each model is estimated once on X_t and once on X_t^{lag} of Equation (5) and evaluated on the same dates, and the same-day part of its accuracy is

$$\Delta_j^{same} = R^2_{oos}\big(X_t\big) - R^2_{oos}\big(X_t^{lag}\big), \qquad \Delta_j^{(\ell)} = R^2_{oos}\big(X_t\big) - R^2_{oos}\big(X_t^{-\ell}\big), \tag{26}$$

where $X_t^{-\ell}$ replaces only today's quote of leg $\ell \in \{m, n\}$ by the previous day's (for the VAR, by its projection given the other quote). Under the noise model of Equation (4), the accuracy decomposes approximately into a noise part Δ_j^{same} and a persistent part $R^2_{oos}(X_t^{lag})$, with the caveats stated there. For the VAR, replacing today's quotes by their projection from $t-1$ gives its forecast from lags $t-1$ to $t-5$. The first roll's rate is known at t; its share of the same-day signal is

$$\phi = \frac{1}{\widehat{\beta}}, \qquad y_t = \alpha + \beta\,\psi_t + e_t, \qquad \psi_t = \frac{d_0}{D_t}\big(\hat{i}_t^{(m)} - \hat{i}_t^{(n)}\big), \tag{27}$$

the share of the covariance of the excess return with the first roll's contribution ψ_t to the excess return, known at t, that the first roll itself accounts for.

Without the same-day quotes, the R^2_{oos} of the direct regression falls from 19.9% to 2.0% for ON-1W, from 3.3% to −2.6% for ON-2W, and from 7.6% to 0.0% for 1W-2W (Table 5, Panel D). Ridge falls to similar levels, the network ensemble from 18.6% to 0.9% for ON-1W and from 7.7% to 0.6% for 1W-2W (Table 11), and boosting is less accurate than the EH forecast with and without them (Table 12, Panel A). The Clark–West test still rejects for ON-1W and 1W-2W without the same-day quotes, the bootstrap intervals, which describe sampling uncertainty, include zero (Table 6), and an SPA test over the lagged VAR, direct regression, ridge, and network ensemble does not reject ($p = 0.10$ to 0.16 for ON-1W and 0.34 to 0.40 for 1W-2W; 0.09 to 0.14 and 0.33 to 0.39 without the networks; mean blocks of $\max(20, 2n)$ to 250 days; Table 12, Panel A). For ON-1W, dropping only today's overnight quote lowers the direct regression's R^2_{oos} from 19.9% to 3.1%, dropping only today's one-week quote to 10.3%, and dropping both to 2.0%; the VAR behaves alike (20.5%, 3.0%, 10.6%, 1.8%). The signal is therefore today's overnight rate relative to today's term quote. The first roll accounts for 50% to 58% of the response of the overnight pairs' excess returns to it (Panel C). The rest comes from the later rolls: the deviation of the overnight rate from the one-week quote has a first-order autocorrelation of 0.41, so part of today's deviation carries over into the next rolls, while the one-week quote moves with today's overnight rate by only 0.22 (0.25 since April 2018; Panel E), about the weight of the first roll. The term quote thus prices today's overnight rate but not the persistence of its deviation. Because repo trades mostly before 9 a.m. and the time of the closing quotes is undocumented (Section 2.1), a

participant deciding at the close cannot be shown to capture either component; what such a participant can use is the forecast with lagged predictors.

1W-2W has no overnight leg, and its signal is today's spread between the two closing term quotes: dropping today's one-week quote lowers the direct regression's R^2_{oos} from 7.6% to 1.0%, dropping today's two-week quote to −0.9% (Panel A). Its same-day accuracy comes from windows that contain a month end: in the other windows, about half of the origins, the R^2_{oos} is negative (−1.6%), and the windows that contain a quarter end give 71% of the gain (Panel F). The spread is short-lived (first-order autocorrelation 0.58), so the one-day lag loses it. For 1W-2W the signal is thus a month-end and quarter-end premium in two term quotes observed at the same close, not a difference in timing between the overnight and the term market. Calendar dummies known at the origin add no out-of-sample accuracy to the rates (Section 7.5), so the premium is not a fixed calendar pattern but one that today's quotes reveal.

With every predictor of the market-expectations set lagged by one day, least squares still attains 2.9% to 6.2% for ON-1W, ON-2W, 1W-2W, and 1W-3W, against at most 2.0% for DST's lagged set (Panel D). The gain over DST's lagged set is significant at the 5% level only for 1W-2W (Diebold–Mariano $p = 0.04$; 0.09 to 0.57 for the other pairs), and the SPA test over the lagged forecasts rejects for 1W-2W ($p = 0.03$) and at 10% for ON-1W and 1W-3W.

5.2. Not noise in the overnight quote

Under Equation (4), noise in today's quotes would create a same-day signal of this kind. Table 13 compares the Bloomberg overnight quote with the TGCR, BGCR, and SOFR of the Federal Reserve Bank of New York on 2,066 business days from April 2018. The levels correlate above 0.999, and the Bloomberg quote is 3.9 to 5.7 basis points higher on average. The daily changes correlate less well, 0.79 to 0.80, and the Bloomberg quote changes on 80.7% of days against 48.9% to 56.1% for the reference rates, which is consistent with transitory quote errors.

The overnight leg of the target can be replaced. If the rolled leg of ON-1W uses the TGCR instead of the Bloomberg quote, the excess return correlates with the original at 0.887, and the direct regression, estimated on the TGCR-based excess return from April 2018 only and therefore evaluated from 2020, predicts it about as well as the Bloomberg one estimated on the same window (36.0% against 39.9%) and better for ON-2W and ON-3W (Table 12, Panel B). These values differ from those since April 2018 in Table 14, Panel A, which evaluate the full-sample forecasts from April 2018. These are re-estimated forecasts, not the same forecasts evaluated on another target. The transaction-rate check thus rules

Table 12: **Same-Day Quotes, Transaction Rates, and Lagged Market Expectations**

	ON-1W	ON-2W	ON-3W	1W-2W	1W-3W
Panel A: Dropping one quote of day t, R^2_{oos} (%), full sample					
OLS direct, with both quotes of day t	19.9***	3.3***	−2.0*	7.6***	0.3***
OLS direct, without today's short-rate quote	3.1***	−1.1*	−3.7	1.0***	−1.2*
OLS direct, without today's long-rate quote	10.3***	0.4***	−3.8	−0.9**	−4.3
OLS direct, without both (lags $t-1$ to $t-5$)	2.0***	−2.6	−4.3	0.0**	−2.4
VAR (DST), with both quotes of day t	20.5***	6.3***	1.1**	10.4***	2.3***
VAR (DST), without today's short-rate quote	3.0***	0.3*	−1.7	2.4***	−0.1*
VAR (DST), without today's long-rate quote	10.6***	4.0***	−1.5	1.3**	−2.0
VAR (DST), without both (lags $t-1$ to $t-5$)	1.8***	−0.2	−2.6	1.2**	−1.7
Ridge, with both quotes of day t	19.0	1.3	0.3	7.0	0.1
Ridge, without both (lags $t-1$ to $t-5$)	0.3***	−1.2	−1.4	−0.4*	−0.7
NN ensemble, with both quotes of day t	18.6***	5.4***	2.2***	7.7***	3.0***
NN ensemble, without both (lags $t-1$ to $t-5$)	0.9***	0.7**	−0.3*	0.6**	0.3**
SPA p, lagged VAR, OLS direct, and ridge	0.14	1.00	1.00	0.36	1.00
SPA p, the same and the NN ensemble	0.15	0.71	1.00	0.38	0.78
Panel B: Transaction-based overnight rate (TGCR), estimated from April 2018, evaluated from 2020, OLS direct, R^2_{oos} (%)					
Bloomberg target, with both quotes of day t	39.9***	12.3***	0.5**	–	–
Bloomberg target, lags $t-1$ to $t-5$	1.6***	−6.8**	−16.8	–	–
target with TGCR as rolled overnight rate, with both quotes of day t	36.0***	21.6***	15.3***	–	–
target with TGCR as rolled overnight rate, lags $t-1$ to $t-5$	8.1***	8.1***	3.4***	–	–
Panel C: The first overnight roll, full sample					
Share of the excess return's response to the first roll	0.58	0.57	0.50	–	–
Panel D: Market expectations with all predictors lagged by one day, OLS, R^2_{oos} (%)					
Same-day predictors, full sample	23.1	11.0	–	17.6	11.5
Lagged predictors, full sample	2.9***	3.6***	–	6.2***	5.0***
DST's set lagged, full sample	2.0	−2.6	–	0.0	−2.4
Gain over DST's lagged set, DM p, full sample	0.57	0.28	–	0.04	0.09
SPA p over the lagged forecasts, full sample	0.09	0.12	–	0.03	0.05
Same-day predictors, from 2016	27.8	13.9	–	23.0	17.6
Lagged predictors, from 2016	−1.9**	−3.7***	–	9.9***	9.0***
DST's set lagged, from 2016	9.6	6.6	–	6.6	2.1
Gain over DST's lagged set, DM p, from 2016	0.00	0.03	–	0.18	0.04
SPA p over the lagged forecasts, from 2016	0.00	0.03	–	0.13	0.06
Panel E: Today's overnight rate and the term quote (overnight pairs)					
Slope of the term-quote change on today's overnight change, full sample	0.22 (5.0)	0.14 (4.9)	0.13 (4.4)	–	–
Same slope, from April 2018	0.25 (3.7)	0.23 (3.9)	0.22 (4.2)	–	–
Slope on yesterday's overnight change, full sample	0.05 (2.1)	0.04 (3.1)	0.02 (2.1)	–	–
One-day autocorrelation of the overnight-minus-term deviation	0.41	0.53	0.60	–	–
Panel F: Windows with and without a month end, OLS direct, full sample					
R^2_{oos} (%), windows without a month end	20.5	−1.9	–	−1.6	–
R^2_{oos} (%), windows with a month end	19.1	7.4	–	16.7	–
R^2_{oos} (%), windows with a quarter end	22.1	10.9	–	20.1	–
Share of the gain from windows with a quarter end (%)	31	112	–	71	–
One-day autocorrelation of the pair's closing spread	0.41	0.53	–	0.58	–

Stars: Clark–West test at 10, 5, and 1%. Panel A: Equation (26), on common dates. Panel B: estimated on data from April 2018, evaluated from April or May 2020. Panel C: ϕ of Equation (27). Panel D: every predictor dated $t-1$ or earlier; DM: against DST's lagged set. Panel E: daily change of the term quote on the same (previous) day's change of the overnight quote, Newey–West t in parentheses; deviation: overnight minus term quote. Panel F: gain share: share of the squared-error reduction.

out noise in the overnight quote as the source of the same-day signal since 2020. Noise in the term quote accounts for at most the part that dropping today's term quote removes: about half for ON-1W over the full sample (19.9% to 10.3%), but most of it from 2020, when without today's one-week quote 6.8% of 39.9% remain (12.4% of 36.0% with the TGCR

target). Three candidates remain for the rest: noise in the term quote, closing quotes of the two maturities taken at different times, and a premium or expectation error lasting about a day; closing quotes cannot separate them.

With every predictor lagged, the Bloomberg quotes predict the TGCR-based excess return from 2020 with an R^2_{oos} of 8.1%, 8.1%, and 3.4% for ON-1W, ON-2W, and ON-3W, against 1.6%, −6.8%, and −16.8% for the Bloomberg target. The Bloomberg overnight quote thus carries a short-lived component relative to the TGCR (gap with mean 5.7 basis points, standard deviation 5.9, first-order autocorrelation 0.37) that enters the Bloomberg target but cannot be predicted from yesterday's quotes. The lag still removes most of the predictability (36.0% to 8.1% for ON-1W), and the period lies in the years after 2019 in which the overnight premium keeps falling (Section 7.6); against an EH forecast estimated from April 2018, this lag-robust part reflects that drift, not a separate persistent departure.

The transaction rates validate the overnight leg only. No transaction series exists for term GC repo, so the term quotes, the long legs of every pair and the short legs of 1W-2W to 1M-3M, cannot be checked, nor can the level of the premium: with the TGCR as the rolled rate, the ON-1W excess return since April 2018 is 5.5 basis points lower on average. The mean gap between the two overnight rates pins down the level of the premium only to within several basis points; it is not a measure of the precision of the daily quotes, and the comparison says nothing about 1991–2018.

Table 19, Panel H, trades the same forecasts with every overnight roll at the TGCR instead of the closing quote, on the same dates from May 2018. In both cases the EH forecast and the EWMA are estimated from April 2018 on that excess return, so that only the overnight rate differs; the other forecasts keep their departure from the EH forecast, and the long leg remains the Bloomberg term quote. Perfect sign foresight is then worth 0.73, 0.47, and 0.45 basis points a year before costs for ON-1W, ON-2W, and ON-3W with TGCR rolls, against 2.40, 1.87, and 1.78 with the quote, and no forecast nets a positive fee in either case (the VAR −0.26 and −0.11 for ON-1W). The bound shrinks because of the level of the transaction rate: rolling at the TGCR, about 5.5 basis points below the closing quote, makes the excess return negative on 91% to 94% of days against 74% to 78% with the quote, so the sign of the departure is almost always known in advance; in both cases the EH strategy then always holds the side of the static rule. The term leg remains a quote, so this does not measure value at transaction prices.

5.3. What survives the lag

What remains without same-day quotes is small, concentrated at the zero lower bound, and unlike a slowly moving premium (Table 14). At the zero lower bound the direct regression

Table 13: **Bloomberg Quotes versus Transaction-Based Overnight Rates**

	TGCR	BGCR	SOFR
Observations	2,066	2,066	2,066
Mean difference, Bloomberg minus reference (bp)	5.67	5.65	3.93
Mean absolute difference (bp)	6.17	6.14	4.43
Share of days with absolute difference above 5 bp (%)	60.7	60.4	34.0
Correlation of levels	1.000	1.000	1.000
Correlation of daily changes	0.789	0.789	0.797
Days without change, Bloomberg (%)	19.3	19.3	19.3
Days without change, reference (%)	51.1	50.7	43.9
Standard deviation of the difference (bp)	5.87	–	–
One-day autocorrelation of the difference	0.37	–	–
Five-day autocorrelation of the difference	0.21	–	–
ON-1W excess return with TGCR as rolled leg: correlation	0.887		

Bloomberg overnight general-collateral repo rate against the New York Fed's TGCR, BGCR, and SOFR, 2 April 2018 to 30 September 2026. Last row: correlation of the ON-1W excess return with the one whose rolled overnight leg uses TGCR.

without same-day quotes attains 10.9%, 6.4%, and 13.0% for ON-1W, ON-2W, and 1W-2W, against 1.6%, −3.0%, and −0.3% away from it (Panel A). Quote errors that persist over several days, such as the constant quotes delivered during parts of the zero-lower-bound years (Section 2), would survive the one-day lag.

A slowly moving premium, as slowly changing supply conditions in money markets could create (d'Avernas and Vandeweyer, 2024; Copeland et al., 2025), explains little: its predictability would survive lags of months, but the EWMA attains at most 2.5% and is less accurate than the EH forecast for five pairs (Table 7). With the EWMA as the benchmark instead of the EH forecast (Panel B), the best forecast still attains 20.0% for ON-1W and 8.2% for 1W-2W, and the SPA test rejects that no forecast beats the EWMA for these two pairs ($p = 0.002$ and 0.011) and at the 10% level for 1M-2M.

Measurement error in the pair's own quotes would make its excess return predictable by its own rates but not by other maturities (Cochrane and Piazzesi, 2005). In sample, the other five maturities dated $t-1$ add significantly to DST's set for eight of the ten pairs (not ON-3M and 1M-3M) and raise the R^2 by 1.4 to 9.6 percentage points (Panel C). The diagnostic is not decisive: when a quote is noisy, the other maturities carry information about the true rate behind it and can predict the measured excess return under the EH, and common quote noise across the maturities of one pricing source cannot be excluded.

Table 14: **Predictability after the One-Day Lag**

	Short leg ON						Short leg 1W		Short leg 1M	
	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: OLS direct, R^2_{oos} by regime (%), with and without same-day quotes										
Zero lower bound, with	40.3***	22.8***	13.4***	9.3***	−4.9	−4.1	34.8***	14.6***	4.1*	8.5**
without	10.9***	6.4***	0.4**	0.0**	−8.8	−3.4	13.0**	4.9*	−0.1	10.0**
Away from the zero lower bound, with	19.0***	2.4***	−2.7*	−2.2**	−9.8	−10.6	6.8***	−0.1***	−3.8*	−8.1
without	1.6***	−3.0	−4.5	−3.6	−8.8	−9.9	−0.3**	−2.7	−8.1	−8.1
NBER recessions, with	19.4**	−4.9	−10.0	−7.1	−13.4	−14.0	1.1***	−7.6	−4.2	−4.5
without	1.5	−9.9	−9.0	−6.1	−10.4	−12.4	−4.1	−5.7	−1.6	−2.6
NBER expansions, with	20.2***	10.4***	6.4***	5.0***	−2.7*	−4.0	13.0***	9.5***	−2.9**	−11.2***
without	2.2***	3.7***	0.7***	0.0**	−6.0	−4.9	3.5***	1.3**	−14.4	−13.4**
Since April 2018, with	35.6***	17.7***	9.6***	0.5**	−2.4	−1.8	8.1***	1.6**	−5.0	−6.5**
without	10.1***	4.7***	−0.9*	−1.9	−1.5	−1.4	1.1*	−3.2	−12.6	−7.5*
From 1 January 2016, with	34.2***	18.8***	11.3***	3.7**	−0.6	−1.6	13.7***	6.7**	−4.4	−6.3**
without	9.6***	6.6***	1.9*	1.1*	−0.1	−1.2	6.6**	2.1	−11.9	−7.3**
Panel B: A slowly moving premium (EWMA) as the benchmark										
R^2_{oos} of the EWMA vs the EH forecast (%)	0.7	2.5	1.3	−0.6	−10.6	−14.1	2.1	1.5	−5.5	−6.5
Best forecast, R^2_{oos} vs the EWMA (%)	20.0	3.8	1.2	2.5	3.4	5.8	8.2	1.7	7.3	4.9
SPA p-value: no forecast beats the EWMA	0.002	0.306	0.666	0.575	0.544	0.333	0.011	0.594	0.085	0.432
Forecasts better than the EWMA (Holm)	16/22	0/22	0/22	0/22	0/22	0/22	0/22	0/22	0/22	0/22
Panel C: Other maturities added to DST's set, in sample										
R^2 of DST's set (%)	22.1	8.7	6.2	6.3	2.2	0.6	13.8	6.8	5.5	4.7
R^2 with the other five maturities at $t-1$ (%)	27.2	15.1	11.9	13.8	6.3	4.2	23.4	15.6	11.7	6.1
Wald test of the added coefficients, p-value	< 0.001	< 0.001	< 0.001	< 0.001	0.001	0.192	< 0.001	< 0.001	< 0.001	0.242

Stars: Clark–West test at 10, 5, and 1%. Panel A: OLS direct with lags t to $t-4$ (with) or $t-1$ to $t-5$ (without), on common dates. Panel B: the EWMA as the benchmark; Holm: one-sided Diebold–Mariano tests at 5%. Panel C: DST's set plus the other five maturities at $t-1$, on dates with all seven maturities (so the first row differs from Table 9); Newey–West Wald test.

6. Economic Value

The economic stakes are small: perfect sign foresight is worth 2.55 to 5.55 basis points a year before costs, the forecasts capture at most 45% of it, and over the full sample the only net gain that is significant after the search, 0.86 basis points a year from an ex-post band rule for ON-1W, needs same-day quotes.

6.1. The perfect-foresight bound

Table 15 reproduces the bound of Equation (22) from the strategy quantities for ON-1W, the pair with the largest bound, and 1W-2W, the pair with the smallest. For ON-1W the long repo lasts $n = 5$ business days, 7.31 calendar days on average, so per-tranche returns are annualised by $360/7.31 = 49.25$. Over the 7,258 out-of-sample origins from April 1994 to October 2026, $\widehat{\sigma}_t$ averages 0.271 basis points of notional and σ^*_t 0.264, so the position averages 0.98 units of notional (0.84 to 1.10 between the 5th and 95th percentiles). The excess return over the life of the long repo averages 0.119 basis points of notional in absolute value, 5.84 basis points a year. The EH strategy, long the roll on 81.9% of origins, is on the wrong side on 47.4%; perfect sign foresight gains $2|w^{PF}_t x_t|$ there and nothing elsewhere, a mean gain of 5.55 basis points a year, and the quadratic-utility adjustment leaves the fee at 5.55. Perfect foresight changes the position of a tranche by 0.86 units of notional on

average, the EH strategy by 0.06, so a spread of 4 basis points costs 3.45 and 0.22 basis points a year and leaves a net fee of 2.32 (1.85 after the quoted spread of 4.4 basis points); the break-even spread is 6.87 basis points. For 1W-2W the same steps give 2.55 before costs and a break-even spread of 3.04 basis points, below the quoted spread.

Table 15: **The Perfect-Foresight Bound Step by Step**

	ON-1W	1W-2W
Horizon n (business days)	5	10
Mean calendar days of the long repo $\bar{D}$	7.31	14.63
Annualisation factor $360/\bar{D}$	49.25	24.62
Out-of-sample origins (one tranche each)	7,258	6,686
Period	Apr 1994–Oct 2026	Aug 1994–Oct 2026
Conditional volatility $\widehat{\sigma}_t$ of x_t, mean (bp of notional)	0.271	0.313
Target volatility σ_t^*, mean (bp of notional)	0.264	0.314
Position $\lvert w_t\rvert = \sigma_t^*/\widehat{\sigma}_t$, mean (units of notional)	0.98	1.02
5th to 95th percentile	0.84–1.10	0.80–1.22
Excess return over the life of the long repo $\lvert x_t\rvert$, mean (bp of notional)	0.119	0.135
annualised (bp a year)	5.84	3.32
EH strategy long the roll (% of origins)	81.9	0.0
EH strategy on the wrong side (% of origins)	47.4	46.8
Mean gain of perfect foresight, $2\lvert w_t^{PF} x_t\rvert$ on wrong-side origins, annualised (bp a year)	5.55	2.56
Certainty-equivalent fee, relative risk aversion 5 (bp a year)	5.55	2.55
as reported (Table 16)	5.55	2.55
Turnover $\lvert w_t - w_{t-n}\rvert$ per tranche, perfect foresight	0.86	0.91
EH strategy	0.06	0.07
Cost of a 4 bp spread, perfect foresight (bp a year)	3.45	3.64
EH strategy (bp a year)	0.22	0.28
Fee of perfect foresight after 4 bp (bp a year)	2.32	−0.81
Quoted spread of the pair (bp)	4.4	4.8
Fee of perfect foresight after the quoted spread (bp a year)	1.85	−1.23
Break-even spread of perfect foresight (bp)	6.87	3.04

Sign strategy; symbols as in Equations (16) to (22). Perfect foresight sets $\widehat{\mu}_t = xr_t$. Mean gain: the bound before the risk adjustment. Costs: the spread on $\lvert w_t - w_{t-n}\rvert$ over the life of the tranche.

6.2. Fees of the forecasts

Table 16 reports, for each pair, the quoted spread, the gross fee, the net fee, and the break-even spread of the VAR, of the best of the 23 forecasts, and of perfect sign foresight, and the static rule. The perfect-sign-foresight bound, the benchmark of Sarno et al. (2016, Section 5.2.1), is 2.55 to 5.55 basis points a year, close to $E|xr| - |E[xr]|$ and below, for the longer pairs well below, the average absolute deviation of the excess return from the EH forecast, 3.3 to 10.0 basis points a year; perfect sign foresight breaks even at spreads of 3.04 to 8.85 basis points.

Before costs, no forecast is worth more than 2.51 basis points a year (ON-1W), 45% of its pair's bound; in every other pair the best gross fee is at most 19% of the bound. The VAR earns 2.48 for ON-1W, 0.68 for ON-2W, 0.49 for ON-3W, and 0.29 for 1W-2W before costs, about as much as ridge, the networks, and the combination. A significant R^2_{oos} does not guarantee a positive fee: a sign strategy rewards correct signs on large excess returns,

which a small gain in squared error need not deliver (Sarno et al., 2016, Section 5.2.2). The forecasts do take the right side more often (Table 17): for ON-1W the VAR and the network ensemble are on the side of the excess return on 69.1% and 67.2% of the origins, against 52.6% for the EH strategy, and both predict the direction of its departure from the EH forecast (65.8% and 65.2% of non-overlapping origins, Pesaran and Timmermann, 1992, $p < 0.01$). These 16.5 percentage points of correct sides are worth 2.48 basis points a year to the VAR before costs, because the excess return they collect is small. The VAR's break-even spread, at most 3.39 basis points, lies below the quoted spread of every pair, because its frequent changes of sign trade often, and after quoted spreads it loses for all ten pairs. Without the EWMA no forecast nets a positive fee; the EWMA's net fees are assessed in Section 6.3.

The EH strategy takes the sign of the real-time prevailing mean, which is negative, so that the strategy is short the roll (it lends long and borrows short), on almost all dates for most pairs, but positive on 81.9% of the dates for ON-1W. The static rule, which always lends long and borrows short (short the roll), therefore earns at least as much as every forecast after quoted spreads for eight of the ten pairs, all but ON-1W and ON-2W, and strictly positive amounts for ON-3W (0.92), ON-1M (0.03), 1M-2M (1.04), and 1M-3M (1.94), on which the EH strategy does not always lend long. For the overnight pairs it gains from a shift in the premium (Section 7.6); for the one-month pairs, whose mean excess return is stable, from estimation noise in the sign of a small constant premium. Figure 5 plots each forecast's accuracy against its gross fee.

Table 16: **Economic Value of Departures from the EH: Forecasts, a Static Rule, and the Upper Bound**

	Spread	VAR (DST)			Best of 23 forecasts			Without	Lend long,	Perfect sign foresight		
	quoted	Gross	Net quoted	Break-even	Gross	Net 4 bp	Net quoted	EWMA	borrow short	$\lvert xr - \overline{xr}\rvert$	Gross	Break-even
ON-1W	4.4	2.48	−0.86	3.39	2.51	0.81	0.74	−0.08	0.12	5.8	5.55	6.87
ON-2W	4.9	0.68	−1.57	1.59	0.87	0.36	0.27	−0.70	0.12	5.9	4.68	7.25
ON-3W	5.2	0.49	−1.33	1.51	0.85	0.55	0.42	−0.50	0.92	6.2	5.36	8.85
ON-1M	5.2	−0.81	−2.58	0.00	−0.27	−0.95	−1.30	−1.30	0.03	6.7	4.59	7.12
ON-2M	5.1	−1.38	−2.25	0.00	−0.79	−1.64	−2.07	−2.25	0.00	7.8	4.45	5.74
ON-3M	5.5	−1.30	−1.91	0.00	−1.30	−1.60	−1.91	−1.91	0.00	10.0	4.61	6.00
1W-2W	4.8	0.29	−2.72	0.51	0.33	−0.35	−0.43	−0.92	0.00	3.3	2.55	3.04
1W-3W	5.1	−0.16	−2.28	0.00	−0.07	−0.25	−0.31	−0.91	0.00	4.1	2.86	3.51
1M-2M	5.7	−0.06	−1.56	0.00	0.07	−0.41	−0.63	−0.63	1.04	4.4	3.48	4.06
1M-3M	5.9	0.33	−0.35	2.43	0.33	−0.15	−0.13	−0.13	1.94	7.1	5.22	6.69

Fee: Equation (20), bp a year. Net 4 bp: constant spread of 4 bp; net quoted: quoted spread of the day. Without EWMA: best net quoted fee of the other 22 forecasts. Break-even: constant spread at which the fee is zero (0: no positive gross fee). Best: largest fee of the 23 forecasts, ex post per column. Lend long, borrow short: static rule, net of quoted spreads. $\lvert xr - \overline{xr}\rvert$: mean absolute deviation of the excess return from the EH forecast $\overline{xr}$.

6.3. Reliability of net gains

Over the full sample, one net gain is significant after the search: the band rule for ON-1W, which nets 0.86 basis points a year (Table 18, Panel A). The rule of Equation (23) raises

Table 17: **Direction, Value, and Risk Aversion of the Forecasts**

	Right side (%)			Direction of departure (%)		Gross fee (bp/yr)			Net fee at 4 bp (bp/yr)			
	EH strategy	VAR	NN ensemble	VAR	NN ensemble	VAR	NN ensemble	Perfect foresight	VAR RRA 2	VAR RRA 10	NN RRA 2	NN RRA 10
ON-1W	52.6	69.1	67.2	65.8***	65.2***	2.48	2.41	5.55	−0.45	−0.45	−0.19	−0.19
ON-2W	62.7	64.7	65.4	62.8***	62.4***	0.68	0.84	4.68	−1.03	−1.03	−0.83	−0.83
ON-3W	61.1	62.3	64.1	54.0**	56.2***	0.49	0.83	5.36	−0.81	−0.80	−0.61	−0.61
ON-1M	54.6	58.5	62.0	61.3***	59.4***	−0.81	−0.34	4.59	−2.01	−2.01	−1.71	−1.71
ON-2M	57.6	56.5	58.9	47.1	60.1**	−1.38	−1.95	4.45	−1.84	−1.84	−2.91	−2.89
ON-3M	59.2	58.7	54.8	59.2	59.2	−1.30	−3.13	4.61	−1.59	−1.61	−4.07	−4.00
1W-2W	53.2	61.6	59.0	67.0***	62.2***	0.29	−0.02	2.55	−2.00	−2.00	−2.13	−2.12
1W-3W	54.2	59.7	59.0	67.7***	60.7***	−0.16	−0.29	2.86	−1.69	−1.69	−1.83	−1.82
1M-2M	55.4	58.2	56.3	67.1***	51.3	−0.06	−0.23	3.48	−1.11	−1.12	−1.48	−1.50
1M-3M	56.1	56.7	56.8	56.0	54.0	0.33	−0.39	5.22	−0.18	−0.27	−1.10	−1.22

Right side: share of origins on which the position has the sign of the excess return (EH strategy: sign of the real-time mean). Direction of departure: share of non-overlapping origins on which the forecast's departure from the EH forecast has the sign of the realised departure; stars: Pesaran and Timmermann (1992) test at 10, 5, and 1%. Fees: Equation (20), bp a year; RRA: relative risk aversion (otherwise 5).

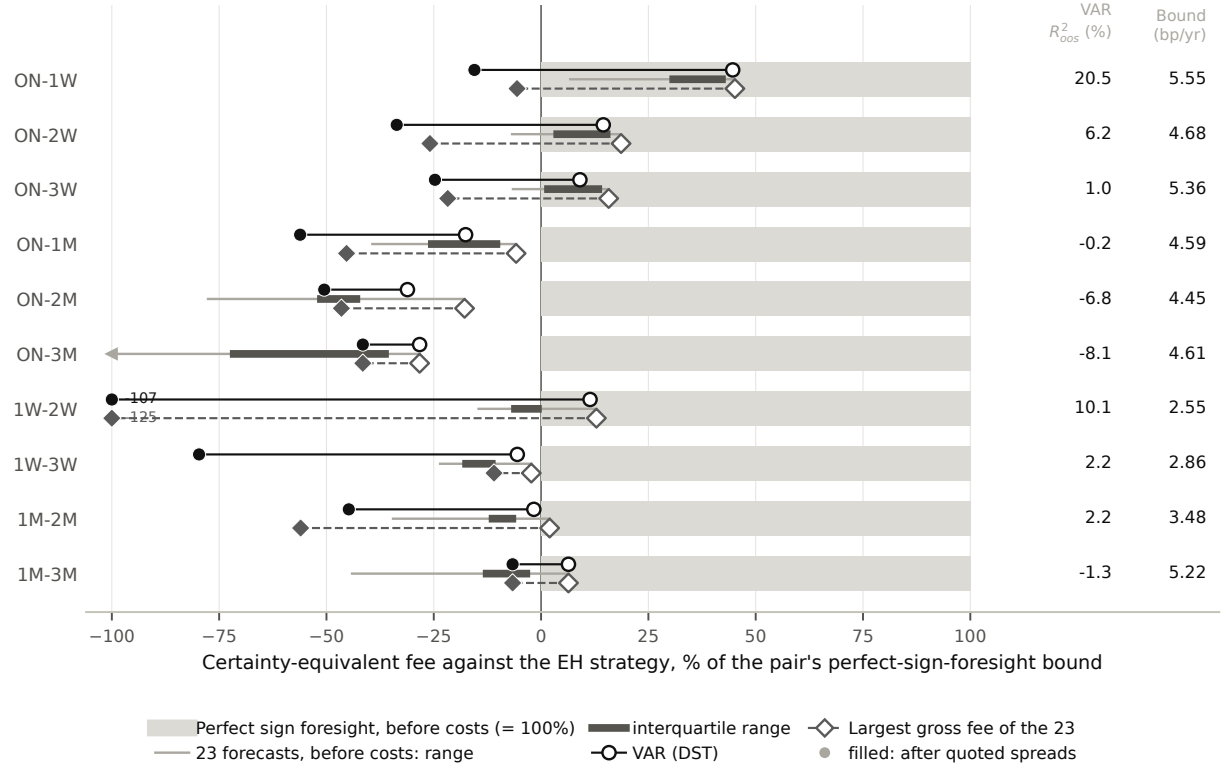


Figure 5: **Fees of the 23 Forecasts Relative to the Bound.** Fee of the sign strategy in percent of the pair's perfect-foresight bound (light grey bar; bound in bp a year on the right). Thin line: range of the 23 forecasts' gross fees (arrow: clipped at −100%); dark bar: interquartile range. Circles: DST's VAR; diamonds: largest gross fee; open before costs, filled after quoted spreads. Right: the VAR's R^2_{oos}.

the VAR's net fee for ON-1W from −0.86 to 0.79 basis points a year, trading on 17% of the dates, and that of the best forecast without the EWMA to 0.86 ($t = 3.16$ with 250

Newey–West lags). It is significant after the search over the 23 banded forecasts ($p = 0.003$), after the Holm adjustment across the ten pairs ($p = 0.03$; 0.05 to 0.07 with mean blocks of 50 to 250 days), and, with 50,000 draws, after the Holm adjustment over the ten pairs and the widths 1, 1.5, and 2 jointly (0.037 to 0.067; Panel A). Its significance, not its magnitude, depends on the time variation of spreads: with a constant spread, equal to the pair's mean (4.4 basis points for ON-1W) or to 4 basis points, the best net fee is almost unchanged (0.80 to 0.84, not tabulated), but the SPA p-value rises to 0.04 to 0.05 within the pair and to 0.38 to 0.50 after the Holm adjustment across the ten pairs; that time variation reflects the quoted widths, which differ across pricing sources and regimes and are fixed since 2015 (Section 2.4). The gain needs same-day quotes: on common dates, the best of the VAR, the direct regression, ridge, and the network ensemble under the band rule nets 0.80 basis points a year with same-day quotes (SPA $p = 0.005$ over these four and the EWMA) and 0.20 with lags $t - 1$ to $t - 5$ ($t = 1.36$, SPA $p = 0.07$; Panel A; the band itself still uses the spread of day t), so the band rule's gain is part of the same-day signal of Section 5. For every other pair no forecast nets more than 0.42 under the band rule (the EWMA, which the rule leaves unchanged), no banded forecast more than 0.18, and the search does not reject. The magnitude is stable across band widths: with 1.5 and 2 spreads the best forecast nets 0.85 and 0.80 (Holm across pairs 0.015 and 0.016), and with half a spread the best is the EWMA at 0.74, not significant. The threshold is set ex post, so significance depends on the rule; the magnitude, below one basis point for every forecast of the search family, does not. From 2016, a subsample chosen after the break was dated, the static rule and the EWMA net more (Section 7.6).

Under the sign rule, the largest net fees after quoted spreads come from the EWMA: 0.74, 0.27, and 0.42 basis points a year for ON-1W, ON-2W, and ON-3W (Table 16; mean net returns of 0.73, 0.27, and 0.41 in Panel B). They are not significant after the search ($p \geq 0.08$ at the paper's block length, 0.17 to 0.62 with blocks of 126 and 250 days), with a Newey–West t-statistic of at most 1.50 (Panel B). The pointwise 95% intervals end at 1.69, 1.81, and 2.20 basis points, and simultaneously over the 23 forecasts at 2.05, 2.48, and 3.04; the SPA test would detect net fees of 1.5 to 2.9 basis points on these pairs with 80% power. No forecast's net gain is thus significant under the sign rule.

The net ceiling of Equation (24), the most that a strategy knowing the excess return can net after quoted spreads, is 1.3 to 4.0 basis points a year with positions of ± 1 unit and 1.5 to 4.4 with a flat position allowed (Panel C). Perfect sign foresight itself loses after quoted spreads for 1W-2W, 1W-3W, and 1M-2M, because it changes position whenever the sign changes; net fees of perfect sign foresight are therefore not bounds, while the ceiling is. Figure 6 shows when the VAR's gross gain accrues and how costs absorb it.

Table 18: **Reliability of Net Gains**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: Band rule, after quoted spreads (bp a year)										
VAR (DST), sign rule	−0.86	−1.57	−1.33	−2.58	−2.25	−1.91	−2.72	−2.28	−1.56	−0.35
VAR (DST), band rule	0.79	0.01	−0.11	−0.24	−0.16	−0.58	−0.16	−0.22	−0.11	−0.15
Share of dates outside the band, VAR	0.17	0.09	0.05	0.06	0.03	0.07	0.06	0.04	0.02	0.06
Best of the 23 band forecasts	0.86	0.27	0.42	−0.24	−0.16	−0.56	−0.06	−0.11	−0.09	−0.06
Newey–West t	3.16	0.34	0.46	−2.43	−2.61	−1.53	−1.29	−1.73	−1.00	−0.13
Best band forecast without the EWMA	0.86	0.18	0.03	−0.24	−0.16	−0.56	−0.06	−0.11	−0.09	−0.06
Newey–West t	3.16	0.91	0.53	−2.43	−2.61	−1.53	−1.29	−1.73	−1.00	−0.13
SPA p, block max$(20, 2n)$	0.003	0.473	0.829	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Holm across the ten pairs	0.026	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
SPA p, block 250	0.007	0.597	0.837	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Holm across the ten pairs	0.072	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Band of 0.5 spreads: best fee	0.74	0.27	0.42	−1.08	−0.89	−1.10	−0.17	−0.28	−0.39	−0.14
Holm p, block max$(20, 2n)$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Band of 1.5 spreads: best fee	0.85	0.27	0.42	−0.11	−0.08	−0.10	−0.03	−0.09	0.04	0.07
Holm p, block max$(20, 2n)$	0.015	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Band of 2 spreads: best fee	0.80	0.27	0.42	−0.06	−0.02	−0.03	−0.01	−0.02	0.04	0.06
Holm p, block max$(20, 2n)$	0.016	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Five forecasts, same day: best band forecast without the EWMA	0.80	0.02	−0.11	–	–	–	−0.13	−0.22	–	–
SPA p over the five	0.005	0.609	0.607	–	–	–	1.000	1.000	–	–
Five forecasts, lags $t-1$ to $t-5$: best band forecast without the EWMA	0.20	−0.08	−0.03	–	–	–	−0.10	−0.03	–	–
Newey–West t	1.36	−1.38	−1.09	–	–	–	−1.87	−2.25	–	–
SPA p over the five	0.070	0.596	0.492	–	–	–	1.000	1.000	–	–
Band of 1 spreads: Holm p over pairs and widths	0.067	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Band of 1.5 spreads: Holm p over pairs and widths	0.037	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Band of 2 spreads: Holm p over pairs and widths	0.053	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Panel B: Precision of the largest net fee, sign rule (bp a year)										
Largest mean net return of the 23 forecasts	0.73	0.27	0.41	−1.30	−2.11	−1.90	−0.43	−0.32	−0.63	−0.04
Newey–West t	1.50	0.34	0.46	−1.35	−2.33	−1.38	−2.08	−1.52	−1.12	−0.03
upper end of its 95% interval	1.69	1.81	2.20	0.59	−0.34	0.79	−0.02	0.09	0.48	2.53
upper end, simultaneous over the 23 forecasts	2.05	2.48	3.04	1.18	0.15	1.56	0.13	0.25	0.91	3.43
EWMA mean net return	0.73	0.27	0.41	−1.49	−2.11	−3.68	−0.43	−0.32	−0.77	−0.92
Newey–West t	1.50	0.34	0.46	−1.89	−2.33	−2.39	−2.08	−1.52	−1.65	−1.20
SPA p, block max$(20, 2n)$	0.085	0.254	0.664	1.000	1.000	1.000	1.000	1.000	1.000	1.000
block 126	0.173	0.506	0.623	1.000	1.000	1.000	1.000	1.000	1.000	1.000
block 250	0.193	0.451	0.609	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Net fee detectable with 80% power after the search	1.51	2.07	2.91	2.45	1.94	3.20	0.18	0.48	1.64	4.02
Panel C: Perfect foresight after quoted spreads (bp a year)										
Perfect sign foresight, sign strategy	1.85	1.58	2.28	1.24	0.65	0.56	−1.23	−1.19	−0.71	1.16
Ceiling: trading around the spread, positions ±1	3.85	3.27	4.02	2.90	2.38	2.40	1.34	1.37	2.29	4.02
Ceiling: trading around the spread, flat position allowed	3.90	3.39	4.16	3.05	2.55	2.64	1.52	1.56	2.71	4.44
Panel D: Participant who pays no extra spread, gross gain over the EH strategy (bp a year)										
Best of the 23 forecasts	2.51	0.87	0.85	−0.27	−0.82	−1.29	0.33	−0.07	0.09	0.37
Newey–West t	4.94	1.68	1.34	−0.44	−1.16	−1.08	1.45	−0.49	0.21	0.54
SPA p, block max$(20, 2n)$	0.000	0.124	0.318	1.000	1.000	1.000	0.092	1.000	0.835	0.688
Holm across the ten pairs	0.000	0.990	1.000	1.000	1.000	1.000	0.827	1.000	1.000	1.000
Five forecasts, lags $t-1$ to $t-5$: best	1.00	0.61	0.84	–	–	–	−0.12	−0.06	–	–
Newey–West t	2.39	0.84	1.04	–	–	–	−1.04	−0.43	–	–
SPA p, block max$(20, 2n)$	0.006	0.272	0.308	–	–	–	1.000	1.000	–	–
SPA p, block 250	0.028	0.345	0.362	–	–	–	1.000	1.000	–	–

Fees after quoted spreads, bp a year. Panel A: band rule of Equation (23), one spread unless stated; best: largest of the 23 band forecasts, ex post; Holm: across the ten pairs; five forecasts: DST's VAR, direct regression, ridge, and NN ensemble (banded) and the EWMA, with and without both same-day quotes, on common dates, SPA with 10,000 draws (–: not computed). Panel B: net returns of the 23 forecasts minus the EH strategy's, 250 Newey–West lags; simultaneous: valid jointly for all forecasts; detectable: 80% power of the SPA test. Panel C: net ceiling, Equation (24). Band rule over pairs and widths: Holm over the ten pairs and the widths 1, 1.5, and 2 jointly, SPA with 50,000 draws; the lagged band rule uses the spread of day t as its threshold. Panel D: gross returns of the sign strategy minus the EH strategy's, SPA with 10,000 draws; five forecasts as in Panel A.

6.4. Costs, participants, and position rules

Who pays the costs changes the sign of the smallest fees; for participants who pay no extra spread, the size of the gross values settles the verdict (Table 19). With a pair spread 0.5 basis points lower the VAR still loses for every pair, and the best forecast without the EWMA nets at most 0.00 (ON-1W). A participant who must lend and borrow anyway and only chooses the maturities pays no extra spread; its gain is significant after the search, 2.51

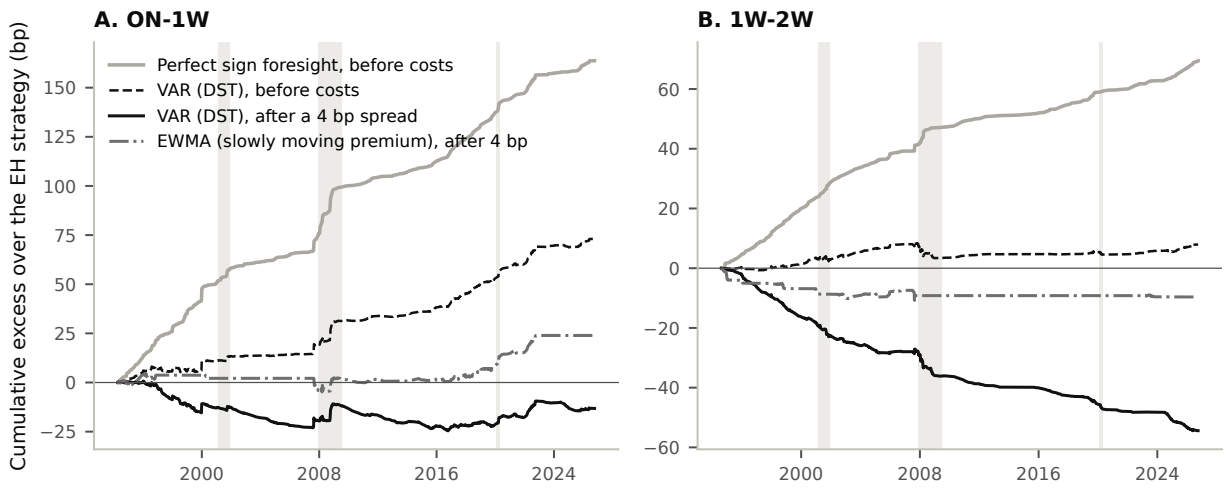


Figure 6: **Cumulative Excess Returns over the EH Strategy.** Cumulative daily excess returns of sign strategies over the EH strategy, bp per unit of notional: perfect sign foresight before costs (light grey), DST's VAR before costs (dashed) and after a constant spread of 4 bp, DST's value (solid), and the EWMA after 4 bp (dash-dot). Shaded: NBER recessions.

basis points a year for ON-1W with same-day quotes (SPA $p < 0.001$) and 1.00 with quotes up to the previous day (SPA $p = 0.006$; Holm across five pairs 0.03), and each is smaller than one quoted spread (Table 18, Panel D); a participant who only lends earns half of them (1.24 for the VAR on ON-1W); at half the spread, an approximation for a dealer, the VAR nets 0.81 for ON-1W and loses for the other pairs; and with DST's convention, which charges the spread on the daily change of position, it loses for every pair at the quoted spreads (-0.06 for ON-1W). At DST's 4 basis points under that convention it nets 0.28 for ON-1W (Newey–West $t = 0.52$) and loses for the other nine pairs; ridge, the network ensemble, the combination, and the EWMA then also net positive fees for ON-1W (0.29, 0.47, 0.44, and 0.89; the EWMA also 0.57 and 0.81 for ON-2W and ON-3W), all below one basis point. As a stand-alone position against holding no position, every tranche pays the spread, and for none of the ten pairs does any forecast cover its costs.

Costs decide the sign because the most accurate forecasts trade most. The VAR changes the position of a tranche by 0.24 to 0.79 units of notional on average, and its break-even spread is 0.51 to 3.39 basis points where it is positive; the EWMA changes sign rarely (0.08 to 0.44 units) and breaks even at 33.8, 9.8, and 11.6 basis points for the three shortest overnight pairs. The EWMA's net fee rises with its half-life, to 0.87 for ON-1W and 1.22 for ON-3W at 500 days, a choice the search does not adjust for. With market expectations the best forecast after quoted spreads is again the EWMA on the overnight pairs; without it, no forecast nets more than 0.06 on the pairs up to three weeks, and 0.69 on 1M-3M. The bound holds in both policy regimes: perfect sign foresight earns at most 3.13 basis points a

year at the zero lower bound and 6.91 away from it.

The conclusions do not depend on the position rule (Table 20). With the mean–variance weights of Equation (17) the VAR earns at most 3.77 basis points a year before costs (ON-1W), and after a spread of 4 basis points only the EWMA earns a positive fee among the tabulated forecasts, for the three shortest overnight pairs (at most 1.18). Applying DST's 1% volatility target to the per-tranche excess return of this paper, rather than per unit of notional, scales positions 8 to 62 times and fees roughly in proportion, as it would scale the perfect-foresight bound: before costs the VAR earns 133.2 basis points a year for ON-1W, and after 4 basis points it loses on every pair, while the EWMA nets 41.7, 16.0, and 19.2 for the three shortest overnight pairs (Newey–West t of 1.65, 0.46, and 0.65, close to the t of 1.71 of the EWMA per unit of notional for ON-1W at 4 basis points), because the scaling multiplies the fees and their standard errors roughly alike. A target of 2% doubles every fee. Values are per unit of notional, the scale on which a repo position uses balance sheet, and the fees are almost invariant to risk aversion between 2 and 10 (Table 17), because the excess returns of the strategies are so small that the variance term of quadratic utility barely matters.

6.5. Magnitudes relative to the literature

Profits need not track statistical accuracy (Leitch and Tanner, 1991); in the sign strategy, the stakes are set by how much the excess return varies, not by R^2_{oos}. For a conditional mean μ_t that explains a share R^2 of the variance of the excess return,

$$E\big|\mu_t - E[\mu_t]\big| \;\le\; \sqrt{E\big[(\mu_t - E[\mu_t])^2\big]} \;=\; \sqrt{R^2}\,\sigma(xr), \tag{28}$$

which bounds the gain of a forecast over an EH strategy with the correct sign of the mean, while perfect sign foresight earns about $E|xr| - |E[xr]|$ (Equation (22)). With $\sigma(xr)$ of 7 to 18 basis points a year (Table 2), the VAR's R^2_{oos} of 20.5% for ON-1W implies a predictable component with a standard deviation of about 5.4 basis points; the VAR earns 2.48 before costs. For one-year excess returns on Treasury bonds the mean absolute value is 130 to 400 basis points (Thornton and Valente, 2012, Table 1), so accuracy like that found for Treasury bonds (Bianchi et al., 2021b,a) is worth one to two orders of magnitude less here. With a gross bound this small, the departures are economically minor whatever share a forecast captures, a stronger form of the comparison with perfect sign foresight in Sarno et al. (2016, Section 5.2.1).

A risk premium moves with persistent state variables, so its predictive power survives lagged predictors: Cochrane and Piazzesi (2005) lag their forward rates against measurement error, and their R^2 is 35% with current and 31% to 35% with one- to three-month-old

Table 19: **Economic Value: Costs, Participants, and Information**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: Quoted spread ±0.5 bp, net fee (bp a year)										
VAR (DST), spread −0.5 bp	−0.50	−1.36	−1.17	−2.43	−2.19	−1.88	−2.44	−2.09	−1.42	−0.28
VAR (DST), spread +0.5 bp	−1.23	−1.79	−1.49	−2.73	−2.30	−1.95	−3.01	−2.47	−1.69	−0.42
Best forecast without the EWMA, spread −0.5 bp	0.00	−0.61	−0.45	−1.23	−2.19	−1.88	−0.88	−0.86	−0.58	−0.09
Best forecast without the EWMA, spread +0.5 bp	−0.12	−0.79	−0.55	−1.37	−2.30	−1.95	−0.97	−0.96	−0.68	−0.17
Panel B: Who pays the costs, VAR (DST) and best forecast (bp a year)										
VAR, lends and borrows anyway, chooses the maturities	2.48	0.68	0.49	−0.81	−1.39	−1.31	0.29	−0.16	−0.06	0.33
VAR, half the spread	0.81	−0.45	−0.42	−1.69	−1.81	−1.61	−1.22	−1.22	−0.81	−0.01
VAR, spread of the cheaper maturity	−0.12	−0.94	−0.82	−2.06	−1.95	−1.69	−2.17	−1.83	−1.46	−0.20
VAR, daily convention of DST	−0.06	−1.28	−1.48	−2.39	−2.14	−1.64	−2.33	−2.00	−2.12	−0.72
VAR, daily convention of DST, 4 bp	0.28	−0.75	−0.87	−1.85	−1.79	−1.44	−1.57	−1.40	−1.30	−0.20
Best forecast, stand-alone against no position	−1.59	−2.61	−3.26	−3.18	−2.39	−1.81	−3.69	−3.84	−4.58	−3.69
Always lend long, borrow short, stand-alone	−4.01	−3.41	−3.45	−2.91	−1.60	−0.50	−4.02	−3.78	−4.15	−2.94
Panel C: EWMA by half-life, after quoted spreads (bp a year)										
Half-life 63 days	0.56	0.09	0.18	−1.16	−3.23	−5.87	−0.80	−0.97	−1.27	−2.03
Half-life 126 days	0.74	0.27	0.42	−1.48	−2.07	−3.54	−0.43	−0.31	−0.76	−0.91
Half-life 250 days	0.64	0.44	0.55	−0.87	−1.92	−1.79	−0.05	−0.05	−0.10	−0.22
Half-life 500 days	0.87	0.63	1.22	−0.58	−2.10	−0.13	−0.04	0.01	−0.19	0.09
Panel D: Market expectations, best forecast (bp a year)										
Before costs	2.47	1.09	1.36	0.76	−0.79	−0.42	0.67	0.48	0.70	0.72
After quoted spreads	0.74	0.27	0.42	−1.47	−2.07	−1.19	−0.38	−0.31	−0.22	0.69
After quoted spreads, without the EWMA	0.06	−0.61	0.03	−1.47	−2.11	−1.19	−0.38	−0.34	−0.22	0.69
Panel E: Break-even bid–ask spread (bp)										
VAR (DST)	3.39	1.59	1.51	0.00	0.00	0.00	0.51	0.00	0.00	2.43
Ridge	3.51	1.15	1.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Random forest	2.20	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Boosting	1.63	0.89	1.71	0.00	0.00	0.00	0.00	0.00	0.33	0.00
NN ensemble	3.71	2.01	2.31	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Combination	3.69	1.51	0.98	0.00	0.00	0.00	0.01	0.00	0.00	0.00
EWMA	33.84	9.76	11.63	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Panel F: Change of the tranche position, $\|w_t - w_{t-n}\|$ (units of notional)										
VAR (DST)	0.785	0.515	0.455	0.401	0.246	0.237	0.643	0.462	0.437	0.372
Ridge	0.734	0.440	0.408	0.358	0.303	0.266	0.450	0.363	0.404	0.294
Random forest	0.404	0.352	0.366	0.290	0.387	0.509	0.264	0.210	0.375	0.392
Boosting	0.310	0.264	0.242	0.248	0.357	0.307	0.259	0.169	0.284	0.328
NN ensemble	0.703	0.508	0.493	0.442	0.375	0.397	0.595	0.462	0.485	0.422
Combination	0.676	0.466	0.419	0.363	0.339	0.365	0.470	0.325	0.356	0.321
EWMA	0.082	0.151	0.205	0.259	0.347	0.440	0.123	0.124	0.310	0.402
Panel G: Fee before costs by policy regime (bp a year)										
VAR (DST), zero lower bound	2.00	0.31	0.14	0.04	−0.02	−0.04	0.15	−0.04	0.10	0.08
VAR (DST), away from it	2.67	0.83	0.63	−1.15	−1.96	−1.85	0.35	−0.21	−0.12	0.43
EWMA, zero lower bound	1.08	−0.02	−0.10	−0.39	−0.28	−0.10	0.00	0.00	0.00	0.00
EWMA, away from it	0.86	0.86	1.22	−0.59	−1.01	−2.42	−0.20	−0.09	−0.35	−0.22
Perfect sign foresight, zero lower bound	3.13	1.67	1.65	1.55	1.46	1.57	0.85	0.88	1.17	1.88
Perfect sign foresight, away from it	6.51	5.87	6.91	5.84	5.71	5.93	3.20	3.63	4.35	6.50
Panel H: Overnight rolls at the closing quote and at TGCR, from May 2018 (bp a year)										
Perfect sign foresight, before costs, quote	2.40	1.87	1.78	–	–	–	–	–	–	–
Perfect sign foresight, before costs, TGCR	0.73	0.47	0.45	–	–	–	–	–	–	–
Perfect sign foresight, after quoted spreads, quote	0.60	0.39	0.01	–	–	–	–	–	–	–
Perfect sign foresight, after quoted spreads, TGCR	0.11	−0.06	−0.17	–	–	–	–	–	–	–
VAR (DST), before costs, quote	0.68	−0.04	−0.26	–	–	–	–	–	–	–
VAR (DST), before costs, TGCR	−0.02	−0.13	−0.07	–	–	–	–	–	–	–
VAR (DST), after quoted spreads, quote	−0.11	−0.65	−0.62	–	–	–	–	–	–	–
VAR (DST), after quoted spreads, TGCR	−0.26	−0.20	−0.10	–	–	–	–	–	–	–
Best forecast, after quoted spreads, quote	0.00	0.00	0.00	–	–	–	–	–	–	–
Best forecast, after quoted spreads, TGCR	0.00	0.00	0.00	–	–	–	–	–	–	–

Fee: Equation (20), bp a year; best: largest of the search family, ex post, per row. Panel B: maturity chooser: lends and borrows anyway; half the spread: a dealer; DST's convention: spread on the daily change of position; stand-alone: against no position. Combination: the equal-weighted combination of all models. Panel H: overnight rolls at the closing quote or at TGCR, long leg at the quote; EH forecast and EWMA estimated from April 2018; 0.00: the best forecast takes the EH strategy's position. –: not computed.

forward rates. Here a one-day lag, a fifth of the ON-1W horizon, removes most of the predictability. The signal is noise in the term quote, asynchronous closing quotes, or a

Table 20: **Economic Value under Other Position Rules**

	VAR (DST)	Ridge	Random forest	Boosting	NN ensemble	Combination	EWMA
Panel A: Mean–variance weights, before costs							
ON-1W	3.77	3.64	1.09	0.62	3.65	3.47	1.33
ON-2W	1.13	0.56	−0.27	0.27	1.42	0.89	1.13
ON-3W	0.73	0.39	−0.07	0.27	1.35	0.37	1.38
ON-1M	−1.31	−1.91	−1.63	−0.66	−0.54	−1.74	−0.81
ON-2M	−2.17	−3.34	−3.93	−2.49	−3.03	−3.64	−1.07
ON-3M	−2.28	−2.54	−5.48	−4.53	−5.14	−4.28	−2.22
1W-2W	0.37	−0.16	−0.65	−0.44	−0.10	−0.07	−0.18
1W-3W	−0.35	−1.04	−0.64	−0.50	−0.51	−0.52	−0.08
1M-2M	−0.16	−0.38	−0.53	−0.03	−0.42	−0.46	−0.39
1M-3M	0.30	−0.38	−1.47	−1.13	−0.91	−0.51	−0.29
Panel B: Mean–variance weights, after a spread of 4 bp							
ON-1W	−0.66	−0.50	−0.99	−0.92	−0.29	−0.31	1.18
ON-2W	−1.60	−1.70	−2.08	−0.87	−1.25	−1.52	0.73
ON-3W	−1.24	−1.31	−1.56	−0.43	−0.84	−1.38	0.93
ON-1M	−3.18	−3.55	−2.96	−1.68	−2.68	−3.43	−1.94
ON-2M	−2.87	−4.36	−5.57	−3.94	−4.52	−4.93	−2.44
ON-3M	−2.86	−3.22	−7.78	−5.51	−6.66	−5.55	−3.89
1W-2W	−2.99	−2.46	−1.82	−1.53	−3.14	−2.40	−0.55
1W-3W	−2.68	−2.73	−1.51	−1.09	−2.79	−2.00	−0.42
1M-2M	−1.69	−1.73	−1.75	−0.67	−2.23	−1.52	−1.24
1M-3M	−0.52	−0.74	−2.34	−1.62	−2.11	−1.03	−1.26
Panel C: Volatility target of 1% a year, literal reading of DST, before costs							
ON-1W	133.2	127.8	39.7	19.6	129.8	123.1	47.5
ON-2W	30.6	18.8	−5.6	7.4	38.8	26.2	27.4
ON-3W	17.9	11.0	−0.4	6.8	30.3	10.8	28.9
ON-1M	−22.1	−32.4	−27.3	−10.2	−9.5	−29.3	−15.4
ON-2M	−25.3	−39.1	−46.8	−29.4	−36.0	−42.8	−14.2
ON-3M	−17.5	−20.7	−47.6	−39.3	−43.9	−38.4	−23.5
1W-2W	18.9	−4.9	−24.6	−14.0	−1.8	−0.3	−8.3
1W-3W	−8.0	−29.4	−17.6	−12.5	−14.6	−14.9	−3.1
1M-2M	−2.1	−5.2	−9.4	0.6	−6.7	−7.1	−7.1
1M-3M	4.8	−1.2	−13.7	−9.0	−6.6	−2.9	−2.2
Panel D: Volatility target of 1% a year, literal reading of DST, after a spread of 4 bp							
ON-1W	−22.6	−16.7	−35.0	−34.9	−8.6	−9.1	41.7
ON-2W	−44.3	−42.2	−51.4	−23.5	−34.1	−39.0	16.0
ON-3W	−25.4	−25.8	−31.1	−7.5	−17.5	−26.9	19.2
ON-1M	−54.3	−59.7	−47.5	−25.7	−45.6	−57.0	−32.4
ON-2M	−33.5	−51.0	−64.8	−45.4	−52.8	−57.5	−28.9
ON-3M	−21.4	−26.3	−66.2	−47.5	−56.2	−49.6	−38.0
1W-2W	−127.0	−102.2	−73.2	−61.5	−136.2	−102.6	−21.6
1W-3W	−79.6	−82.4	−42.4	−29.1	−86.1	−61.0	−11.9
1M-2M	−31.9	−32.0	−32.3	−12.0	−42.3	−28.1	−22.5
1M-3M	−3.7	−5.2	−23.0	−14.6	−18.6	−8.5	−11.8

Fee: Equation (20), bp a year. Panels A and B: mean–variance weights, Equation (17). Panels C and D: sign strategy with DST's volatility target of 1% applied to the excess return over the life of the long repo (positions of 8 to 62 times the notional). Combination: equal-weighted.

premium or expectation error lasting about a day; the data do not separate the three, but each is unlike a slowly moving premium.

DST considered misspecification of their linear VAR and "tiny data imperfections" as explanations of their rejections (p. 166) and read the gap to their economic results as doubt about the linear model (p. 172). Without a parametric model of expectations, search-

adjusted rejections hold at one to three weeks within pairs and for ON-1W across pairs, and under the tranche convention the VAR loses after a spread of 4 basis points for every pair, also for the four pairs with small in-sample gains after costs in DST (0.31 to 0.49 basis points; their Table 6, Panel A, Case 2). Bianchi et al. (2021b) find large gains from nonlinear forecasts with macroeconomic information, computed without trading costs; the gains rise with maturity and are insignificant for a mean–variance investor in two-year bonds (-0.002% a year with macroeconomic information, $p = 0.98$; their Table 5, before the revisions of Bianchi et al., 2021a). In repo, macroeconomic information lowers accuracy (Section 7.5), consistent with the short end of this gradient. Kopchak (2013) estimates, in sample with a state-space model, a time-varying realized forward premium that is positive and statistically and economically significant over 1991–2011 once the spread between repurchase and reverse repurchase rates is counted; the spread that creates his premium is the cost that, here, no forecast traded on its sign earns out of sample.

With a premium of known sign, always lending long and borrowing short is the optimal sign strategy under the EH. That it earns at least as much as every forecast after quoted spreads for eight pairs is consistent with the EH with a constant premium of 0.2 to 5.0 basis points a year (Table 2, Panel C), the order of the 0.56 to 3.19 basis points of Longstaff (2000, p. 405). The shift of the overnight premium around 2016 (Section 7.6) is a departure from a constant premium, but a small one, about the size of a bid–ask spread.

7. Robustness

None of the checks changes the three results: the ON-1W rejection is robust to the bootstrap block length, overlapping returns, extreme periods, the loss function, and the classification of stale quotes, while the evidence for ON-2W and 1W-2W is weaker on non-overlapping dates; no network design improves on the VAR by more than 1.5 points and no flexible forecast beats it significantly, and the predictability is larger from 2016.

Table 21 collects, for each of the three findings, the checks of this section and of Sections 5, 6, and 8. Sections 7.1 to 7.5 check the statistical rejection; the checks of the same-day finding and of the economic verdict are reported with those results.

7.1. Bootstrap block length

Table 22 recomputes the SPA test and the Holm adjustment with 10,000 draws for mean block lengths of $\max(20, 2n)$, 50, 100, and 200 days, and with 50,000 draws for ON-1W at the longer blocks (Monte Carlo standard error $\sqrt{p(1-p)/N_b}$ for N_b draws). On DST's set the Holm-adjusted p-value of ON-1W is 0.016 at the paper's block length and 0.032 to 0.051 with 50,000 draws and blocks of 50 to 200 days, so the rejection is robust to blocks of up to

Table 21: **Robustness by Finding**

Check	Result	Exhibit
Finding 1: of the ten pairs, only ON-1W rejects after the search and the Holm adjustment ($p = 0.016$)		
Bootstrap blocks of 50 to 200 days	Holm-adjusted p 0.032 to 0.051	Table 22
DST's period, to 2005	ON-1W 15.5%, SPA $p = 0.077$; no pair survives Holm	Table 10
Origins before 2016	17.4%	Section 7.6
Real-time EH forecast as benchmark	20.6%; ON-1M and 1M-3M turn positive	Section 3.5
Non-overlapping dates, five offsets	VAR R^2_{oos} 5.9% to 30.2%; Clark–West $p \leq 0.002$; ON-2W and 1W-2W: largest p 0.104 and 0.259	Table 23, A
Bootstrap interval of R^2_{oos}	$[13.9, 27.2]$	Table 23, B
Without the 1% largest errors; absolute loss	17.1%; Diebold–Mariano rejects at 1%	Table 23, D
Volatile against calm windows	17.0% against 29.6%	Table 24, A
Without crisis 2007–09, September 2019, Covid, and year ends	15.4%; Clark–West $p < 0.001$	Section 7.2
Stale quotes classified in real time	20.0% against 20.5%	Table 25, A
Clark–West size under a simulated EH	4.0% to 6.5% (ON pairs to 2M and 1W-2W); VAR in the data above the 95th percentile	Table 31
Market-expectations set	Holm 0.054 to 0.10 for ON-1W; no pair at 5%	Table 22
Flexible forecasts against the VAR	none significantly more accurate	Table 26, A
Network design, 12 network variants	ON-1W 35.8% to 38.9% (10 of 12) against the VAR's 37.4% (2017/18–2026); all below it on 2007–2013	Table 28
Finding 2: the rejection lives in same-day quotes (VAR, ON-1W: 20.5% to 1.8% with a one-day lag)		
Direct regression and ridge, lagged	19.9% to 2.0%; ridge alike	Table 12, A
NN ensemble, lagged	18.6% to 0.9%	Table 12, A
SPA over the lagged forecasts with the NN ensemble	$p = 0.10$ to 0.16 (0.09 to 0.14 without), blocks $\max(20, 2n)$ to 250	Table 12, A
DST's period, to 2005, lagged VAR	negative for all five short pairs (ON-1W −0.5%)	Table 10
Direct regression: stale threshold 3 or 10; without source switches	lagged 1.5% to 2.9%; 1.8%	Table 25, B
Overnight leg at the TGCR, from 2020	same day 36.0% against 39.9%; lagged 8.1%	Table 12, B
Market expectations, lagged	2.9% to 6.2%; significant gain only for 1W-2W	Table 12, D
Finding 3: perfect sign foresight bounds the gross value at 2.55 to 5.55 bp; in the sign strategy, no forecast nets a significant fee after quoted spreads over 1994–2026		
Pair spread 0.5 bp lower	VAR loses for every pair; best without EWMA ≤ 0.00	Table 19
Participant who only chooses maturities or only lends	gross fees (VAR ≤ 2.48) or half of them	Table 19
Search-adjusted net gains, sign rule	EWMA $p \geq 0.08$	Table 18, B
Band rule	0.86 for ON-1W, Holm 0.03 (0.07 with 250-day blocks; 0.38 to 0.50 with constant spreads); over pairs and widths 0.037 to 0.067; with lags $t-1$ to $t-5$ 0.20, SPA $p = 0.07$	Table 18, A
Participant who pays no extra spread	2.51 bp same day (SPA $p < 0.001$), 1.00 with lags (SPA $p = 0.006$); each below one quoted spread	Table 18, D
From 2016 (subsample chosen ex post)	static rule 2.49, EWMA 2.39 after 4 bp, VAR 1.94 at quoted spreads: a shift in the premium's level	Section 7.6
Mean–variance positions, 4 bp	only the EWMA positive, ≤ 1.18	Table 20
Risk aversion from 2 to 10	net fees at 4 bp unchanged to 0.12 bp	Table 17
Overnight rolls at the TGCR, from 2018	bound 0.73 for ON-1W; no positive net fee	Table 19, H
Decomposition: extreme and normal outcomes	VAR nets −1.56 on the 95% normal outcomes, 0.70 on the extreme 5%	Table 24, B

R^2_{oos} of DST's VAR for ON-1W unless stated; fees in bp a year per unit of notional after quoted spreads unless stated. Network design: static test blocks, Section 7.5.

about 100 days and borderline at 200. ON-2W stays between 0.074 and 0.090, and 1W-2W rises from 0.093 to 0.290 as the blocks lengthen. With market expectations no pair survives the adjustment at 5% at any block length.

7.2. Overlap, intervals, and loss

Overlapping returns do not drive the result for ON-1W (Table 23, Panel A): evaluated only on every fifth date, for each of the five offsets, the VAR's R^2_{oos} lies between 5.9% and 30.2%, and the largest Clark–West p-value is 0.002. For ON-2W and 1W-2W it ranges from 1.5% to 12.0% and from −2.9% to 27.0% across offsets, with largest p-values of 0.104 and 0.259, so the evidence for these pairs relies partly on the overlapping dates. The 95% stationary-bootstrap intervals of the VAR's R^2_{oos} lie above zero for ON-1W $[13.9, 27.2]$, ON-2W $[3.1, 10.7]$, and 1W-2W $[4.4, 17.0]$, include zero for the other three pairs with a pos-

Table 22: **Search-Adjusted Inference by Bootstrap Block Length**

	max(20, 2n)		50 days		100 days		200 days	
	SPA p	Holm	SPA p	Holm	SPA p	Holm	SPA p	Holm
Panel A: DST's information set								
ON-1W	0.002	0.016	0.002	0.024	0.004	0.040	0.005	0.046
ON-1W, 50,000 draws	–	–	0.003	0.032	0.004	0.041	0.005	0.051
ON-2W	0.008	0.074	0.010	0.090	0.010	0.086	0.008	0.075
ON-3W	0.772	1.000	0.769	1.000	0.732	1.000	0.680	1.000
ON-1M	0.910	1.000	0.906	1.000	0.882	1.000	0.851	1.000
ON-2M	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
ON-3M	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1W-2W	0.012	0.093	0.019	0.155	0.029	0.232	0.036	0.290
1W-3W	0.362	1.000	0.341	1.000	0.335	1.000	0.237	1.000
1M-2M	0.628	1.000	0.598	1.000	0.642	1.000	0.612	1.000
1M-3M	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Panel B: With market expectations								
ON-1W	0.005	0.054	0.008	0.083	0.010	0.100	0.011	0.103
ON-2W	0.016	0.110	0.022	0.153	0.020	0.161	0.015	0.117
ON-3W	0.100	0.599	0.083	0.496	0.067	0.376	0.057	0.284
ON-1M	0.118	0.599	0.108	0.538	0.063	0.376	0.036	0.251
ON-2M	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
ON-3M	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
1W-2W	0.006	0.055	0.016	0.127	0.031	0.218	0.043	0.260
1W-3W	0.008	0.064	0.012	0.109	0.012	0.109	0.008	0.085
1M-2M	0.679	1.000	0.660	1.000	0.699	1.000	0.706	1.000
1M-3M	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

SPA over the 23 forecasts by mean block length; 10,000 draws, for ON-1W also 50,000; Holm across the ten pairs. Monte Carlo standard error: $\sqrt{p(1-p)/B}$. Panel B: with market expectations.

itive R^2_{oos}, and lie below zero only for ON-2M (Panel B; the intervals describe sampling uncertainty, the tests are those of Section 3.5); without same-day quotes no interval of the direct regression lies above zero. The network ensemble behaves alike: with all quotes its interval for ON-1W is $[11.8, 24.5]$ and its R^2_{oos} on non-overlapping dates 12.4% to 25.3% (largest Clark–West p-value 0.001), and without same-day quotes its interval includes zero for every pair.

A set of five standard forecasts (the EH forecast, the VAR, the term-spread regression, ridge, and the EWMA) checks whether the full model confidence set retains the EH forecast for lack of power (Panel C). It excludes the EH forecast for ON-1W ($p = 0.001$) and, at the 10% level, for ON-2W and 1W-2W, and keeps the VAR for every pair; the set was chosen after the full-set results were known, so it is a check on power only. Excluding the crisis of 2007–09, September 2019, the Covid months, and year ends together leaves the VAR's R^2_{oos} for ON-1W at 15.4% (Clark–West $p < 0.001$). Removing the 1% of origins with the largest errors of the EH forecast leaves the VAR's R^2_{oos} at 17.1%, 9.5%, and 10.0% for ON-1W, ON-2W, and 1W-2W, and with absolute errors the one-sided Diebold–Mariano test rejects at the 1% level for ON-1W, ON-2W, ON-3W, 1W-2W, and 1W-3W (Panel D).

Table 23: **Overlap, Intervals, and Loss**

	Short leg ON						Short leg 1W		Short leg 1M	
	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: VAR (DST) on non-overlapping origins only										
R^2_{oos} (%), minimum over offsets	5.9	1.5	−6.2	−8.2	−27.7	−22.3	−2.9	−14.0	−10.6	−26.0
R^2_{oos} (%), median over offsets	21.9	6.6	0.7	−0.8	−5.4	−7.0	9.3	2.6	1.0	−0.5
R^2_{oos} (%), maximum over offsets	30.2	12.0	7.6	8.1	2.0	0.9	27.0	18.2	33.7	10.0
Largest Clark–West p-value	0.002	0.104	0.585	0.669	0.998	0.957	0.259	0.738	0.872	0.910
Observations per offset	1,451	687	453	322	152	97	668	429	157	99
NN ensemble, R^2_{oos} (%), minimum over offsets	12.4	2.3	−5.1	–	–	–	−4.1	−5.2	–	–
maximum over offsets	25.3	9.7	7.9	–	–	–	25.2	14.9	–	–
largest Clark–West p-value	0.001	0.044	0.356	–	–	–	0.226	0.313	–	–
Panel B: 95% bootstrap interval of R^2_{oos} (%)										
VAR (DST), lower bound	13.9	3.1	−2.5	−3.5	−11.4	−14.8	4.4	−1.4	−1.8	−6.3
upper bound	27.2	10.7	6.3	5.1	−1.7	0.5	17.0	8.1	9.4	2.5
OLS direct, with same-day quotes, lower	12.5	−1.8	−6.9	−7.0	−15.5	−18.1	1.7	−5.5	−13.8	−21.9
upper	26.5	9.1	5.5	5.4	−0.7	−0.5	13.8	8.8	4.4	0.8
OLS direct, without same-day quotes, lower	−0.9	−7.3	−8.4	−7.5	−14.1	−16.8	−4.1	−6.1	−20.7	−23.3
upper	5.2	2.9	1.3	1.3	−1.8	−0.8	4.7	2.6	−0.1	1.2
NN ensemble, with same-day quotes, lower	11.8	−2.1	−4.0	–	–	–	1.3	−3.3	–	–
upper	24.5	11.3	7.3	–	–	–	13.4	8.2	–	–
NN ensemble, without same-day quotes, lower	−1.3	−3.9	−4.7	–	–	–	−3.8	−3.3	–	–
upper	3.4	4.4	3.1	–	–	–	4.5	3.6	–	–
Panel C: Model confidence set of five standard forecasts										
MCS p-value of the EH forecast	0.001	0.062	0.574	0.943	1.000	1.000	0.080	0.652	0.441	1.000
MCS p-value of the VAR (DST)	1.000	0.900	0.946	0.943	0.420	0.263	1.000	1.000	1.000	0.839
Panel D: VAR (DST), largest errors and loss function, R^2_{oos} (%)										
Squared-error loss, all origins	20.5***	6.2***	1.0**	−0.2*	−6.8	−8.1	10.1***	2.2***	2.2***	−1.3*
Largest 1% of EH errors removed	17.1***	9.5***	3.9***	2.0***	−6.8	−7.3	10.0***	4.8***	1.5***	−3.3
Absolute-error loss, all origins	13.9***	5.6***	2.6***	1.4	−4.4	−1.5	7.7***	5.4***	1.6**	0.3

Stars: Clark–West test at 10, 5, and 1%. Panel A: every n-th origin, for each of the n offsets. NN ensemble: the five pairs with horizons up to three weeks (–: not computed). Panel B: 1,000 draws. Panel C: 90% MCS of the EH forecast, the VAR, the term-spread regression, ridge, and the EWMA; below 0.10: outside. Panel D: absolute loss $1-\sum|e|/\sum|e_{EH}|$; stars: one-sided Diebold–Mariano test.

7.3. Extreme and normal periods

Statistical accuracy comes from normal periods, economic value from extreme outcomes (Table 24). Each rule splits the out-of-sample origins into an extreme subset $\mathcal{A}$ and its complement, using the whole sample, so the split describes when the forecasts work and is not a trading rule. The *volatility* rule marks origins whose investment window $[t, t+n]$ contains a day with 21-day volatility of the short rate in its top 10% (13% to 27% of origins, depending on the pair); the *jump* rule marks windows that contain a daily change of the short or the long rate in its top 1% (7% to 36%); and the *outcome* rule marks the 1% or 5% of origins with the largest realised deviation from the EH forecast, $|y_t - \widehat{\mu}_t^{EH}|$. On each subset,

$$R^2_{oos}(\mathcal{A}) = 1 - \frac{\sum_{t\in\mathcal{A}}\left(y_t - \widehat{\mu}_{j,t}\right)^2}{\sum_{t\in\mathcal{A}}\left(y_t - \bar{y}_{t-1}\right)^2}, \qquad C_j(\mathcal{A}) = \mathcal{F}_j(\mathcal{A})\,\frac{|\mathcal{A}|}{T}, \tag{29}$$

where $\mathcal{F}_j(\mathcal{A})$ is the fee of Equation (20) on the subset, so that the contributions C_j of a subset and its complement add up approximately to the full-sample fee.

The VAR is more accurate in calm than in stressed windows (Panel A). Its R^2_{oos} for

ON-1W is 29.6% outside and 17.0% inside the high-volatility windows (25.6% and 18.5% with the jump rule, not tabulated), for ON-2W 16.7% against 2.1%, and for 1W-2W 21.6% against 7.2%. The EWMA is the mirror image: mildly positive in stressed windows (up to about 4% for ON-2W and 1W-2W) and negative in calm windows for most pairs. The direct regression without same-day quotes is also more accurate in calm windows (5.5% against 0.7% for ON-1W).

The economic value, in contrast, comes from extreme outcomes (Panel B). Of the VAR's gross fee for ON-1W, 2.48 basis points a year, the 5% most extreme outcomes contribute 1.10 and the other 95% 1.38 (not tabulated); after quoted spreads they contribute 0.70 and −1.56, so on normal days the VAR loses money because it trades on many small departures. Perfect sign foresight nets 1.73 of its 1.85 on the extreme 5%. For ON-2W and ON-3W the EWMA's whole net fee comes from the extreme 5% (0.72 and 0.78 there, −0.44 and −0.36 elsewhere), as does that of the static rule for ON-2W (1.21 against −1.10); for ON-1W the EWMA earns most of its net fee on normal days. The statistical rejection is thus a property of calm markets, while the little economic value there is accrues in a few extreme episodes, as it must for a sign strategy, whose gains scale with the size of the excess return; in these episodes the measured excess return is largest and the quoted spread widest (8.9 against 4.2 basis points for the 5% most extreme ON-1W outcomes).

7.4. Stale-quote classification

The baseline classifies stale runs ex post (Section 2.2). Table 25 repeats the key results for the three pairs that survive the search with a classification that uses only past quotes: a quote is masked from the sixth identical value of a run on, both when the pricing sources are combined and in the cleaning (Panel A). The real-time series keep more quotes, so the samples differ. The VAR's R^2_{oos} is 20.0%, 4.9%, and 11.6% for ON-1W, ON-2W, and 1W-2W, against 20.5%, 6.2%, and 10.1% with the ex-post classification, a change of at most 1.5 percentage points. The direct regression and ridge change by up to 2.4 and 1.9 percentage points (1W-2W: 7.4% to 9.8% and 6.8% to 8.7%), and the EWMA, the subsample from 2016, and the VAR's gross fee change little. The ex-post classification therefore does not create the predictability.

Panel B re-runs the direct regression with other thresholds. Classifying runs of more than 3, 5 (the baseline), or 10 identical quotes as stale gives an ON-1W same-day R^2_{oos} of 18.6%, 19.9%, and 20.5% and a lagged R^2_{oos} of 1.5%, 2.0%, and 2.9%, and excluding the origins whose predictor or target window contains a switch of pricing source leaves 18.7% and 1.8%. With the threshold of 10, which keeps more stale quotes, 1W-2W gains same-day accuracy (16.0%), consistent with stale term quotes creating predictable structure.

Table 24: **Accuracy and Value in Stressed and Calm Periods**

	ON-1W	ON-2W	ON-3W	1W-2W	1W-3W
Panel A: R^2_{oos} (%) in stressed and calm investment windows					
Share of origins in windows with high volatility (%)	12.8	14.9	15.9	14.3	16.1
VAR (DST)					
high volatility	17.0^{***}	2.1^{***}	-3.5	7.2^{***}	-0.9^{*}
other windows	29.6^{***}	16.7^{***}	11.1^{***}	21.6^{***}	14.4^{***}
Most accurate forecast					
high volatility	17.0^{***}	2.1^{***}	-0.6	7.2^{***}	1.6^{**}
other windows	29.6^{***}	16.7^{***}	9.4^{***}	21.6^{***}	9.3^{***}
which forecast	VAR (DST)	VAR (DST)	NN (32-16)	VAR (DST)	NN (32-16)
EWMA					
high volatility	0.0	3.9^{*}	4.5^{*}	4.1^{**}	4.7^{*}
other windows	2.4^{***}	-1.2^{***}	-5.8^{**}	-5.4^{**}	-11.1^{*}
OLS direct without same-day quotes					
high volatility	0.7^{*}	-6.4	-8.3	-1.7	-4.7
other windows	5.5^{***}	7.0^{***}	4.8^{***}	6.8^{***}	6.4^{***}
Panel B: Contribution to the net fee after quoted spreads (bp a year)					
VAR (DST)					
5% most extreme outcomes	0.70	0.37	0.20	-0.46	-0.73
other 95%	-1.56	-1.94	-1.52	-2.26	-1.55
EWMA					
5% most extreme outcomes	0.19	0.72	0.78	-0.25	-0.25
other 95%	0.55	-0.44	-0.36	-0.18	-0.07
Always lend long, borrow short					
5% most extreme outcomes	0.48	1.21	1.60	0.00	0.00
other 95%	-0.36	-1.10	-0.67	0.00	0.00
Perfect sign foresight					
5% most extreme outcomes	1.73	1.94	2.08	0.37	0.34
other 95%	0.12	-0.36	0.20	-1.60	-1.53

Pairs with horizons up to three weeks. Subsets: Equation (29), classified with the whole sample. Stars: Clark–West test at 10, 5, and 1% (indicative). Panel A: high volatility: window with a day of 21-day short-rate volatility in its top 10%; most accurate: highest R^2_{oos} in Table 7. Panel B: extreme outcomes: the 5% largest $|y_t - \widehat{\mu}_t^{EH}|$; contribution: C_j of Equation (29).

7.5. Functional form and information

No forecast improves on the VAR on DST's information set (Table 26). Where the flexible forecasts predict, they find the VAR's signal (Table 27): for ON-1W the departures of the network ensemble from the EH forecast correlate at 0.92 with the VAR's and those of ridge at 0.91, with 0.90 and 0.89 of their dispersion, while the tree-based models, whose departures correlate at 0.10 to 0.19 with the VAR's and vary up to 1.8 times as much, are less accurate than the EH forecast. In two-sided Diebold–Mariano tests (Panel A), no forecast is significantly more accurate than the VAR for any pair, and the VAR is significantly more accurate at the 5% level than 10 to 13 of the 24 other forecasts for ON-1W, ON-2W, and 1W-2W. In the encompassing test, after a Holm adjustment within each pair, no forecast adds information to the VAR on any pair (Panel C). With market expectations (Panel B) least squares, the linear SVR, two single networks, and the network ensemble are significantly more accurate than DST's VAR for 1W-2W, the SPA test rejects that no forecast with expectations beats the VAR for this pair ($p = 0.036$), and after the Holm adjustment 11,

Table 25: **Stale-Quote Classification: Real Time, Thresholds, and Source Switches**

	ON-1W		ON-2W		1W-2W	
	Ex post	Real time	Ex post	Real time	Ex post	Real time
Panel A: Ex-post against real-time classification						
VAR (DST), R^2_{oos} (%)	20.5***	20.0***	6.2***	4.9***	10.1***	11.6***
OLS direct, R^2_{oos} (%)	19.9***	19.6***	3.3***	2.9***	7.4***	9.8***
Ridge, R^2_{oos} (%)	18.9***	18.7***	1.3***	2.2***	6.8***	8.7***
EWMA, R^2_{oos} (%)	0.7***	1.0***	2.5**	1.9**	2.1**	1.3**
VAR (DST), R^2_{oos} from 2016 (%)	38.8***	39.0***	22.1***	21.6***	14.8***	15.1***
VAR (DST), fee before costs (bp a year)	2.48	2.35	0.68	0.73	0.29	0.30
Out-of-sample origins	7,258	7,839	6,878	7,547	6,686	7,438
Panel B: Stale-quote thresholds and pricing-source switches, R^2_{oos} (%)						
	Same day	Lagged	Same day	Lagged	Same day	Lagged
VAR (DST), run of more than 3 identical values	20.6***	2.5***	6.0***	0.8**	10.9***	1.4**
VAR (DST), more than 5 (baseline)	20.5***	1.8***	6.3***	−0.2	10.4***	1.2**
VAR (DST), more than 10	22.6***	3.6***	9.2***	2.4***	17.9***	4.3***
VAR (DST), baseline without windows with a source switch	20.9***	1.8***	6.0***	−0.7	10.7***	1.0**
OLS direct, run of more than 3 identical values	18.6***	1.5***	2.7***	−2.0	8.4***	−1.6
OLS direct, more than 5 (baseline)	19.9***	2.0***	3.3***	−2.6	7.6***	0.0**
OLS direct, more than 10	20.5***	2.9***	5.8***	0.1***	16.0***	3.7***
OLS direct, baseline without windows with a source switch	18.7***	1.8***	2.5***	−3.7	6.6***	−0.8**

Stars: Clark–West test at 10, 5, and 1%. Panel A: real time: a quote is masked only from the sixth identical value of a run on; the samples differ. Panel B: series rebuilt with the stated threshold; same day: lags t to $t-4$; lagged: $t-1$ to $t-5$; last rows: without windows that contain a source switch; common dates per pair.

5, and 18 forecasts add information to the VAR for ON-1W, ON-2W, and 1W-2W. The gain again comes from the information. In simulations the flexible forecasts recover little of a nonlinear departure (Section 8.2), so their failure to beat the VAR is weak evidence of linearity.

The network's design does not drive this (Table 28). Re-run on the current data with a static split, one factor at a time, the paper's network attains a test-block R^2 of 37.5% for ON-1W on 2017/18–2026, against 37.4% for DST's VAR, and 10 of 12 network variants lie between 35.8% and 38.9%; only ReLU units (21.3%) and min-max scaled inputs (30.3%) do worse. On the earlier test block 2007–2013 every variant stays below the VAR (−0.9% to 15.8% against 19.4%). In the main run the five architectures attain 17.8% to 18.8% for ON-1W (Table 7).

In stationary form (Table 29, Panel A), the trees improve markedly: the random forest attains 18.0% for ON-1W and 11.0% for 1W-2W, against −13.9% and −10.2% with the rates in levels, and boosting 13.5% for 1W-2W against −47.5%. The SPA test against the EH forecast rejects for ON-1W, ON-2W, and 1W-2W ($p \leq 0.033$), but no forecast with stationary inputs beats DST's VAR on these pairs ($p \geq 0.650$). Calendar dummies known at the origin add no out-of-sample accuracy (Panel B): added to the direct regression they change its R^2_{oos} by at most 1.6 percentage points, never significantly, the dummies alone attain at most 1.5%, and in boosted trees they reduce the losses without making boosting

Table 26: **Forecasts versus DST's VAR**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: Diebold–Mariano statistic, DST's information set										
OLS (direct)	−0.44	−2.02**	−2.13**	−0.81	−1.02	−1.17	−1.48	−1.06	−1.09	−1.88*
Ridge	−0.89	−2.98***	−0.54	−0.72	−1.55	−0.85	−1.71*	−1.77*	−1.47	−2.20**
Lasso	−0.92	−2.74***	−0.62	−0.40	−0.87	0.02	−1.34	−1.06	−1.76*	−2.67***
Elastic net	−0.90	−2.74***	−0.54	−0.34	−0.81	−0.08	−1.25	−1.03	−1.68*	−2.75***
PCR	−1.49	−2.60***	−1.02	−1.30	−1.65*	−0.94	−1.98**	−1.88*	−1.94*	−2.42**
PLS	−0.97	−3.18***	−1.25	−1.47	−1.70*	−1.29	−1.91*	−2.26**	−1.62	−2.20**
Linear SVR	−1.29	−2.86***	−0.69	−1.22	−0.41	0.14	−0.38	−1.03	−2.11**	−1.83*
RBF SVR	−2.60***	−3.53***	−2.84***	−2.64***	−1.70*	−1.17	−1.82*	−1.68*	−2.53**	−2.07**
KNN	−3.74***	−1.65*	−2.41**	−3.69***	−3.91***	−4.23***	−3.11***	−3.56***	−5.66***	−3.43***
Tree	−3.41***	−2.34**	−2.13**	−1.84*	−2.01**	−1.81*	−4.28***	−3.77***	−3.24***	−2.29**
Random forest	−3.22***	−2.14**	−2.21**	−2.18**	−1.83*	−2.21**	−4.77***	−4.25***	−3.58***	−2.76***
Boosting	−2.06**	−1.92*	−1.97**	−2.09**	−2.23**	−1.63	−2.26**	−2.25**	−2.81***	−1.90*
Bagging	−2.82***	−2.37**	−2.31**	−2.23**	−2.19**	−2.39**	−3.61***	−2.85***	−6.23***	−4.50***
Random features	−1.49	−1.29	−1.32	−1.57	−1.78*	−1.36	−1.92*	−2.89***	−1.94*	−2.41**
NN (16)	−1.43	−0.37	0.10	0.23	−0.55	−1.66*	−2.40**	−0.17	−2.14**	−2.75***
NN (32)	−1.28	−0.65	0.11	0.08	−1.48	−1.69*	−1.85*	−0.25	−1.90*	−3.10***
NN (16-8)	−1.57	−0.15	0.29	0.25	−1.36	−1.50	−2.20**	0.01	−2.18**	−1.56
NN (32-16)	−1.73*	−0.56	0.34	0.04	−1.30	−1.72*	−2.50**	0.24	−2.22**	−2.17**
NN (32-16-8)	−1.43	−0.43	0.11	0.38	−1.47	−1.33	−2.53**	0.08	−2.19**	−1.16
NN ensemble	−1.37	−0.34	0.26	0.26	−1.20	−1.41	−2.34**	0.17	−2.05**	−1.97**
Combination	−2.13**	−1.83*	−1.76*	−1.96*	−1.27	−1.49	−1.19	0.36	−1.72*	−2.06**
Combination, 50% to EH	−2.91***	−2.40**	−1.52	−0.74	0.60	0.68	−2.01**	0.33	−1.44	−0.77
Combination, DMSPE	−2.02**	−1.91*	−1.89*	−1.80*	−1.21	−1.36	−1.08	0.58	−1.60	−1.84*
EWMA	−3.68***	−1.19	0.06	−0.07	−0.57	−0.86	−2.58***	−0.16	−1.63	−0.78
Panel B: Diebold–Mariano statistic, DST's set and market expectations										
OLS	0.94	1.11	0.80	0.91	−0.47	−1.01	2.41**	1.79*	−0.39	−1.02
Ridge	0.46	0.42	0.36	0.57	0.07	0.61	1.67*	1.45	−0.85	−0.32
Lasso	0.22	−0.03	0.20	0.53	1.42	2.42**	1.71*	1.24	−1.10	−0.93
Elastic net	0.11	−0.04	0.22	0.30	1.11	2.21**	1.68*	1.27	−0.97	−0.86
PCR	−4.39***	0.22	0.54	0.27	0.50	0.04	−0.67	0.63	−1.67*	−0.56
PLS	−3.24***	−0.41	0.47	−1.01	0.04	−0.13	−0.36	1.13	−0.70	−0.73
Linear SVR	0.26	0.13	0.89	1.19	0.89	0.60	2.50**	1.89*	−0.39	−0.64
RBF SVR	−2.53**	−1.08	−1.15	−1.22	−1.68*	−0.02	−1.49	−1.16	−1.62	−0.75
KNN	−3.70***	−2.48**	−2.20**	−1.80*	−1.63	−1.05	−2.86***	−1.65*	−3.16***	−3.15***
Tree	−3.89***	−2.16**	−2.06**	−1.90*	−1.75*	−1.48	−3.59***	−2.13**	−2.22**	−2.66***
Random forest	−3.13***	−1.70*	−2.21**	−2.96***	−1.45	−1.47	−3.31***	−1.98**	−2.65***	−2.21**
Boosting	−2.86***	−1.83*	−1.39	−1.26	−1.61	−2.08**	−2.40**	−1.53	−1.89*	−1.67*
Bagging	−3.13***	−2.40**	−1.92*	−2.32**	−1.69*	−2.28**	−3.72***	−2.94***	−2.70***	−2.47**
Random features	−3.14***	−1.20	−1.21	−1.03	−1.32	−2.02**	−1.63	−1.42	−3.27***	−2.65***
NN (16)	−1.09	1.16	0.87	1.12	0.76	−1.30	2.03**	1.41	−0.47	−1.76*
NN (32)	−1.21	−0.26	0.43	0.67	−1.26	−1.55	2.09**	1.18	−0.59	−2.57**
NN (16-8)	−1.57	0.18	0.21	1.16	−0.06	−0.31	1.55	1.26	−0.29	−1.60
NN (32-16)	−1.39	0.67	0.46	1.01	0.16	−1.39	1.91*	1.34	−1.03	−2.10**
NN (32-16-8)	−0.58	0.53	0.68	0.89	−0.44	0.85	1.87*	1.27	−0.81	−2.15**
NN ensemble	−0.89	0.73	0.72	1.21	0.27	−0.65	2.20**	1.41	−0.23	−1.97**
Combination	−0.93	0.37	1.48	1.71*	−0.20	0.20	1.51	2.34**	0.10	−0.56
Combination, 50% to EH	−2.55**	−0.38	1.34	1.70*	1.99**	1.94*	−0.04	2.01**	0.44	0.59
Combination, DMSPE	−0.93	0.53	1.57	1.82*	−0.02	0.59	1.82*	2.38**	0.20	−0.38
EWMA	−3.68***	−1.19	0.06	−0.07	−0.57	−0.86	−2.58***	−0.16	−1.63	−0.78
Panel C: Learners not encompassed by DST's VAR (encompassing test)										
DST's set, unadjusted	0/20	0/20	0/20	0/20	0/20	0/20	0/20	1/20	0/20	0/20
DST's set, Holm within the pair	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20	0/20
With market expectations, unadjusted	13/20	15/20	15/20	14/20	12/20	13/20	18/20	18/20	15/20	5/20
With market expectations, Holm within the pair	11/20	5/20	2/20	3/20	5/20	4/20	18/20	4/20	11/20	0/20

Each forecast against DST's VAR on DST's set, on common dates. Panels A and B: Diebold–Mariano statistic, positive: more accurate than the VAR; stars: two-sided test at 10, 5, and 1%. Panel B: OLS is a learner. Panel C: learners for which the encompassing test rejects at 5%.

competitive.

Adding the 126 FRED-MD predictors, a panel like the 128 series of Bianchi et al. (2021b), to DST's set lowers accuracy for almost every forecast, and the SPA test against the EH forecast rejects for no pair at the 5% level (smallest $p = 0.078$, ON-1W; Panel C). Adding one of the eight FRED-MD groups at a time to ridge lowers its R^2_{oos} in 77 of 80 cases; the three gains are at most 2.7 percentage points. Figure 7 sets this against the market expectations,

Table 27: **Departures of the Forecasts from the VAR**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: Dispersion of departures relative to the VAR										
Ridge	0.89	0.95	0.95	1.10	1.11	0.91	0.88	1.01	0.96	1.40
Lasso	0.84	0.88	0.86	0.95	0.97	0.81	0.87	0.95	0.88	1.27
Linear SVR	0.76	0.83	0.87	1.03	0.91	0.68	0.76	0.88	0.82	1.16
RBF SVR	0.88	0.89	1.11	1.26	1.70	2.41	0.75	0.94	1.05	1.84
KNN	0.65	1.80	2.59	2.55	4.05	3.92	1.12	1.98	2.25	3.33
Random forest	0.95	1.76	2.34	2.39	2.92	3.20	1.38	2.10	1.37	2.02
Boosting	1.80	2.52	2.97	3.11	2.41	3.01	2.31	3.46	1.08	2.16
Random features	0.93	1.48	1.71	1.73	2.10	4.07	0.73	1.01	0.93	2.06
NN ensemble	0.90	1.13	1.31	1.39	1.48	1.44	0.99	1.16	1.29	1.53
Combination	0.73	0.93	1.09	1.14	1.32	1.27	0.81	1.02	0.81	1.26
EWMA	0.37	1.01	1.46	1.54	2.16	2.31	0.62	1.06	1.18	1.92
Panel B: Correlation of departures with the VAR										
Ridge	0.91	0.78	0.67	0.69	0.66	0.64	0.84	0.75	0.33	0.73
Lasso	0.89	0.75	0.59	0.54	0.49	0.59	0.84	0.74	0.29	0.67
Linear SVR	0.89	0.79	0.67	0.68	0.55	0.37	0.86	0.75	0.71	0.59
RBF SVR	0.68	0.58	0.40	0.39	0.35	0.12	0.59	0.49	0.19	0.29
KNN	0.20	0.14	0.23	0.18	0.14	0.15	0.18	0.17	0.06	0.14
Random forest	0.19	0.14	0.25	0.25	0.25	0.27	0.21	0.19	0.09	0.11
Boosting	0.10	0.13	0.17	0.18	0.30	0.30	0.14	0.14	0.13	0.10
Random features	0.70	0.43	0.38	0.35	0.47	0.19	0.58	0.45	0.07	0.33
NN ensemble	0.92	0.80	0.73	0.73	0.53	0.29	0.87	0.77	0.41	0.78
Combination	0.86	0.72	0.68	0.69	0.58	0.48	0.79	0.69	0.45	0.65
EWMA	0.12	0.09	0.03	0.02	0.06	0.12	0.05	−0.01	−0.06	0.13

Departure: forecast minus the EH forecast. Panel A: standard deviation of the departures relative to the VAR's. Panel B: correlation with the VAR's departures.

Table 28: **Sensitivity of the Network to Its Design**

	Test block 2017/18–2026				Test block 2007–2013			
	ON-1W	ON-2W	1W-2W	Mean	ON-1W	ON-2W	1W-2W	Mean
DST's VAR	37.4	20.2	7.5	6.1	19.4	6.5	11.5	3.8
Ridge	37.3	20.2	7.6	7.3	19.4	4.4	11.4	2.0
Paper's network	37.5	19.0	7.0	7.8	12.9	0.8	9.7	−0.6
With bias terms	36.3	18.7	7.0	6.8	15.8	2.3	8.1	1.6
ReLU activation	21.3	13.4	2.7	1.8	1.5	−0.8	3.3	−1.6
ELU activation	38.9	19.6	6.5	5.9	1.9	−1.5	9.6	−2.5
One layer of 32	36.8	18.1	6.3	8.0	14.1	0.0	10.6	−2.2
Three layers 32-16-8	38.0	19.2	8.3	8.6	12.6	−0.1	9.0	−0.3
Adam instead of Nadam	37.2	18.9	7.1	7.9	14.1	−0.9	9.7	−0.6
Learning rate 0.0005	37.1	18.8	6.8	7.9	11.6	0.7	9.6	−0.7
Learning rate 0.002	37.3	19.3	7.1	7.9	14.1	0.9	8.7	−0.2
No dropout	35.8	17.6	6.6	6.4	15.2	0.8	0.5	−0.5
Inputs min-max scaled	30.3	18.2	3.2	8.0	−0.9	−0.3	0.8	−2.0
Paper's network, ten seeds	37.4	18.6	7.0	7.8	12.4	0.8	9.3	−0.5

R^2 (%) on the test block against the training-block mean of the excess return; static chronological split per pair (training, validation, test; gaps of $n-1$ days), DST's information set. Paper's network: tanh, no bias terms, 16-8, Nadam at 0.001, dropout 0.1, average of five seeds; each other row changes one factor. Mean: over the ten pairs.

which raise the accuracy of the same model for every pair: information about the expected policy path predicts, broad macro-financial information does not. Real-time vintages do not

change this verdict. Repo premia at horizons of days to months are predicted, if at all, by the repo and bill curves themselves, unlike one-year bond returns (Ludvigson and Ng, 2009).

Table 29: **Other Information Sets**

	ON-1W	ON-2W	ON-3W	ON-1M	ON-2M	ON-3M	1W-2W	1W-3W	1M-2M	1M-3M
Panel A: Stationary repo inputs (term spread and change of the short rate), R^2_{oos} (%)										
OLS	18.8***	4.2***	0.4***	1.5***	−5.2	−6.3	7.0***	2.7***	0.0***	0.4**
Ridge	17.9***	3.8***	1.0***	1.8***	−3.9	−5.8	7.1***	1.9***	−0.2***	0.5**
Random forest	18.0***	−3.1*	−9.2	−10.1	−12.9	−14.2	11.0***	2.2***	−1.3***	2.1**
Boosting	15.0***	−4.9**	−4.9	−4.0	−8.8	−9.9	13.5***	−4.2**	−1.6**	−0.5*
RBF SVR	15.7***	3.5***	1.6***	−0.7***	−7.3	−7.2	10.7***	2.8***	2.6***	2.0***
Combination	19.0***	2.9***	−0.3*	−0.3*	−4.4	−5.3	11.9***	3.9***	1.7***	3.0***
Best of the 13 learners	18.0***	4.1***	1.8***	2.7**	−1.9	−3.7	13.5***	4.4***	2.6***	3.2**
SPA p-value: no forecast beats the EH forecast	0.003	0.033	0.417	0.783	1.000	1.000	0.013	0.083	0.131	0.288
SPA p-value: no forecast beats the VAR on DST's set	1.000	1.000	0.902	0.787	0.164	0.273	0.650	0.630	0.730	0.216
Largest performance fee before costs (bp a year)	2.61	0.87	0.83	0.20	−0.73	−0.25	0.48	0.21	0.31	0.84
VAR on DST's set, on the dates of this panel, R^2_{oos} (%)	20.5	6.2	1.1	−0.2	−6.8	−8.1	10.4	2.3	2.1	−1.3
Panel B: Calendar dummies, R^2_{oos} (%)										
OLS (direct)	19.9***	3.3***	−2.2*	−1.6**	−9.5	−10.4	7.4***	0.1***	−3.2**	−7.4
OLS (direct) + calendar	20.4***	4.9***	−1.4***	−1.8***	−9.7	−9.9	7.8***	−0.2***	−3.9**	−7.7*
Calendar only	−0.7***	1.5***	1.3***	0.6***	−1.4*	−1.6	−0.5***	0.3***	−1.3	−1.7
Boosting	−55.4	−30.4	−44.8	−40.0	−26.8	−40.1	−47.5*	−52.3*	−5.9	−23.8
Boosting + calendar	−17.1	−11.4	−9.9*	−17.7	−23.0	−39.1	−7.5***	−21.9**	−5.4	−19.9
DM statistic: OLS + calendar vs OLS	0.40	0.72	0.27	−0.05	−0.10	0.40	0.65	−0.16	−1.09	−0.29
Panel C: DST's information set and 126 macro-financial predictors										
Best of the 13 learners, R^2_{oos} (%)	5.5***	0.9**	−1.5***	−0.2***	−13.3*	−14.9	0.0***	−1.9**	−4.4	2.7*
SPA p-value: no forecast beats the EH forecast	0.078	0.265	0.924	1.000	1.000	1.000	0.653	0.626	1.000	0.862
SPA p-value: no forecast beats the VAR on DST's set	1.000	1.000	0.957	0.942	0.818	0.972	1.000	1.000	1.000	0.829
Largest performance fee before costs (bp a year)	1.37	1.06	0.54	0.79	−0.16	−0.48	−0.05	0.08	0.30	0.92

Robustness sets of Section 3.3. Stars: Clark–West test at 10, 5, and 1%. Panels A and C: dates common with DST's VAR; SPA: no forecast beats the EH forecast, or DST's VAR; largest fee: gross. Panel B: DM: Diebold–Mariano, with against without dummies. Panel C: with the 126 FRED-MD predictors (Table A.34).

7.6. Supporting subsample from 2016

The forecasts from 1 January 2016 use the specification of the full sample (Section 3.3) and show whether its evidence persists in the most recent decade. The rejection does not depend on that decade: on origins before 2016 the VAR attains 17.4% for ON-1W. Predictability is larger (Table 5, Panel E): the VAR attains 38.8%, 22.1%, 14.8%, and 13.6% for ON-1W, ON-2W, 1W-2W, and ON-3W, the direct regression without same-day quotes 9.6%, 6.6%, and 6.6% for the first three, and the combination with market expectations shrunk half-way to the EH forecast beats the EH forecast for every pair (5.5% to 15.6%).

The mean excess return of the overnight pairs shifts (Table 30). A sup-Wald test (Andrews, 1993) dates one break in the mean for ON-1W, ON-2W, and ON-3W in August 2007 to January 2008 on the full sample, driven by the crisis, and in September and October 2016 once 2007–09 is removed; it finds no break for 1W-2W and 1M-2M. Before 2016 the ON-1W excess return averages 0.68 basis points a year, from 2016 −2.22 (t of 1.8 and −5.8), and those of ON-2W and ON-3W fall by 2.8 and 3.0 basis points. The Sharpe ratio of always lending long and borrowing short is 1.46 to 1.63 from 2016 for the three overnight pairs, against below zero before.

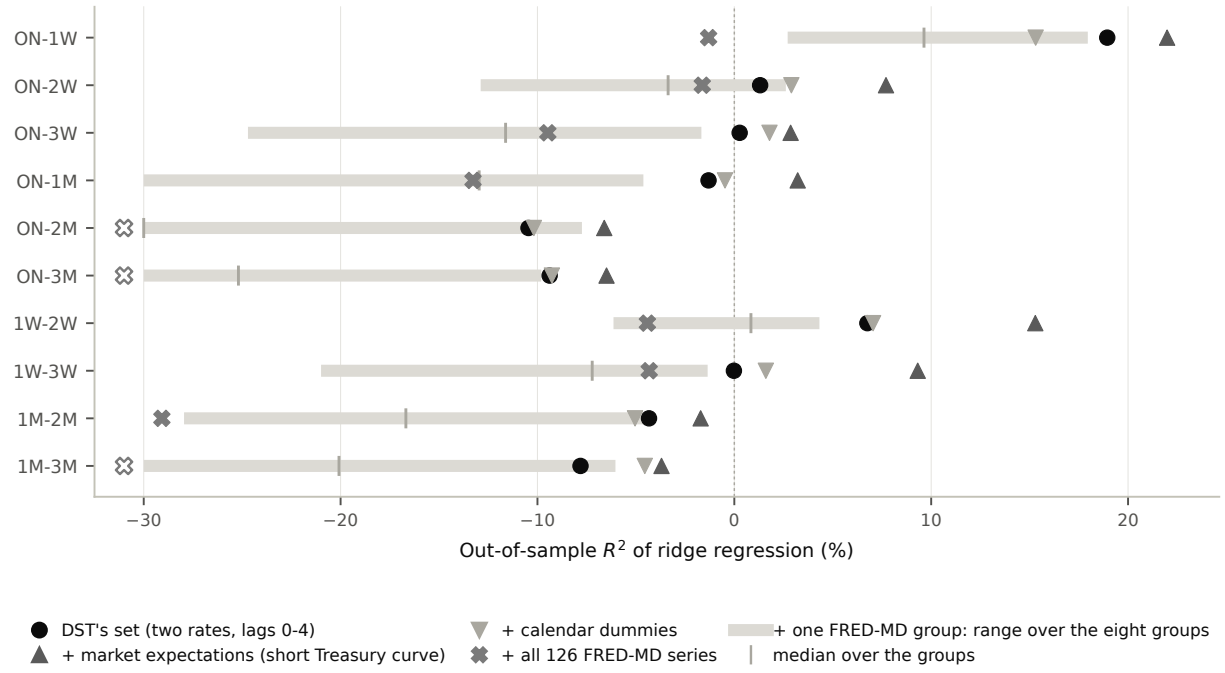


Figure 7: **Which Information Predicts.** Ridge, R^2_{oos} (%) on DST's set and with one additional set at a time; grey bar: range over the eight FRED-MD groups added one at a time, median marked. Hollow symbols at the left edge: below −30%.

From 2016, a subsample chosen after the break was dated, this static rule earns 2.49 basis points a year for ON-1W after a spread of 4 basis points (Newey–West $t = 2.85$, 250 lags), more than its 2.47 before costs because the EH strategy pays the spread when it switches and the static rule never trades, and the EWMA 2.39 ($t = 2.73$). In this period perfect sign foresight nets 2.56 after the same spread (5.40 before costs), which is not a bound. The largest fee of the 23 forecasts in this period is 3.77 basis points a year before costs (ON-1W) and 2.39 after a spread of 4 basis points (EWMA; the VAR 0.95), so from 2016 the premium's level shifts by about one spread: the static rule nets 2.49 basis points a year after a 4 basis-point spread and the EWMA, which tracks the level, 2.39, and no forecast of its variation does better; at the quoted spreads of this period, 2.5 basis points on average for ON-1W, the VAR nets 1.94 ($t = 2.29$), the EWMA 2.40, and the static rule 2.48. The shift is a departure from a constant premium in its level, about one spread in size, and the quotes pin down that level only to within several basis points (Section 5.2).

Most of the shift falls after 2019: the ON-1W excess return averages 0.68 basis points a year in 1994–2015, −1.06 in 2016–2019, and −3.10 in 2020–2026. The period covers ample reserves with the overnight reverse repo facility, the repo-rate spike of 2019, and the tightening of 2022–23; before it, short-term repo spreads moved with reserves and bill supply (Copeland et al., 2025; d'Avernas and Vandeweyer, 2024). The data change as well:

the more frequently updated pricing source covers the maturities beyond one week only from June 2013 (Table 1), so the quotes after 2016 are less stale.

Table 30: **The Shift of the Premium and the Subsample from 2016**

	Full sample		Without 2007–09		Mean excess return (bp a year)				Sharpe ratio, static rule	
	sup-W	Break	sup-W	Break	to 2015	t	from 2016	t	to 2015	from 2016
Panel A: Break in the mean excess return										
ON-1W	28.6	Jan 2008	51.1	Oct 2016	0.68	1.8	−2.22	−5.8	−0.44	1.63
ON-2W	18.6	Aug 2007	35.6	Sep 2016	−0.17	−0.2	−2.94	−5.9	−0.10	1.56
ON-3W	14.5	Oct 2007	28.2	Sep 2016	−0.42	−0.5	−3.39	−5.6	−0.03	1.46
1W-2W	5.8†	–	5.5†	–	−0.89	−2.9	−0.77	−3.5	0.28	0.50
1M-2M	3.0†	–	2.4†	–	−1.56	−2.8	−1.36	−2.2	0.29	0.34

Panel B: Supporting subsample, forecast origins from 1 January 2016

	R^2_{oos} (%)			Fee (bp a year)						
				Largest fee		VAR (DST)	Lend long, borrow short		Perfect sign foresight	
	VAR (DST)	Combination	NN ensemble	Gross	4 bp	4 bp	Gross	4 bp	Gross	4 bp
ON-1W	38.8***	33.7***	33.1***	3.77	2.39	0.95	2.47	2.49	5.40	2.56
ON-2W	22.1***	17.3***	17.8***	0.24	−0.18	−1.36	0.00	0.00	2.31	−0.27
ON-3W	13.6***	10.6***	9.6***	−0.05	−0.10	−1.57	0.00	0.00	2.15	−0.37
ON-1M	5.7***	5.6*	4.0**	−0.15	−0.29	−1.39	0.00	0.00	2.34	0.00
ON-2M	0.0	0.1	0.3	0.00	0.00	−0.53	0.00	0.00	1.68	−0.96
ON-3M	−2.5	−2.3	−2.8	0.00	0.00	−1.00	0.00	0.00	2.39	−0.51
1W-2W	14.8***	13.1**	13.4***	0.35	0.00	−1.53	0.00	0.00	1.88	−1.56
1W-3W	8.4***	8.1**	7.7**	0.00	0.00	−1.49	0.00	0.00	2.05	−1.09
1M-2M	7.9**	−2.8	−10.7	−0.13	−0.42	−1.23	0.00	0.00	3.21	−0.58
1M-3M	−1.5**	−1.3*	−6.7*	0.51	0.15	−0.84	0.00	0.00	4.46	0.87

Panel A: sup-Wald test (Andrews, 1993) for one break in the mean, 15% trimming (5% critical value 8.85); †: not significant; –: no break dated; without 2007–09: origins August 2007 to June 2009 dropped; Sharpe ratio: always lending long and borrowing short after 4 bp. Panel B: Stars: Clark–West test at 10, 5, and 1%. Fees: gross and after 4 bp.

8. Size and Power

Asymptotic tests over-reject a true EH at long horizons, and the forecasts recover a linear departure but a nonlinear one of the same size only weakly, so the failure of flexible forecasts to beat the VAR is weak evidence of linearity.

8.1. Size under the EH

The null model is a two-factor model of the short rate in which the EH holds up to a small compounding term and the calendar weighting of the rolls,

$$q_{t+1} = \varphi_q\, q_t + v_t + \varepsilon_{t+1}, \qquad v_{t+1} = \varphi_v\, v_t + \xi_{t+1}, \qquad i_t^{(n)} = \bar{\imath} + \pi^{(n,m)} + \frac{1}{k}\sum_{j=0}^{k-1} E_t\big[q_{t+jm}\big], \quad (30)$$

with q_t the short rate minus its mean $\bar{\imath}$, v_t an information factor that participants observe, and $\pi^{(n,m)}$ a constant premium. The persistence parameters are estimated by maximum likelihood for each pair, shocks are drawn by a wild bootstrap (Mammen, 1993), and the

pattern of missing and masked quotes of the data is imposed; 200 samples are simulated per pair, and the Wald test and DST's VAR forecast are computed as in the data. The in-sample Wald test rejects the true EH at the 5% level in 14% to 79% of the samples (Table 31, Panel A). Out of sample, the Clark–West test rejects a true EH in 4.0% to 6.5% of the samples for the overnight pairs up to two months and for 1W-2W, and in 9.0% to 13.0% for ON-3M, 1W-3W, 1M-2M, and 1M-3M (Panel B). For the VAR, the R^2_{oos} in the data exceeds the 95th percentile of the simulated values for the six pairs with a positive R^2_{oos} (simulated $p \leq 0.025$). The near-integrated rates and the overlap of the returns drive the size distortions; because the model is stylised, with several persistence estimates at the upper bound, these numbers show that the asymptotic tests overstate the evidence rather than measure the exact size.

Table 31: **Size of the Tests under the EH**

	Persistence		A: In-sample Wald test					B: VAR R^2_{oos}		B: Clark–West	
	φ_q	φ_v	Size	95th	Data	p	Matched	95th	p	Size	p
ON-1W	0.999	0.351	52.0	466.3	383.7	0.055	52.0	−1.2	< 0.005	4.5	< 0.005
ON-2W	0.999	0.525	43.0	167.0	164.4	0.065	43.0	−0.7	< 0.005	5.0	< 0.005
ON-3W	1.000	0.599	41.0	152.0	105.9	0.085	41.0	−0.1	0.005	5.5	0.005
ON-1M	1.000	0.653	38.5	115.9	88.4	0.080	38.5	−0.5	0.040	5.0	0.055
ON-2M	1.000	0.741	24.5	34.5	120.4	< 0.005	24.5	0.2	0.875	6.5	0.785
ON-3M	1.000	0.803	14.0	25.9	35.8	0.010	14.0	0.6	0.900	9.0	0.765
1W-2W	1.000	0.716	79.0	1,924.6	546.7	0.225	79.0	−0.7	< 0.005	4.0	0.005
1W-3W	0.729	1.000	66.0	822.3	203.7	0.255	66.0	0.5	0.005	12.5	0.010
1M-2M	0.907	1.000	16.0	26.0	241.1	< 0.005	17.9	0.4	0.025	13.0	0.055
1M-3M	0.931	1.000	15.0	25.4	117.5	< 0.005	15.0	1.1	0.220	11.5	0.160

200 samples per pair from the null model of Equation (30); φ_q, φ_v: persistence of the short rate and the information factor. Size: rejection rate of a true EH at 5%, in percent; 95th: 95th percentile of the simulated statistic; p: share of simulated statistics at least as large as in the data. Panel A: matched: samples whose term-spread volatility is 0.75 to 1.33 times the data's.

8.2. Power against known departures

The simulations keep the predictors, the sample gaps, and the design of the ON-1W pair and simulate only the target,

$$y_t = \sqrt{\mathrm{SNR}}\, f(X_t) + \epsilon_t, \qquad \mathrm{Var}(f) = \mathrm{Var}(\epsilon) = 1, \qquad R^2 = \frac{\mathrm{SNR}}{1 + \mathrm{SNR}}, \tag{31}$$

so that the population R^2 of the true conditional mean, the oracle forecast, is 4.8%, 13.0%, 28.6%, and 50.0% for signal-to-noise ratios of 0.05, 0.15, 0.40, and 1.00. The linear signal is the full-sample projection of the actual ON-1W excess return on the predictors X_t of Equation (5). The nonlinear signal is the product of the bounded (hyperbolic tangent) standardised 1W–ON spread and the bounded daily change of the ON rate, orthogonalised on the predictors so that no linear forecast can capture it. The noise ϵ_t is a moving average of

five independent normal shocks, which reproduces the overlap of the five-day excess returns. Six forecasts compete with the EH forecast: least squares, ridge, random forest, boosting, random features, and the network (32-16-8), with the design of the paper; each design is replicated with five noise draws, so the rejection shares are illustrative, not precise estimates of power.

A linear departure is recovered (Table 32): least squares, ridge, and the network attain 27.9%, 27.3%, and 26.8% at a population R^2 of 28.6%, and the Clark–West test rejects for them in every draw; random features recover somewhat less, and the trees much less (4.7% and 6.4%), because they predict a constant outside the range of rate levels in their window. A nonlinear departure is not: at a population R^2 of 50% the random forest attains 4.8% and boosting -0.1%, while least squares, ridge, and the network stay below zero, and at 28.6% all forecasts lie between -2.9% and -0.5%. The Clark–West test nevertheless detects the nonlinear signal with the trees and random features in 80% to 100% of the draws from a ratio of 0.40 on, because it adjusts for the estimation noise of the larger model, but never with the network, which, as an odd function of its inputs (Section 3.4), cannot represent this signal: the signal is an even function of the departures of the inputs from their means less its linear projection, so its best odd approximation is that projection, which carries about 7% of its variance. A nonlinear departure of this form that explains up to half of the variance would therefore produce the pattern of the data, in which no flexible forecast beats the VAR.

With the simulated target scaled to the mean and volatility of the actual ON-1W excess return, the oracle's fee before costs is at most 4.62 basis points a year, below the 5.55 of perfect sign foresight for ON-1W in the data, and with the nonlinear signal the flexible forecasts earn at most 0.91. The economic bound does not rely on these simulations: it is computed on the actual data and covers any departure, linear or not, that the information at t could reveal, for sign strategies with the position size of Equation (16).

Table 32: **Recovery of a Known Signal: Signal-to-Noise Simulation**

SNR	Population R^2	Oracle	OLS	Ridge	Random forest	Boosting	Random features	NN (32-16-8)
Panel A: linear signal								
0.05	4.8	4.8	4.0 (0.6)	3.8 (0.6)	−2.9 (1.2)	−2.5 (1.2)	1.4 (0.3)	3.6 (0.4)
0.15	13.0	12.5	11.8 (1.3)	11.3 (1.4)	−0.9 (0.8)	−0.3 (1.1)	8.3 (1.1)	11.0 (1.4)
0.40	28.6	28.5	27.9 (0.3)	27.3 (0.6)	4.7 (1.5)	6.4 (3.0)	22.9 (0.9)	26.8 (0.3)
1.00	50.0	48.2	47.8 (1.0)	47.6 (0.8)	14.5 (2.2)	19.3 (3.0)	41.6 (1.0)	46.4 (1.0)
Panel B: nonlinear signal								
0.05	4.8	4.2	−1.0 (0.2)	−0.6 (0.2)	−3.0 (1.1)	−3.6 (1.8)	−1.2 (0.6)	−1.0 (0.4)
0.15	13.0	12.4	−1.1 (0.1)	−0.6 (0.2)	−3.2 (1.0)	−2.8 (1.2)	−1.1 (0.4)	−0.8 (0.2)
0.40	28.6	27.9	−1.7 (0.4)	−0.5 (0.2)	−1.2 (2.5)	−2.9 (4.4)	−0.8 (0.8)	−1.6 (0.8)
1.00	50.0	47.6	−2.3 (0.2)	−0.7 (0.2)	4.8 (1.4)	−0.1 (1.7)	0.5 (0.8)	−2.3 (1.0)
Panel C: linear signal, performance fee before costs (bp a year)								
0.05		1.26	0.97	0.91	0.17	0.15	0.65	1.03
0.15		1.80	1.75	1.72	0.44	0.38	1.35	1.71
0.40		2.92	2.83	2.79	1.03	1.14	2.39	2.76
1.00		4.01	3.90	3.88	1.83	1.86	3.46	3.86
Panel D: nonlinear signal, performance fee before costs (bp a year)								
0.05		1.19	−0.25	−0.27	0.01	0.03	−0.14	−0.32
0.15		2.06	−0.26	−0.19	0.00	−0.10	−0.13	−0.32
0.40		3.32	−0.37	−0.21	0.33	0.21	0.05	−0.36
1.00		4.62	−0.51	−0.11	0.91	0.59	0.06	−0.34
Panel E: linear signal, Clark–West rejections at 5% (% of draws)								
0.05			100	100	40	60	100	100
0.15			100	100	100	100	100	100
0.40			100	100	100	100	100	100
1.00			100	100	100	100	100	100
Panel F: nonlinear signal, Clark–West rejections at 5% (% of draws)								
0.05			0	0	40	20	20	0
0.15			0	0	80	40	20	0
0.40			0	0	100	80	100	0
1.00			0	0	100	100	100	0

Simulated ON-1W targets, Equation (31); f linear (Panels A, C, E) or nonlinear without a linear part (B, D, F). Oracle: the true conditional mean. Panels A and B: mean over 5 noise draws (standard deviation). Panels C and D: target scaled to the mean and volatility of the ON-1W excess return.

9. Conclusion

This paper tests whether the EH fails in an economically meaningful way at the short end once no parametric model of expectations is imposed. Ten repo maturity pairs are forecast out of sample over 1994–2026 by 23 linear and nonlinear models; a forecast is economically meaningful only if it earns its bid–ask spread.

Three results emerge. First, linear and nonlinear forecasts agree: after the search, only overnight versus one week rejects the EH. Second, this rejection is confined to same-day quotes; without them, the out-of-sample R^2 on DST's information is at most 2%. Third, perfect sign foresight bounds the gross value of any sign forecast at 5.55 basis points a year, and no such forecast earns its spread.

The EH therefore does not fail in an economically meaningful way. The remaining rejec-

tion is a statistical property of same-day quotes, not a source of value for a participant who pays the spread; the EH remains, in DST's words, "a useful theory for practitioners in the repo market."

Two limitations remain. The evidence rests on closing dealer quotes taken at an undocumented time, and transaction rates validate only the overnight leg from 2018, so quote timing cannot be separated from a one-day premium. From 2016, a subsample chosen after the break was dated, the premium's level shifts by about one spread, so that a static position, not a forecast, then earns the spread. Transaction-level term rates and a longer sample after the shift would address both.

Declaration of competing interest

The author is employed by Quintet Private Bank (Europe) S.A.; the views expressed are the author's own and do not necessarily reflect those of Quintet Private Bank. The author declares no other known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The repo quotes (Bloomberg tickers USRG1T, USRG1Z, USRGCG2Z, USRGCG3Z, USRGCGA, USRGCGB, and USRGCGC; pricing sources PREB and ICUS) and their bid–ask spreads are proprietary Bloomberg data obtained under a paid licence and cannot be shared. The transaction-based rates (secured overnight financing rate, broad and tri-party general collateral rates; Federal Reserve Bank of New York), the Treasury yields, the federal funds rates, the NBER recession indicator, and the FRED-MD predictors (Federal Reserve Bank of St. Louis), and the FOMC meeting calendars (Board of Governors of the Federal Reserve System) are publicly available. The code (Appendix Appendix A) is available on request and will be made publicly available upon publication.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

The author used an AI tool to support programming and text editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the

content of the published article.

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Appendix A. Replication Details

A holder of a Bloomberg licence can re-run all results from the code, the stated software versions, fixed seeds, and the tuning grids and predictor list below; network training is reproducible up to floating-point differences across hardware.

Appendix A.1. Code, software, and tests

The computations use Python 3.8 with NumPy 1.23, pandas 1.5, SciPy 1.10, statsmodels 0.14, scikit-learn 1.3, and TensorFlow 2.12, and all random elements (initialisations of the networks, random features, tree ensembles, bootstrap draws, and simulations) use fixed seeds. The code is checked by unit tests (among them tests that no information after the forecast origin enters a predictor, a forecast combination, or a position, apart from the ex-post classification of stale quotes described in Section 2.2), and code reviews were run before the final computations. The code will be deposited in a public repository on acceptance and is available on request, as are the construction of the repo series, a log of the pricing source used on each day, a report by maturity and year, the FRED-MD transformation codes, and augmented Dickey–Fuller tests of the levels and transformed series. Elements of the design taken from the literature are cited where they are defined in Section 3; all other elements are the author's own implementation.

Appendix A.2. Models and tuning grids

Table A.33 lists, for each model, its class and tuning grid; the design common to all forecasts is in Table 3.

Appendix A.3. FRED-MD predictors

Table A.34 lists the 126 monthly series of FRED-MD (McCracken and Ng, 2016), vintage 2026-09, January 1959 to August 2026, in the eight groups of the database: output and income (16 series); consumption, orders, and inventories (10); labour market (31); housing (10); money and credit (13); prices (20); interest and exchange rates (22); and the stock market (4: the S&P 500 index, its dividend yield and price–earnings ratio, and the VIX). The groups define the analysis of Figure 7.

The series are used in levels, without a stationarity transformation, and standardised within each estimation window like all predictors; the penalised and tree-based models are designed to handle the resulting collinearity, and least squares is not estimated on this set. Macroeconomic series enter with a publication lag of three months: a value for reference month M is used from the business day after the end of month $M+3$. Interest and exchange rates, the S&P 500 index, and the VIX are known at the end of their month and enter from the business day after it, so they are never used on the same day as the repo quotes; the dividend yield and price–earnings ratio take the macroeconomic lag. These lags cover the publication delays visible in the 326 vintages from August 1999 to September 2026, so every value is available in real time when it is used. Two months are missing in every vintage: October 2025 for 21 series (CPI, the household survey, and real M2, because the BLS surveys were not run during the government shutdown), filled by interpolation within the three-month lag, and April 2020 for two commercial-paper series, filled with the last

Table A.33: **Model Specification: Models and Tuning Grids**

Model	Class	Tuning grid and fixed settings
EH forecast	Benchmark	not tuned; prevailing mean of past excess returns (Table 3)
VAR (DST)	DST's VAR	not tuned; bivariate VAR(5) in levels, bias correction of Pope (1990) with the stationarity adjustment of Kilian (1998), iterated to the expected rolling return
OLS (direct)	Linear	not tuned
Ridge	Regularised linear	`alpha` $\in \{10^{-2}, 10^{-1.5}, \ldots, 10^{3}\}$ (11 points)
Lasso	Regularised linear	`alpha` $\in \{10^{-4}, 10^{-3.5}, \ldots, 10^{0}\}$ (9 points)
Elastic net	Regularised linear	`alpha` $\in \{10^{-4}, 10^{-3.5}, \ldots, 10^{0}\}$ (9 points); `l1_ratio` $\in$ {0.1, 0.5, 0.9}
PCR	Regularised linear	`n_components` $\in$ {1, 2, 3, 5, 8}
PLS	Regularised linear	`n_components` $\in$ {1, 2, 3, 5}
Linear SVR	Regularised linear	`C` $\in \{10^{-3}, 10^{-2}, \ldots, 10^{1}\}$ (5 points); `epsilon` $\in$ {0, 0.1}
RBF SVR	Kernel and local	`C` $\in$ {0.1, 1, 10}; `gamma` $\in$ {scale, 0.01}; `epsilon` $\in$ {0.05, 0.2}
KNN	Kernel and local	`n_neighbors` $\in$ {5, 10, 20}; `weights` $\in$ {uniform, distance}
Tree	Tree-based	`max_depth` $\in$ {2, 3, 5}; `min_samples_leaf` $\in$ {5}
Random forest	Tree-based	`max_depth` $\in$ {3, 6, ∞}; `max_features` $\in$ {0.33, 1}; `min_samples_leaf` $\in$ {5}; fixed: `n_estimators`=300
Boosting	Tree-based	`n_estimators` $\in$ {100, 300}; `learning_rate` $\in$ {0.01, 0.1}; `max_depth` $\in$ {1, 2, 3}; fixed: `subsample`=0.8
Bagging	Tree-based	not tuned; fixed: `n_estimators`=100
Random features (ridge)	Random features	`n_features` $\in$ {100, 1000, 5000}; `gamma` $\in$ {0.02, 0.1}; `alpha` $\in$ {0.1, 10, 1000}
Networks (five) and ensemble	Neural networks	architectures (16), (32), (16-8), (32-16), (32-16-8); `l2` $\in$ {1e-05, 0.0001, 0.001}; `dropout` $\in$ {0, 0.1, 0.2}; `batch_size` $\in$ {64, 256}; Nadam with learning rate 0.001, early stopping with patience 20, at most 500 epochs, 10 seeds; ensemble: equal-weighted average of the five
Combination	Reference	not tuned; equal-weighted average of all forecasts
Combination, 50% to the EH	Reference	not tuned; weight fixed at one half (not in the SPA family)
DMSPE combination	Reference	not tuned; discount factor one, equal weights in the first 250 out-of-sample days (not in the SPA family)
EWMA	Reference	not tuned; half-life 126 business days, realised excess returns only

Hyperparameters minimise the mean squared error on the validation block. Networks: L2 penalty and dropout in each hidden layer, early stopping on the validation error. Seeds are fixed for every random element. Software: Python 3.8, NumPy 1.23, pandas 1.5, SciPy 1.10, statsmodels 0.14, scikit-learn 1.3, TensorFlow 2.12.

value. Series renamed or replaced during the sample are spliced as documented by FRED-MD. As a robustness check, on each day of month M the vintage of month $M-1$ is used with the last available value of each series (Wan et al., 2022); ridge with these predictors is less accurate than the EH forecast with either vintage for most pairs.

Table A.34: **Macro-Financial Predictors**

No.	Series	FRED-MD mnemonic
Macro Indicator - Output and Income		
1	Real Personal Income	RPI
2	Real personal income ex transfer receipts	W875RX1
3	IP: Total Index	INDPRO
4	IP: Final Products and Nonindustrial Supplies	IPFPNSS
5	IP: Final Products (Market Group)	IPFINAL
6	IP: Consumer Goods	IPCONGD
7	IP: Durable Consumer Goods	IPDCONGD
8	IP: Nondurable Consumer Goods	IPNCONGD
9	IP: Business Equipment	IPBUSEQ
10	IP: Materials	IPMAT
11	IP: Durable Materials	IPDMAT
12	IP: Nondurable Materials	IPNMAT
13	IP: Manufacturing (SIC)	IPMANSICS
14	IP: Residential Utilities	IPB51222S
15	IP: Fuels	IPFUELS

No.	Series	FRED-MD mnemonic
16	Capacity Utilization: Manufacturing	CUMFNS
Macro Indicator - Consumption, Orders and Inventories		
17	Real personal consumption expenditures	DPCERA3M086SBEA
18	Real Manufacturing and Trade Industries Sales	CMRMTSPLx
19	Retail and Food Services Sales	RETAILx
20	New Orders for Consumer Goods	ACOGNO
21	New Orders for Durable Goods	AMDMNOx
22	New Orders for Nondefense Capital Goods	ANDENOx
23	Unfilled Orders for Durable Goods	AMDMUOx
24	Total Business Inventories	BUSINVx
25	Total Business: Inventories to Sales Ratio	ISRATIOx
26	Consumer Sentiment Index	UMCSENTx
Macro Indicator - Labour Market		
27	Help-Wanted Index for United States	HWI
28	Ratio of Help Wanted to Number Unemployed	HWIURATIO
29	Civilian Labor Force	CLF16OV
30	Civilian Employment	CE16OV
31	Civilian Unemployment Rate	UNRATE
32	Average Duration of Unemployment (Weeks)	UEMPMEAN
33	Civilians Unemployed - Less Than 5 Weeks	UEMPLT5
34	Civilians Unemployed for 5-14 Weeks	UEMP5TO14
35	Civilians Unemployed - 15 Weeks & Over	UEMP15OV
36	Civilians Unemployed for 15-26 Weeks	UEMP15T26
37	Civilians Unemployed for 27 Weeks and Over	UEMP27OV
38	Initial Claims	CLAIMSx
39	All Employees: Total nonfarm	PAYEMS
40	All Employees: Goods-Producing Industries	USGOOD
41	All Employees: Mining and Logging: Mining	CES1021000001
42	All Employees: Construction	USCONS
43	All Employees: Manufacturing	MANEMP
44	All Employees: Durable goods	DMANEMP
45	All Employees: Nondurable goods	NDMANEMP
46	All Employees: Service-Providing Industries	SRVPRD
47	All Employees: Trade, Transportation & Utilities	USTPU
48	All Employees: Wholesale Trade	USWTRADE
49	All Employees: Retail Trade	USTRADE
50	All Employees: Financial Activities	USFIRE
51	All Employees: Government	USGOVT
52	Avg Weekly Hours: Goods-Producing	CES0600000007
53	Avg Weekly Overtime Hours: Manufacturing	AWOTMAN
54	Avg Weekly Hours: Manufacturing	AWHMAN
55	Avg Hourly Earnings: Goods-Producing	CES0600000008
56	Avg Hourly Earnings: Construction	CES2000000008
57	Avg Hourly Earnings: Manufacturing	CES3000000008
Macro Indicator - Housing		
58	Housing Starts: Total New Privately Owned	HOUST
59	Housing Starts, Northeast	HOUSTNE
60	Housing Starts, Midwest	HOUSTMW
61	Housing Starts, South	HOUSTS
62	Housing Starts, West	HOUSTW
63	New Private Housing Permits (SAAR)	PERMIT
64	New Private Housing Permits, Northeast (SAAR)	PERMITNE
65	New Private Housing Permits, Midwest (SAAR)	PERMITMW
66	New Private Housing Permits, South (SAAR)	PERMITS
67	New Private Housing Permits, West (SAAR)	PERMITW
Macro Indicator - Money and Credit		
68	M1 Money Stock	M1SL
69	M2 Money Stock	M2SL
70	Real M2 Money Stock	M2REAL
71	Monetary Base	BOGMBASE
72	Total Reserves of Depository Institutions	TOTRESNS
73	Nonborrowed Reserves of Depository Institutions	NONBORRES
74	Commercial and Industrial Loans	BUSLOANS

No.	Series	FRED-MD mnemonic
75	Real Estate Loans at All Commercial Banks	REALLN
76	Total Nonrevolving Credit	NONREVSL
77	Nonrevolving Consumer Credit to Personal Income	CONSPI
78	Consumer Motor Vehicle Loans Outstanding	DTCOLNVHFNM
79	Total Consumer Loans and Leases Outstanding	DTCTHFNM
80	Securities in Bank Credit at All Commercial Banks	INVEST
Financial Indicator - Stock Market		
81	S&P's Common Stock Price Index: Composite	S&P 500
82	S&P's Composite Common Stock: Dividend Yield	S&P div yield
83	S&P's Composite Common Stock: Price-Earnings Ratio	S&P PE ratio
84	CBOE Volatility Index (VIX)	VIXCLSx
Financial Indicator - Interest and Exchange Rates		
85	Effective Federal Funds Rate	FEDFUNDS
86	3-Month AA Financial Commercial Paper Rate	CP3Mx
87	3-Month Treasury Bill	TB3MS
88	6-Month Treasury Bill	TB6MS
89	1-Year Treasury Rate	GS1
90	5-Year Treasury Rate	GS5
91	10-Year Treasury Rate	GS10
92	Moody's Seasoned Aaa Corporate Bond Yield	AAA
93	Moody's Seasoned Baa Corporate Bond Yield	BAA
94	3-Month Commercial Paper Minus FEDFUNDS	COMPAPFFx
95	3-Month Treasury Bill Minus FEDFUNDS	TB3SMFFM
96	6-Month Treasury Bill Minus FEDFUNDS	TB6SMFFM
97	1-Year Treasury Constant Maturity Minus FEDFUNDS	T1YFFM
98	5-Year Treasury Constant Maturity Minus FEDFUNDS	T5YFFM
99	10-Year Treasury Constant Maturity Minus FEDFUNDS	T10YFFM
100	Moody's Aaa Corporate Bond Minus FEDFUNDS	AAAFFM
101	Moody's Baa Corporate Bond Minus FEDFUNDS	BAAFFM
102	Trade Weighted U.S. Dollar Index (Advanced Foreign Economies)	TWEXAFEGSMTHx
103	Switzerland / U.S. Foreign Exchange Rate	EXSZUSx
104	Japan / U.S. Foreign Exchange Rate	EXJPUSx
105	U.S. / U.K. Foreign Exchange Rate	EXUSUKx
106	Canada / U.S. Foreign Exchange Rate	EXCAUSx
Macro Indicator - Prices		
107	PPI: Final Demand: Finished Goods	WPSFD49207
108	PPI: Final Demand: Finished Consumer Goods	WPSFD49502
109	PPI: Intermediate Materials	WPSID61
110	PPI: Crude Materials	WPSID62
111	Crude Oil, Spliced WTI and Cushing	OILPRICEx
112	PPI: Metals and Metal Products	PPICMM
113	CPI: All Items	CPIAUCSL
114	CPI: Apparel	CPIAPPSL
115	CPI: Transportation	CPITRNSL
116	CPI: Medical Care	CPIMEDSL
117	CPI: Commodities	CUSR0000SAC
118	CPI: Durables	CUSR0000SAD
119	CPI: Services	CUSR0000SAS
120	CPI: All Items Less Food	CPIULFSL
121	CPI: All items less shelter	CUSR0000SA0L2
122	CPI: All items less medical care	CUSR0000SA0L5
123	Personal Cons. Expend.: Chain Index	PCEPI
124	Personal Cons. Exp: Durable goods	DDURRG3M086SBEA
125	Personal Cons. Exp: Nondurable goods	DNDGRG3M086SBEA
126	Personal Cons. Exp: Services	DSERRG3M086SBEA

Source: FRED-MD (McCracken and Ng, 2016), vintage 2026-09, final revised data. All series enter in levels, without a stationarity transformation, standardised with the mean and standard deviation of each estimation window. Publication lags: Section Appendix A.3.